

University of Western Australia

Master of Physics (Computational Physics)

Quantum Circuits for Solving Differential Equations via Physics-Informed Effective Hamiltonians and GQSP Quantum Imaginary Time Evolution

Dhyan Patel

School of Physics, Mathematics and Computing

Supervised by

Jingbo Wang

12 October 2026

Declaration

This is to certify that:

1. This thesis comprises of my original work.
2. Due acknowledgment has been made in the text to all other materials used.
3. This thesis consists of less than 60 pages inclusive of tables and figures but exclusive of references and appendices.

I authorise the Head of the Department of Physics to make a copy of this thesis to any person judged to have an acceptable reason for access to the information, i.e, for research, study or instruction.



Dhyan Patel
Student ID: 23340229
Date: 12 October 2026



Jingbo Wang
The University of Western Australia
Date: 12 October 2026

Acknowledgments

Thank you to my supervisor, Jingbo Wang, for her guidance and support throughout my research, particularly for her continual encouragement and belief in me.

I would also like to thank my family and friends for their support and encouragement throughout my studies.

Summary of Student Achievement

Abstract

Brief summary.....

Contents

1 Introduction **1**

1.1 Differential Equations as a Target for Quantum Computation 1
1.2 Physics-Informed Hamiltonians and Where Their Cost Lies 1
1.3 Contributions and Scope 1
1.4 Outline 1

2 Background and Literature Review **2**

2.1 Quantum Computation Preliminaries 2
2.1.1 Quantum Gates and Circuits 3
2.1.2 Complexity of Quantum Algorithms 4
2.1.3 Noisy Intermediate-Scale Quantum (NISQ) Era 4
2.2 Quantum Algorithms for Differential Equations 5
2.2.1 Linear-Systems and Linear-Combination Routes 5
2.2.2 Nonlinear Equations: Carleman and Related Linearisations 5
2.2.3 Variational Routes 5
2.2.4 Ground-State Routes and the Physics-Informed Lineage 5
2.2.5 Critical Assessment 5
2.3 Spectral Methods and Their Conditioning 5
2.3.1 Chebyshev Series, Galerkin Truncation and Differentiation 5
2.3.2 Banded Differentiation: the Ultraspherical Method 5
2.3.3 First-Order Reformulation and Least Squares 5
2.3.4 Fourier–Galerkin Discretisation of Periodic Flows 5
2.4 Block Encoding, Linear Combinations of Unitaries and Qubitisation 5
2.5 Quantum Signal Processing and Its Generalisation 5
2.6 Ground-State Preparation by Filtering 5
2.7 Readout and Amplitude Amplification 5
2.8 Synthesis 5

3 The Physics-Informed Effective Hamiltonian Method **6**

3.1 The Chebyshev Latent-Space Model 6
3.2 Hamiltonian Construction 7
3.3 Circuits, Filtering and Readout 8
3.4 Doubled-Space Extension to Nonlinearity 9
3.5 Critical Assessment 9

4 The Standard Form and Where Its Cost Lies **11**

4.1 The Model 11
4.2 Differentiation and Multiplication 12
4.3 Constraints and Data 13
4.4 The Standard Form 15

4.5 Encoding the Derivative . . . 16
4.5.1 The Paper's Encoding . . . 16
4.5.2 An Exact Factorisation . . . 18
4.5.3 Circuits and Cost . . . 18
4.6 What Remains: The Standard-Form Hamiltonian . . . 20

5 The Descriptor Reformulation 22
5.1 The Idea . . . 22
5.2 The Banded Factorisation . . . 23
5.3 Two Layouts and Kernel Equivalence . . . 24
5.4 Constraints and Partial Differential Equations . . . 27
5.5 The Gap of a Carried Derivative . . . 27
5.6 Block Rescaling . . . 28
5.7 The Block Encoding: One Flat Linear Combination . . . 30
5.8 Cost of the Descriptor Encoding . . . 32
5.9 Consequences for the Spectrum and the Filter . . . 34
5.10 Summary . . . 35

6 Beyond a Single Real Field: Systems, Complex Fields and Nonlinearity 36
6.1 Several Fields on One State . . . 36
6.2 Data for Several Fields: Ratio Rows . . . 37
6.3 Complex-Valued Fields . . . 37
6.4 The Doubled Space: Exact Encoding, Degenerate Kernel . . . 38
6.5 The Carleman Lift in Space–Time . . . 39
6.6 Couplings and Products on a Circuit . . . 40
6.7 Fourier–Galerkin Fluids: Burgers and Navier–Stokes . . . 41
6.8 Preparation Versus Projection . . . 42

7 Preparing and Verifying the Ground State 44
7.1 Problems and Model Inputs . . . 44
7.2 The Classical Reference . . . 44
7.3 From Stacked Residual to Quantum Walk . . . 45
7.4 The Filter and Its Parameters . . . 45
7.5 Verification and the Error Budget . . . 46
7.6 Reading Out the Solution . . . 47
7.7 Design of the Numerical Experiment . . . 47
7.8 Software and Computation . . . 47

8 Numerical Study 48
8.1 Reproducing Wu et al. . . . 48
8.2 The Pipeline Is Faithful . . . 51
8.3 Where the Cost Lies . . . 51

8.4 The Descriptor's Gap and Its Rescaling 51
8.5 Systems and Nonlinear Problems 51
8.6 Case Study: Two-Dimensional Navier–Stokes 51
8.7 Summary of Findings 51

9 Discussion **53**
9.1 Why the Descriptor Form Wins, and What It Costs 53
9.2 End-to-End Complexity 53
9.3 What the Reproduction Says About the Published Method 53
9.4 Where the Construction Applies 53
9.5 Feasibility on Fault-Tolerant Hardware 53
9.6 Limitations 53

10 Conclusions and Future Work **54**
10.1 Conclusions 54
10.2 Future Work 54

11 References **55**

Appendices **60**

List of Figures

1 **The odd-offset shifts are one subtraction.** PREPARE (left) puts the offset register d in a uniform superposition over the $N/2$ odd offsets, by an X on its lowest qubit and Hadamards on the rest. SELECT is a ripple-carry subtractor [10] that replaces the system register p by $p - d$ and sets the borrow flag when $p < d$. PREPARE† (right), with post-selection of d and the borrow flag on $|0\rangle$, keeps $\frac{2}{N}\,|k\rangle\langle p|$ for every odd $d = p - k \geq 1$, and the wrapped terms vanish exactly. The block is $U_{\text{odd}}/(N/2)$, so $\alpha = N/2$, on n system and $n + 2$ ancilla qubits. 19

2 **Dense against banded: the two residuals of the same equation.** Sparsity of the stacked residuals of $f'' + 4f' + 4f = 0$ (R2a) at $n = 4$, constraint rows included. Left, the standard form's R_{std}, whose equation block $A_{\text{std}} = \mathbb{G}^2 + 4\mathbb{G} + 4I$ has a full upper triangle. Right, the descriptor's R_{desc}, whose recurrence blocks, equation row and padding are banded; the only dense rows are the rank-one constraint rows, which both forms share. The descriptor's residual has four times as many columns and, at $n = 7$, six times fewer non-zero entries (§ 5.1). 23

3 **The whole descriptor residual is one preparation, one SELECT and one unpreparation.** PREPARE$_R$ loads the signed amplitudes $\text{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ over the joint index of the factored selector registers, and PREPARE$_L^\dagger$ unloads the unsigned $\sqrt{|c_t|/\alpha_A}$, so that the signs of the terms survive. SELECT applies each term's field word on the block register (n_f qubits), then on the system register (n qubits, widened by one overflow qubit) its diagonal, a Z-string or a reflection, and its truncated shift. Post-selecting the selectors, the overflow qubit and the flag on $|0\rangle$ leaves the system rows of A_{desc} divided by α_A, with the block rescaling carried entirely in the coefficients c_t. 31

4 Model configuration: from problem specification to decoded field. 44

5 The GQSP filter circuit. 46

6 Printed choices distort the paper's solutions at the printed resolution. 48

7 Every encoding is verified to its mode's tolerance at every size tested. 49

8 The prepared state reproduces the analytic solution to within the predicted error budget. 49

9 The encoder wall and the Hamiltonian wall are separate gaps. 50

10 Three representative problems: where the comparison, quoted as built, reverses at the ideal encoding bound. 50

11 Block rescaling restores an O(1) gap on one axis, and stops at a double zero or on two axes. 51

12 Three representative problems, solved in both forms. 51

13 Carleman truncation converges geometrically once time is resolved; the doubled space only projects; on Burgers the error tracks a/ν, not ρ. 52

14 Navier–Stokes is reached by exact emulation and a resource estimate, not by circuit simulation. 52
15 The layers of the `pihm` package and the one direction in which they may import. 105

List of Tables

1 Quantum approaches to differential equations: what each assumes, what it costs, and where it fails. 5

2 **Every departure from the printed method removes a stated defect or adds a stated capability.** The method of Wu et al. [1] against the standard form and pipeline of this thesis. Rows above the rule change the residual, and none of them changes the differentiation matrix $\mathbb{G}$, the lifts or the Gram structure of the Hamiltonian except the substitution for R3c and R3d, which changes the equation. Rows below the rule change how the residual is encoded, filtered, solved or read. 17

3 **The structured encoding stays within a constant of the norm; the paper's prefactor does not.** Subnormalisation of the two encodings of the differentiation matrix, relative to $\|\mathbb{G}\|_2$. The structured encoding of Proposition 2 rises towards its limit $2j_{-3/4,1} \approx 2.117$; the paper's prefactor (Equation (3.5)) grows as N^3. The full table to $n = 10$ is Table 15. Exact classical computation of the norms and subnormalisations. 20

4 **The descriptor lowers the degree's exponent where the equation controls every carried derivative.** Norm, gap and filter degree of each form by class of problem, with the block weight ζ set by the a priori rule of § 5.6. Norms follow from the block weights and are proved (Appendix G.7); gaps are measured. A vanishing leading coefficient lowers the standard form's norm too, so the third row is no advantage over the standard form on the same problem (§ 5.6). 34

5 **The Carleman lift prepares; the doubled space projects.** For each nonlinear route: the forms it is run in, its kernel, what a filter returns from an initial state ψ_{init}, and what a trusted kernel certifies. $\rho < 1$ is sufficient for the lift's truncation error to fall as ρ^K; the convergence is measured on every build, not inferred from ρ (§ 6.5). 42

6 The shared constraint layer: every constraint row, in either form, is one of four kinds. 44

7 The benchmark ladder. Each problem adds one capability to those below it; R2–R6 are the paper's Figs 3–7. 45

8 Operator-verification modes: what each compares, up to what size, and to what tolerance. 46

9 Acceptance criteria for every state-preparation run. 47

10 The paper's printed scale factors against the exact solution's $\|b\|^2$ (Equation (4.1)) and the paper-faithful ground state at the printed n. †: the ground state is not trusted (a degenerate kernel, or a gap ratio below 20). Classical computation. 48

11 Where the cost lies: the standard form under the paper's encoding and under the structured exact encoding, against the descriptor form, for Fig. 3b ($\{1, 4, 4\}$, $f(-1) = 0$). Tier: T2 where measured, T3 beyond. 49
12 The gates used in this thesis. 63
13 **The paper's encodings hold their gate and error claims but not their efficiency.** The claims of Wu et al. for their block encodings [1], tested against the rebuilt circuits. Counts are for the rebuilt circuits; the error is to leading order in μ. 74
14 The published encoding of $\mathbb{G}$ against the paper's claims. Classical exact-mode measurement. 76
15 **The structured encoding rises to $2j_{-3/4,1} \approx 2.117$ times the norm; the paper's prefactor grows as $j_{-3/4,1}N^3$.** Subnormalisation of the structured encoding (Proposition 2) and the paper's prefactor (Equation (3.5)) of $\mathbb{G}$, relative to $\|\mathbb{G}\|_2$, for $n = 2$ to 10. Exact classical computation. 80
16 **The chain's row blocks.** Each row block of the descriptor's stacked residual, the blocks of the state it reads and its content. $\mathbb{M}_{a_j}$ is the standard form's multiplication operator (§ 4.2), lifted where the standard form lifts it; the constraint rows are the standard form's (Definition 10) with each power of $\mathbb{G}$ replaced by the block that carries the derivative. 81
17 **A zero of the leading coefficient closes the unscaled descriptor's gap at the rate the Chebyshev resolution at the zero predicts.** Unscaled gap of the descriptor from $n = 5$ to $n = 9$ on controlled equations, against the standard form's. Planning probes, exact classical computation. 84
18 **The block weight need only be right to about a factor of two.** The window of ζ within which the cost $\sqrt{\|H\|/\Delta}/\gamma$ stays within twice its least, at $n = 7$ on one axis and $n = 4$ per axis on two (one weight shown per two-axis problem). Planning probes. 86
19 **The a priori rule lands near the best weight on a grid.** The rule's cost as a multiple of the cheapest ζ on a grid of powers of two, with the gap on the physical subspace and the geometric initial state on the field block. A ratio below one is the grid's spacing. Planning probes, in-sample. 86
20 **The matchings are cheaper only on small lifts; the tensor layout's term count does not grow.** The coupling $\mathbb{A}_K$ of the Carleman lift of Burgers (R9c, $a/\nu = 1$) and of the Navier–Stokes vortex pair (R10c, $a/\nu = 10$) on M modes, encoded alone: in the symmetric layout as matchings, and in the tensor layout as $2K - 1$ terms on its physical states. $\alpha/\|\mathbb{A}_K\|$ is the encoding's subnormalisation over the coupling's norm; CNOTs are per query. Every entry is verified by intermediate representation (IR) to 10^{-10}; "–" marks a symmetric layout not built. Classical construction and verification. 92

21 **The doubled space's kernel grows as N^2.** Dimension of the kernel of $H_\otimes$ for the paper's two quadratic equations, R6a ($4f'' + 2(f')^2 + f = 0$) and R6b ($f'' - 2f^2 + x = 0$), against the state's dimension N^2 and the paper's lower bound 2^{2n-1}. Classical computation (singular value decomposition (SVD) of the stacked residual). 93
22 Accuracy of the classical reference, $1 - |\cos|$ against the exact coefficients, for the standard form by $\texttt{eigh}(R_{\text{std}}^\top R_{\text{std}})$ and by SVD of R_{std}, and for the descriptor form. 100
23 The linear benchmark problems, R1–R5, with the paper's printed invariant constraints. 101
24 The nonlinear and coupled benchmark problems, R6–R10. 101
25 State preparation by problem and form. Tier as marked. 102
26 The paper's solutions at its printed resolution: L^∞ error of the paper-faithful and corrected variants, and the corrected variant's best over n. Classical computation. 103
27 Which printed choice causes the Fig. 5 error: the paper-faithful variant over $n = 4, \ldots, 10$, each correction alone, and both together. 103
28 The claims of Wu et al. tested in this thesis, and their verdicts. 104
29 Dropping the top k rows of the equation block (the Chebyshev tau method) gives an exact kernel with accuracy comparable to the paper's least squares. . . 104
30 Resource classes for the Setonix campaigns. 105
31 The test tiers of the `pihm` package. 105
32 Deviations from the research proposal of May 2025. 106

Notation and Conventions

A quick reference for symbols used across the thesis; each is properly introduced at its first use in the text below.

Typeface. The typeface carries a symbol's role and its subscript carries the form. Fixed matrices that discretise an operator or a constraint are blackboard bold: $\mathbb{G}$, $\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$, $\mathbb{M}$, $\mathbb{n}$, $\mathbb{B}$, $\mathbb{D}^{(0)}$, $\mathbb{S}$, $\mathbb{U}$, $\mathbb{A}$, $\mathbb{F}$. Assembled objects are italic: the residuals A_{std} and A_{desc}, the stacked residuals R_{std} and R_{desc}, the Hamiltonians H_{std} and H_{desc}, and V, W and H'. The subscripts std and desc name the standard and descriptor forms (§ 4.2); bare A, H and R are used only in statements true of both forms, and in § 3, where they are the paper's own residual and Hamiltonian. Wu et al. print their residual and Hamiltonian in blackboard bold; this thesis writes them A and H throughout. Calligraphic letters are used for the Gram map $\mathcal{T}$, the Hilbert space $\mathcal{H}$ and the order symbol $\mathcal{O}$.

Discretisation

Symbol	Meaning	Introduced
n	Qubits per Chebyshev axis (the resolution).	§ 3.1
$N = 2^n$	Truncation dimension of one Chebyshev axis.	§ 3.1
k	Order of an ordinary differential equation.	§ 3.2
p	Degree of a polynomial nonlinearity.	§ 6.5
n_f	Qubits in the field (block) register of the descriptor form.	§ 5.3
$\dim w$	Padded descriptor state dimension, $2^{\lceil \log_2 K \rceil} \prod_a N_a$ with K physical blocks ($2^{\lceil \log_2 (k+1) \rceil} N$ on one axis).	Appendix G.2

The Chebyshev model

Symbol	Meaning	Introduced
T_ℓ	Chebyshev polynomial of the first kind, degree ℓ.	§ 3.1
$\lvert\tau(x)\rangle$	The raw Chebyshev feature state.	§ 3.1
φ_k	Normalised Chebyshev basis functions, $\varphi_0 = T_0/\sqrt{N}$, $\varphi_k = T_k/\sqrt{N/2}$; $\lvert\tau(x)\rangle$ holds their values at x.	§ 4.1
ψ, η, η_e	Normalised coefficient state; its scale; $\eta_e = \lVert b \rVert^2$.	§ 3.1
ψ_e	The ideal state $b/\lVert b \rVert$ of a truncated solution f_N with coefficients b; the model represents f_N exactly at (ψ_e, η_e).	§ 4.1
$\hat{f}^{(j)}$	Coefficients of $f^{(j)}$: one block of the descriptor state.	§ 5.3
w	The descriptor state $(\hat{f}^{(0)}; \ldots; \hat{f}^{(k)})$.	§ 5.3

Operators and constraints

Symbol	Meaning	Introduced
$\mathbb{G}$	Differentiation matrix, $f \mapsto f'$, in coefficient space: dense, strictly upper triangular, $\|\mathbb{G}\|_2 = \Theta(N^2)$ (Wu et al. print it as $\mathbb{G}_n^\top$).	§ 3.2
$\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$	Banded factors of the differentiation matrix, $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$.	§ 5.2
$\mathbb{M}_{x^p}$, $\mathbb{M}_a$	Multiplication by x^p (the exact lift, $2N \times N$) and by a smooth $a(x)$.	§ 3.2
$\mathfrak{n}_1$, $\mathfrak{n}_x$	Product-to-sum folds.	§ 3.4
$\mathbb{B}(x)$, $\mathbb{D}^{(0)}(x_s)$	Rank-one zero (invariant) and collapse (regular) constraint rows.	§ 3.2
x_z, x_m, x_s	Constraint points: a zero, an extremum, a source datum.	§ 3.2
R_0, U_{odd}, Λ	Factors of the structured exact encoding of $\mathbb{G}$.	§ 4.5
$\mathbb{S}_a(x^*)$, $\hat{g}$	Datum-slice row on axis a at x^*; the coefficients of the datum function.	§ 4.3
$\mathbb{U}$	The Chebyshev–Gauss transform used by the nodal fold.	§ 6.6
$\|\mathbb{G}\|_S$	The paper's norm $\max\big(\|\mathbb{G}\mathbb{G}^\top\|_1, \|\mathbb{G}^\top\mathbb{G}\|_1\big)$.	§ 3.3

Residuals and Hamiltonians

Symbol	Meaning	Introduced
A_{std}, A_{desc}	Residual of the standard form; of the descriptor form.	§ 4.2
H_{std}, H_{desc}	Effective Hamiltonian of each form; bare H only in regime-independent statements.	§ 4.2
$R = [R_1; \ldots; R_L]$	Stacked residual, with $H = R^\dagger R$.	§ 7.3
$H_\otimes$	The doubled-space Hamiltonian.	§ 6.4

Block encoding, filtering and state preparation

Symbol	Meaning	Introduced
α, α_R, $\alpha_H = \alpha_R^2$	Subnormalisation: generic, of R, of H.	§ 2.4
V, W	Stacked-residual reflection; qubitised walk.	§ 7.3
$H' = 2H/\alpha_R^2 - I$	The normalised Hamiltonian that V block-encodes.	§ 7.3
Δ	Spectral gap $\lambda_1 - \lambda_0$.	§ 7.4
β	Imaginary time.	§ 7.4
d, q_k	GQSP degree; filter coefficients.	§ 7.4
ε	Target infidelity (10^{-6}).	§ 7.4
ψ_{init}, γ	Initial state; its overlap $\lvert \langle \psi_{\text{ref}} \vert \psi_{\text{init}} \rangle \rvert$ with the reference.	§ 7.4
p_G	GQSP success probability.	§ 7.4
ζ	Block-rescaling constant.	§ 5.6
S	Diagonal of block weights; the rescaled residual is $R_{\text{desc}}S$.	§ 5.6
E	Map from a field's coefficients to the descriptor state that represents it.	§ 5.3
w_f	Weight of the field block in the descriptor's ground state; $\gamma^2 \leq w_f$.	§ 5.5
α_A, Q_m	Subnormalisation of the descriptor's system rows; the reflection $I - 2\lvert m \rangle \langle m \rvert$.	§ 5.7
$\epsilon_{\mathbb{G}}$, μ	Entry error and per-bit coefficient of the published encoding of $\mathbb{G}$.	Appendix D.2

Nonlinearity

Symbol	Meaning	Introduced
K	Carleman truncation order.	§ 6.5
M	Dimension of the base state u.	§ 6.5
ρ	Nonlinearity-to-dissipation ratio; $\rho < 1$ is sufficient, not necessary, for the Carleman truncation error to fall as ρ^K.	§ 6.5
F_1, F_2, $\mathbb{A}_K$	Linear and quadratic terms; the lifted Carleman generator.	§ 6.5

Systems and complex coefficients

Symbol	Meaning	Introduced
C	Number of components (fields) of a system.	§ 6.1
z	A system's state: one block per component.	§ 6.1
$\mathbb{A}$	A system's coupling matrix, $u_i' = \sum_j \mathbb{A}_{ij} u_j$.	§ 6.1
$\mathbb{F}$	Ratio row: fixes the direction of a vector datum and leaves its scale free.	§ 6.2
y, y_a, $y_\perp$	The vector datum; its anchor component; its off-anchor part.	§ 6.2
w_0	A system's stated initial data; it weights the initial state.	§ 6.2
$c = \operatorname{Re} c + i \operatorname{Im} c$	A complex coefficient, in the realification example.	§ 6.3

Acronyms

DNS direct numerical simulation.

GQSP generalised quantum signal processing.

IR intermediate representation.

IVP initial-value problem.

LCU linear combination of unitaries.

NLFT nonlinear Fourier transform.

ODE ordinary differential equation.

PDE partial differential equation.

PINN physics-informed neural network.

QFT quantum Fourier transform.

QITE quantum imaginary-time evolution.

QSVT quantum singular value transformation.

SVD singular value decomposition.

1 Introduction

1.1 Differential Equations as a Target for Quantum Computation

1.2 Physics-Informed Hamiltonians and Where Their Cost Lies

1.3 Contributions and Scope

1.4 Outline

2 Background and Literature Review

2.1 Quantum Computation Preliminaries

In classical computing, the fundamental unit of information is the bit, which can take one of two values: 0 or 1.

The fundamental unit of information in quantum computing is the qubit, which can exist in a superposition of states, allowing it to encode both 0 and 1 simultaneously. This property enables quantum computers to perform certain computations more efficiently than classical computers.

Definition 1 (State Space): The set of all quantum states that a quantum system can occupy is a complex vector space with inner product structure known as a Hilbert space (denoted $\mathcal{H}$). A finite dimensional Hilbert space is isomorphic to $\mathbb{C}^N$ ($\mathcal{H} \cong \mathbb{C}^N$) for some $N \in \mathbb{N}$, where N is the dimension of the Hilbert space. A quantum state is a unit vector in this Hilbert space, represented in dirac notation as $|\psi\rangle \in \mathcal{H}$, with $\langle\psi|\psi\rangle = 1$ (normalisation condition). A quantum state $|\psi\rangle$ can be expressed as a linear combination of basis states $\{|i\rangle\}_{i=0}^{N-1}$, where $|i\rangle$ is the ith basis state of the Hilbert space.

$$|\psi\rangle = \sum_{i=0}^{N-1} \psi_i |i\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{N-1} \end{pmatrix} \tag{2.1}$$

Its hermitian conjugate (or adjoint) is denoted as $\langle\psi| = |\psi\rangle^\dagger$. Note that for some $e^{i\theta}$ for $\theta \in [0, 2\pi)$, the states $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ are physically indistinguishable, and thus correspond to the same physical quantum state. This is known as a global phase factor.

Definition 2 (The Qubit): A quantum bit (qubit) is a two-state quantum system, characterised by $|\psi\rangle \in \mathbb{C}^2$.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \tag{2.2}$$

where $\alpha, \beta \in \mathbb{C}$, $|0\rangle$ and $|1\rangle$ are the computational basis states, and $|\alpha|^2, |\beta|^2$ are the probabilities of measuring the qubit in the $|0\rangle$ and $|1\rangle$ states, respectively. The normalisation condition requires that $|\alpha|^2 + |\beta|^2 = 1$.

Definition 3 (Composite Systems): The state space of a composite quantum system is the tensor product of the state spaces of its subsystems. Consider two quantum states $|\psi\rangle_A \in \mathcal{H}_A$

remove once previous section complete

Write a mini summary: this section will... Add linlin reference here.

and $|\psi\rangle_B \in \mathcal{H}_B$, the composite state $|\psi\rangle_{AB}$ is given by:

$$|\psi\rangle_{AB} = |\psi\rangle_A \otimes |\psi\rangle_B \in \mathcal{H}_A \otimes \mathcal{H}_B \tag{2.3}$$

where $\otimes$ denotes the tensor product and is often hidden by convention. The dimension of the composite state space is the product of the dimensions of the individual state spaces. If a composite system cannot be written as a tensor product of its subsystems, it is said to be entangled. Entanglement is a unique feature of quantum mechanics that has no classical counterpart and plays a crucial role in quantum information processing.

We often consider the state of a composite system of n qubits, which is represented in a 2^n-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^{2^n}$.

2.1.1 Quantum Gates and Circuits

Definition 4 (Quantum Algorithm): A quantum algorithm is a sequence of operations that manipulates qubits to perform a specific computation.

The goal of any quantum algorithm is to manipulate quantum states in a controlled manner to achieve a desired outcome. This is accomplished through the application of quantum gates, which are unitary operations that transform quantum states. Quantum gates can be combined to form quantum circuits, which represent the sequence of operations performed on qubits. Each quantum gate can be implemented using physical operations on hardware, and the design of efficient quantum algorithms is a central challenge in quantum computing research.

May need to re-write

Definition 5 (Unitary Evolution): The evolution of a closed quantum system between two states is described by unitary transformations $U \in \mathbb{C}^{2^n \times 2^n}$, where n is the number of qubits in the system.

$$|\psi\rangle_{final} = U\,|\psi\rangle_{initial}\,, \quad U^\dagger U = UU^\dagger = I \tag{2.4}$$

where $U^\dagger$ is the hermitian conjugate of U and I is the identity operator. Unitary transformations preserve the inner product and thus the norm of quantum states, ensuring that states have a physical interpretation as probability amplitudes.

These unitary transformations are known as quantum gates, and each gate serves a specific purpose in manipulating quantum states.

The quantum circuit diagram offers a succinct and intuitive representation of quantum algorithms, analogous to Feynman diagrams in quantum field theory.

Time flows left to right, and each wire (horizontal line) represents a qubit or a quantum register. Gates are placed sequentially on the wires indicating operations applied to qubits. An example is the Hadamard gate, which creates superposition, and the CNOT gate, which entangles qubits. A full list of quantum gates and their corresponding circuit symbols can be found in .

Create a table in the appendix

2.1.2 Complexity of Quantum Algorithms

Definition 6 (Gate Complexity): The total number of quantum gates used in a quantum circuit.

Definition 7 (Circuit Depth): The maximum number of gates applied sequentially on any qubit in a quantum circuit.

Definition 8 (Ancillary Qubits): Additional qubits introduced into a quantum circuit beyond the primary input qubits, typically initialised to a baseline state (such as $|0\rangle$) to assist with intermediate calculations, temporary storage, or entanglement generation.

Definition 9 (Big-$\mathcal{O}$ Notation): Big-$\mathcal{O}$ notation is a mathematical notation used to describe the asymptotic behaviour of functions.

$$\mathcal{O}(f(n)) = \{g(n) : \exists c > 0, n_0 \in \mathbb{N} \text{ such that } 0 \leq g(n) \leq cf(n) \text{ for all } n \geq n_0\} \tag{2.5}$$

In the context of quantum algorithms, it is used to express the scaling of gate complexity, circuit depth, and space requirements with respect to the number of qubits n.

For example, if a quantum algorithm has a gate complexity of $\mathcal{O}(n^2)$, it means that the number of gates required grows quadratically with the number of qubits.

A quantum circuit is said to be **efficient** if its circuit depth, gate complexity, and the number of ancillary qubits all scale polynomially with the number of qubits n.

2.1.3 Noisy Intermediate-Scale Quantum (NISQ) Era

Write this section. Use Josh's as a guide.

Table 1: Quantum approaches to differential equations: what each assumes, what it costs, and where it fails.

Approach	Assumptions	Query cost	Output	Failure mode
Linear systems (HHL family)				
Linear combination of Hamiltonian simulations				
Carleman linearisation				
Variational				
Physics-informed				
Hamiltonian [1]				
This work				

2.2 Quantum Algorithms for Differential Equations

2.2.1 Linear-Systems and Linear-Combination Routes

2.2.2 Nonlinear Equations: Carleman and Related Linearisations

2.2.3 Variational Routes

2.2.4 Ground-State Routes and the Physics-Informed Lineage

2.2.5 Critical Assessment

2.3 Spectral Methods and Their Conditioning

2.3.1 Chebyshev Series, Galerkin Truncation and Differentiation

2.3.2 Banded Differentiation: the Ultraspherical Method

2.3.3 First-Order Reformulation and Least Squares

2.3.4 Fourier–Galerkin Discretisation of Periodic Flows

2.4 Block Encoding, Linear Combinations of Unitaries and Qubitisation

2.5 Quantum Signal Processing and Its Generalisation

2.6 Ground-State Preparation by Filtering

$$d = \Theta\left(\sqrt{\frac{\alpha_H}{\Delta}}\,\log\frac{1}{\varepsilon}\right), \qquad \alpha_H = \alpha_R^2. \tag{2.6}$$

2.7 Readout and Amplitude Amplification

2.8 Synthesis

remove once previous section complete

3 The Physics-Informed Effective Hamiltonian Method

This chapter presents the physics-informed effective Hamiltonian method of Wu, Paine, Philip, Gentile and Kyriienko [1] and evaluates it. The method turns a differential equation into a ground-state problem in three steps. The unknown function is expanded in Chebyshev polynomials whose coefficients are the amplitudes of an n-qubit state $|\psi\rangle$; the equation and each of its data conditions become matrix operators that annihilate $|\psi\rangle$ when it holds the solution's coefficients, so that the paper's introductory example $f''+f=0$ becomes $(\mathbb{G}^2+I)\,|\psi\rangle=0$, with $\mathbb{G}$ the matrix that differentiates a Chebyshev series; and the Gram matrices of these operators are summed into an effective Hamiltonian H whose ground state is the solution. Preparing that ground state needs a block encoding of H, a filter that removes its excited spectrum, and a measurement that reads the field back. The construction descends from differentiable quantum circuits [2] through their Chebyshev latent space [3].

The paper's case for the method is that it needs no variational training, that its solution can be evaluated at any point of the domain rather than on a grid, and that "necessary terms have $O(\mathrm{poly}(n))$ scaling and can be implemented efficiently" [1]. §§ 3.1 to 3.4 present the model, Hamiltonian, circuits and nonlinear extension in the paper's own terms, and § 3.5 assesses them; the form this thesis runs is stated in § 4.

3.1 The Chebyshev Latent-Space Model

The model is an overlap [1], [3]. On n qubits, with $N=2^n$, the paper defines the quantum Chebyshev state or feature map

$$|\tau(x)\rangle_n = \frac{1}{2^{n/2}}\,T_0(x)\,|0\rangle^{\otimes n} + \frac{1}{2^{(n-1)/2}}\sum_{k=1}^{N-1} T_k(x)\,|k\rangle\,, \tag{3.1}$$

whose components are the first N Chebyshev polynomials of the first kind, and writes the solution as the scaled overlap

$$f_Q(x)=\sqrt{\eta}\,f_q(x), \qquad f_q(x)=\langle\tau(x)|\psi\rangle_n\,, \qquad \langle\psi|\psi\rangle=1, \tag{3.2}$$

where the amplitudes of $|\psi\rangle$ are the series' weighted coefficients and $\eta>0$ is a scaling factor that carries the norm a unit vector cannot (the paper's Eqs 2–3). The N states $|\tau(\bar{x}_j)\rangle_n$ at the Chebyshev nodes $\bar{x}_j=\cos[\pi(j+\frac{1}{2})/N]$ are orthonormal, but elsewhere $|\tau(x)\rangle_n$ is not a unit vector; a circuit prepares its normalised version by post-selection, and the paper discards the norm, "since it can be implicitly incorporated into the scaling factor", so its derivations use the unnormalised state. Every constraint below is linear in $|\psi\rangle$ and fixes only its direction. The scale is recovered after the ground state $|\psi_g\rangle$ has been prepared, from one known value of the

solution, the *regular* datum $f(x_s) = y_s$: $\sqrt{\eta} = y_s / \langle \tau(x_s) | \psi_g \rangle_n$ (the paper's Eq. 11).

3.2 Hamiltonian Construction

Differentiation acts on the coefficients by a fixed matrix, $f'_q(x) = \langle \tau(x) | \mathbb{G} | \psi \rangle_n$, with $\mathbb{G}$ the Chebyshev differentiation matrix (the paper's Eq. 4). The paper defines the lower-triangular $\mathbb{G}_n$ and applies its transpose, $\mathbb{G}_n^\top$, to the coefficient vector; that upper-triangular matrix is written $\mathbb{G}$ throughout. $\mathbb{G}$ is dense, because the derivative of T_p contains every T_k of the opposite parity below it [4]: one high coefficient of f reaches every lower coefficient of f'. Its entries are given in § 4.2. Higher derivatives are its powers, so a homogeneous linear ordinary differential equation (ODE) with constant coefficients, $\sum_j a_j f^{(j)} = 0$, is represented by

$$A \, |\psi\rangle = 0, \qquad A = \sum_j a_j \, \mathbb{G}^j \tag{3.3}$$

(the paper's Eq. 15, there of second order). Every derivative is here a power of $\mathbb{G}$; § 4 restates this representation in full.

A variable coefficient cannot be substituted in the same way, because the Chebyshev coefficients of a product are not the products of the factors' coefficients. The paper uses the lift of Paine et al. [3], $x^p \, \langle \tau(x) |_n = \langle \tau(x) |_{n+1} \, \mathbb{M}_{x^p}$ (its Eq. 5), a constant $2N \times N$ matrix that carries a power of x onto a register one qubit wider, so that every term of the equation shares the $(n+1)$-qubit basis; a polynomial coefficient $a_j(x)$, the only kind the paper's examples use, enters as the sum $\mathbb{M}_{a_j}$ of the lifts of its monomials. A source $r(x)$ does not scale with $|\psi\rangle$ and is brought into the same form through the regular datum: the collapse row $\mathbb{D}^{(0)}(x_s) = \mathbb{B}_n(x_s)/f(x_s)$, with $\mathbb{B}_n$ defined below, maps the solution's coefficients to the constant $1/\sqrt{\eta}$, so a source expanded in a Maclaurin series truncated at order $\bar{p}$ enters the equation's operator as $-\mathbb{M}_r \mathbb{D}^{(0)}(x_s)$, with $\mathbb{M}_r = \sum_{p \le \bar{p}} c_p \mathbb{M}_{x^p}$ and c_p the Maclaurin coefficients of r (the paper's Eqs 11–14). The paper's three examples with transcendental sources take $\bar{p} = 3$, 5 and 7.

Boundary and initial conditions enter as data constraints built from the rank-one row $\mathbb{B}_n(x) = \sqrt{2^n} \, |0\rangle^{\otimes n} \, \langle \tau(x) |_n$ (the paper's Eqs 9–10), which maps the model to the constant function equal to its value at x. An *invariant* constraint has a zero target and is unchanged by scaling: $f(x_z) = 0$ gives $\mathbb{B}_n(x_z) \, |\psi\rangle = 0$, and $f'(x_m) = 0$ gives $\mathbb{B}_n(x_m) \mathbb{G} \, |\psi\rangle = 0$. The paper places these at boundaries where the solution vanishes or at interior zeros and extrema of the known solution, printed to two decimal places for the inhomogeneous examples ($x = -0.2$, -0.73, 0.91 and -0.75), and uses one invariant constraint for each second-order ordinary example, with the regular datum fixing the scale.

Two variables are carried on two n-qubit registers with the product state $\langle \tau(x) |_n \otimes \langle \tau(y) |_n$; derivatives act as $\mathbb{G} \otimes I$ and $I \otimes \mathbb{G}$, and a constraint on a boundary as $\mathbb{B}_n(x_z) \otimes I$ (the paper's Eqs 22–28). The invariant sets of the three partial differential equation (PDE) examples are chosen by hand. Laplace's equation uses its three homogeneous sides, and its non-zero side

$f(x,1) = \sin(\pi(x+1)/2)$ appears in no constraint. The heat equation $\partial_t f = \frac{1}{25}\partial_x^2 f$ uses $f(t,\pm 1) = 0$ and $\partial_x f(t,\pm\frac{3}{4}) = 0$, and the wave equation $\partial_t^2 f = 4\partial_x^2 f$ uses $f(t,\pm 1) = 0$, $f(\frac{1}{8},x) = 0$ and $f(\frac{7}{8},x) = 0$. The points $\pm\frac{3}{4}$, $\frac{1}{8}$ and $\frac{7}{8}$ are zeros of the known solutions.

Each operator is turned into a Gram matrix, and the Gram matrices are summed:

$$H = \mathcal{T}(A) + \sum_i \mathcal{T}(\mathbb{B}_i), \qquad \mathcal{T}(X) = X^\top X, \qquad \langle\psi|H|\psi\rangle = \|A\psi\|^2 + \sum_i \|\mathbb{B}_i\psi\|^2, \tag{3.4}$$

where $\mathbb{B}_i$ runs over the invariant constraints (the paper's Eq. 16). The energy of a state is its total squared violation of the equation and its data. The paper describes H as "a real symmetric positive-definite matrix with distinct eigenvalues" and recasts the equation as the eigenvalue problem $H\,|\psi_g\rangle = \lambda_{\min}\,|\psi_g\rangle$ with $\lambda_{\min} \approx 0$. It calls H physics-informed because $\mathcal{T}(A)$ embeds the differential constraint as the loss of a physics-informed neural network (PINN) embeds it, while the $\mathcal{T}(\mathbb{B}_i)$ embed the data, but nothing is trained: the ground state is prepared. The paper's worked residual at $N = 4$ and the entries of the lifts are given in Appendix C.

3.3 Circuits, Filtering and Readout

Block encodings. The paper block-encodes each term of H separately and combines them by a linear combination of unitaries (LCU) (§ 2.4). The rank-one rows $\mathbb{B}_n(x)$ and $\mathbb{D}^{(0)}(x_s)$ pair the inverse Chebyshev feature map with a reflection about the all-zero state (its Figs 2 and 8), on $n+2$ qubits at subnormalisation $\sqrt{2^{n+1}}$, and run "with high probability". $\mathbb{G}$ is encoded approximately by two conditioned subroutines that load its entries as rotation amplitudes (its Fig. 9). The circuit's top-left block is $\mathbb{G}$ divided by the prefactor

$$\alpha_{\mathbb{G}}^{\text{paper}} = \left(2^{n-1} + 2^n\right)\|\mathbb{G}\|_S, \qquad \|\mathbb{G}\|_S = \max\left(\|\mathbb{G}\mathbb{G}^\top\|_1,\ \|\mathbb{G}^\top\mathbb{G}\|_1\right), \tag{3.5}$$

on $2n+3$ qubits, "with negligible errors in matrix entries that exponentially decrease as n increases", and with $O(n^3)$ gates once the multi-controlled gates are decomposed: "polynomial in n and thus efficient" [1]. The registers, subroutines and angles are given in Appendix D.1.

The imaginary-time filter. The ground state is prepared by quantum imaginary-time evolution (QITE), which multiplies each eigencomponent of an initial state by $e^{-\beta E}$, so that the lowest dominates once β exceeds the inverse gap. The paper applies it to $\tilde{H} = H/\|H\|_F$ by an indefinite-parity quantum singular value transformation (QSVT) sequence whose phases are fitted to $e^{-\beta x}$ (§§ 2.5 and 2.6); the paper's imaginary time t is written β here, since τ names the feature state. The time is set by an "approximate lower bound",

$$\beta \;\geq\; \frac{\lambda_{\max}}{\lambda_2\,(n-1)}, \tag{3.6}$$

with $\lambda_{\max}$ and λ_2 the largest and second-smallest eigenvalues of $\tilde{H}$ (or H), and two examples are reported to need $\beta = 15$ and 8 with even and odd polynomial degrees $(6, 7)$. For the cost,

the paper expects a preparation time of order $\Delta^{-1}\,\mathrm{polylog}(1/\varepsilon)$ by eigenstate filtering, or $\Delta^{-1/2}$ by algorithmic thermalisation, and notes that the gaps "are problem-dependent and in general are not easy to bound". Its phase sets, the dilation of $\tilde{H}$ and the parameters of its examples are given in Appendix K.5.

Readout. The field is read out point by point by an interferometric overlap measurement [5], [6]. Measuring the all-zero outcome after the inverse feature map returns only the squared overlap, so the paper measures two probabilities, one for the prepared state and one for that state superposed with a reference, and combines them into the signed overlap $\langle\tau(x)|\psi_g\rangle$ and, with η, the field value $f_Q(x)$ (its Eqs 36–39). The identity is restated in Equation (7.12).

3.4 Doubled-Space Extension to Nonlinearity

A nonlinear term such as $(f')^2$ cannot be expressed as a linear operator on $|\psi\rangle$. The paper's treatment doubles the space: the unknown becomes $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$ on $2n$ qubits, a product of two model functions is the overlap of $|\Psi\rangle$ with a product of two feature states, and the product-to-sum rule folds that product back into one Chebyshev series on one extra qubit,

$$\langle\tau(x)|_n \otimes \langle\tau(x)|_n = \langle\tau(x)|_{n+1}\ \mathtt{m}_1, \qquad \mathtt{m}_1 \in \mathbb{R}^{2N\times N^2} \tag{3.7}$$

(its Eq. 8), so that $f^{(i)}f^{(j)}$ becomes $\mathtt{m}_1(\mathbb{G}^i \otimes \mathbb{G}^j)$ on $|\Psi\rangle$. Linear terms are carried into the same space by tensoring them with the collapse row $\mathbb{D}^{(0)}(x_s)$, which supplies the constant of one copy (its Eq. 30), and the Hamiltonian is built as in Equation (3.4). The paper demonstrates the construction on two quadratic ordinary equations at $n = 2$ and reports that the Hamiltonian then has at least 2^{2n-1} degenerate zero-energy states, so that solving the equation is "converted to that of seeking a degenerate state" that "closely matches the analytical solution". The construction is quadratic: a term of degree p needs p copies, a register of dimension N^p (Appendix H.2), as Paine's concurrent work also observes [7].

3.5 Critical Assessment

Strengths. The construction is unified: ordinary and partial differential equations and their data constraints take one least-squares form, and nonlinear terms have a route into it. Its kernel is exact whenever the solution is a polynomial within the truncation. The Chebyshev basis needs no spatial grid, so the model and all its derivatives are available at every $x \in [-1, 1]$ from one prepared state, and the Hamiltonian has few terms, each with polynomially many gates.

Weaknesses in the cost. The most serious weakness is that the spectrum of the Hamiltonian, not its circuits, may fix the cost. A polynomial filter must suppress the excited spectrum relative to the ground state, and its degree grows as the square root of the ratio of the encoding's subnormalisation, which is at least $\|H\|$ for any block encoding, to the gap (Equation (2.6)).

Because H is the Gram matrix of the residual, that square root is the residual's own norm over the root of the gap. Every derivative enters the residual as a power of $\mathbb{G}$, whose norm grows as N^2 (§ 4.2), so for an equation whose leading coefficient does not vanish, and a gap that does not grow with N, the number of queries grows with each derivative the equation eliminates through $\mathbb{G}$, whatever the circuit. A least-squares functional built from operators of high order is the setting in which first-order-system least squares carries the derivatives as unknowns instead [8]. The expected time $O(\Delta^{-1}\,\mathrm{polylog}(1/\varepsilon))$ leaves its units open: if Δ is the gap of $\tilde{H}$, it is the gap of H divided by $\|H\|_F$, and the norm re-enters there. The paper's report of "a sizable gap for the studied problems" is therefore not yet evidence about the cost, since a gap is sizable only against the norm. A second weakness sits on top of the first: the prefactor of Equation (3.5) is not compared with $\|\mathbb{G}\|_2$, which it exceeds by a factor that grows with N, so the circuit for $\mathbb{G}$ is loose even against the matrix it encodes. § 4.5.1 and § 4.6 separate the two.

Weaknesses in the evidence. The paper's examples support the method less than they appear to. Each inhomogeneous example combines a rounded constraint point with a truncated Maclaurin source, and either alone sets an error floor that no increase in n removes: a zero imposed where the solution does not vanish contradicts the equation, and a truncated source poses a different equation (§§ 4.2 and 4.3, Appendix K). The PDE examples take their invariant points from the known solution, which a new problem does not supply, and their invariant sets omit the initial or boundary function that fixes the solution, so the kernel degenerates as n grows beyond the printed resolution. For the heat equation the natural completion, its initial function at $t = 0$, lies inside the time domain $[-1, 1]$ and would pose a backward, ill-posed problem on $[-1, 0]$ (§ 4.3). The description of H as positive definite with distinct eigenvalues is stronger than a sum of Gram matrices allows, and the case the method aims at, a solution represented exactly, is a zero eigenvalue (§ 4.4). The time rule (Equation (3.6)) contains no ε, so it cannot certify a target accuracy, and it needs $\lambda_{\max}$ and λ_2, which the method does not supply in advance (Appendix K.4).

Weaknesses in scope. The nonlinear route projects rather than prepares: any function of the Hamiltonian acts as a scalar on its zero-energy eigenspace, so a filter returns the projection of the initial state onto a kernel of dimension at least 2^{2n-1}, and the direction it picks within that kernel is set by the initial state, not by the equation (§ 6.4). The readout identity treats the prepared state as the ground state with certainty, although a filter followed by post-selection succeeds with probability below one, and it returns one point of the field per measurement campaign (§ 7.6). The construction is stated for one real scalar field, and the paper sets complex-valued states aside, since with a Fourier basis "the resulting total energy function is of a complicated form, rendering it difficult to find the ground state" (§ 6).

Open problems. The paper's Discussion leaves three questions open: "studying the spectral gap dependence is an important question for the future work"; the choice of basis, for which "one guiding principle here can be the cost of differentiation, where the goal is minimizing

the depth of quantum circuits to implement derivatives, multiplication etc."; and the data constraints, whose choice it calls "crucial for regularizing solutions and growing the gap" [1]. Paine's concurrent work [7] finds, on a few small instances, that the normalised gap depends on the Hamiltonian's construction and the function encoding, but gives no scaling with resolution or circuits, so it raises the first question without answering it. The first weakness above is that question's quantitative form: the gap must be set against the Hamiltonian's norm, and the norm is set by the differentiation matrix, the subject of the second.

The assessment leaves three questions, which the next three chapters answer in turn:

(i) *Does the method's total number of queries originate in the particular encoder circuits, or in the Hamiltonian itself?* (§ 4)

(ii) *Can the residual be reformulated so that its Hamiltonian is banded and well conditioned without changing its solution, and what does that cost in register size and spectral gap?* (§ 5)

(iii) *Can the solution of a nonlinear equation be prepared as the unique ground state of a Hamiltonian, rather than selected by projection from a degenerate kernel?* (§ 6, which first carries several fields and complex-valued fields as coupled linear systems, since the lift that answers it is one)

Beneath all three lies one question: where the Hamiltonian's norm, and with it the filter's degree, comes from, and whether changing what the residual carries can lower it.

4 The Standard Form and Where Its Cost Lies

Does the total number of queries originate in the particular encoder circuits, or in the Hamiltonian itself? This is the first question of § 3.5, and the two are separable because a filter's degree depends on the encoding only through its subnormalisation α_H (Equation (2.6)), and every block encoding of H has $\alpha_H \geq \|H\|$. We call the ratio $\alpha_H/\|H\| \geq 1$ the *encoder's cost* and $\|H\|$ itself the *Hamiltonian's cost*: the first belongs to the circuit, the second to the operator. The degree is linear in $\alpha_R = \alpha_H^{1/2}$, so the encoder's cost enters it as $\alpha_R/\|R\|$. The construction that separates them is that of Wu et al. [1], built on the Chebyshev latent space and lifts of Paine et al. [3]. We call it the *standard form*: the state holds the field alone, and every derivative is eliminated through the dense differentiation matrix $\mathbb{G}$, so that a term $f^{(j)}$ acts as $\mathbb{G}^j$ on the field's coefficients and the equation is one block of a stacked residual. Each departure from the printed method is argued where it occurs and collected in Table 2. §§ 4.1 to 4.4 build the form, § 4.5 encodes its derivative, and § 4.6 shows what no encoding removes.

4.1 The Model

We use the paper's model, Equations (3.1) and (3.2), unchanged, and write it in the orthonormal basis that makes its scale explicit. The prefactors of the feature map define the basis functions

$\varphi_0 = T_0/\sqrt{N}$ and $\varphi_k = T_k/\sqrt{N/2}$ for $1 \le k < N$, which are orthonormal under the discrete inner product on the N Chebyshev nodes, and $\langle\tau(x)| = (\varphi_0(x), \ldots, \varphi_{N-1}(x))$ is the row of their values at x. A truncated solution $f_N = \sum_{k<N} b_k\varphi_k$ is then $f_N(x) = \langle\tau(x)|b\rangle = \|b\|\,\langle\tau(x)|\psi_e\rangle$ with $|\psi_e\rangle = b/\|b\|$, so the model f_Q of Equation (3.2) represents f_N exactly at $|\psi\rangle = |\psi_e\rangle$ and $\eta = \eta_e$, where, by Parseval's identity under the Chebyshev weight (Appendix B.3),

$$\eta_e = \|b\|^2 = \frac{N}{\pi}\int_{-1}^{1} \frac{f_N(x)^2}{\sqrt{1-x^2}}\,dx. \tag{4.1}$$

The state carries the direction of b and the scale its length, and η_e is fixed by the solution and the resolution alone, growing with N for a smooth solution. This is representability, not a requirement on the prepared state: the ground state $|\psi_g\rangle$ is set by the residual's kernel and approaches $\pm\,|\psi_e\rangle$ only as far as f_N satisfies the constraints. The regular datum recovers the scale and its sign: $y_s/\,\langle\tau(x_s)|\psi_g\rangle$ is negative when the ground state, defined only up to sign, opposes the datum, so the paper's $\eta > 0$ is a convention and the decoded field carries the sign.

The feature vector is kept unnormalised, as the paper's derivation uses it. Its squared norm oscillates about one and reaches $(2N-1)/N$ at $x = \pm 1$ (Appendix B.3), and every constraint row of § 4.3 inherits that weight. Where the residual has no exact kernel the ground state is a least-squares compromise between its rows, and normalising $\tau(x)$ point by point, which gives the paper's $\tilde{\tau}(x)$, would change the rows' relative weights and with them the solution; it would also make collapse rows at different points disagree. The rows are therefore built, and block-encoded, from the unnormalised vector. A function of several variables is carried on one register per axis, each with its own n, with the feature vector $\langle\tau(x_1)| \otimes \langle\tau(x_2)| \otimes \ldots$ in Kronecker order. The running example of this chapter is the paper's Fig. 3b: the critically damped oscillator $f'' + 4f' + 4f = 0$ with $f(-1) = 0$ and $f(0) = \frac{1}{2}$, whose solution is $f = \frac{1}{2}(1+x)e^{-2x}$; in the benchmark set of Table 7 it is labelled R2a.

4.2 Differentiation and Multiplication

In the normalised basis the differentiation matrix has the entries

$$[\mathbb{G}]_{k,p} = \begin{cases} 2p, & p > k \ge 1,\ p-k \text{ odd}, \\ \sqrt{2}\,p, & k = 0,\ p \text{ odd}, \\ 0, & \text{otherwise}, \end{cases} \tag{4.2}$$

and it is exact on the represented space, since the derivative of a polynomial of degree below N stays in it [4]. $\mathbb{G}$ is upper triangular, dense and nilpotent. $\ker \mathbb{G}^j$ is the space of polynomials of degree below j, and about half the positions above the diagonal are non-zero. Its norm grows as N^2: $\|\mathbb{G}\|_2/N^2$ converges to $1/(2j_{-3/4,1}) \approx 0.472$, with $j_{-3/4,1}$ the first positive zero of the Bessel function $J_{-3/4}$, and $\|\mathbb{G}^k\|_2/N^{2k}$ converges to a positive constant for every k. Because the basis is orthonormal at the Chebyshev nodes, $\mathbb{G}$ is orthogonally similar to the matrix that

differentiates nodal values, so the growth is not an artefact of the normalisation. The structure, the similarity and the bounds $N^2/(4\sqrt{2}) \le \|\mathbb{G}\|_2 \le N^2/\sqrt{2}$ are proved in Appendix B.4, and the two limits in Appendix E.4.

A polynomial coefficient $a(x)$ acts by the lift of Paine et al. [3], $x^p \langle\tau(x)|_n = \langle\tau(x)|_{n+1} \mathbb{M}_{x^p}$ (the paper's Eq. 5), extended by linearity to $\mathbb{M}_a = \sum_p a_p \mathbb{M}_{x^p}$. It is exact whenever $\deg a \le N$, a constant enters as $a_0\mathbb{M}_1 = a_0\sqrt{2}\,[\,I;\,0\,]$, and $\|\mathbb{M}_a\|_2 \le \sqrt{2}\sup|a|$ (Proposition 13). The equation $\sum_j a_j(x) f^{(j)} = r(x)$ then has the equation block

$$A_{\text{std}} = \sum_j \mathbb{M}_{a_j}\mathbb{G}^j - \mathbb{M}_r\,\mathbb{D}^{(0)}(x_s), \qquad \mathbb{M}_r\,\mathbb{D}^{(0)}(x_s) = \frac{|\hat{r}\rangle\,\langle\tau(x_s)|}{f(x_s)}, \tag{4.3}$$

in which the source enters through the collapse row of § 4.3, with $\hat{r}$ the coefficients of r on the lifted register. The subscript marks the standard form, defined in Definition 10.

The paper expands a transcendental source in a Maclaurin series truncated at order $\bar{p}$ (§ 3.2). A truncated source poses a different equation, whose solution the ground state then approximates. The order-7 series of the source $x\,e^{-2x}\sin 4x$ of the paper's Fig. 5d (R4d, $f'' + 4f' + 20f = x\,e^{-2x}\sin 4x$) differs from the source by more than the source itself at $x = -1$, and the resulting error is a floor that no increase in n removes. The standard form loads the Chebyshev coefficients of the source instead, through the same collapse row, so the source is as accurate as the series that represents the solution.

The paper treats at most two variables. On any number of axes a term is the Kronecker product over the axes of $\mathbb{M}_a\mathbb{G}^{j_a}$, each axis lifted if any term on it needs the exact product, and a constant coefficient on a lifted axis enters through $\mathbb{M}_1$ (Appendix C.2). The benchmark problems use at most two.

4.3 Constraints and Data

The constraints are built from the paper's rank-one rows (its Eqs 9–11). The zero row $\mathbb{B}(x) = \sqrt{N}\,|0\rangle\,\langle\tau(x)|$ maps a series to the constant function $f_N(x)$, so $\mathbb{B}(x)\mathbb{G}^j$ annihilates exactly the series whose j-th derivative vanishes at x. The collapse row $\mathbb{D}^{(0)}(x_s) = \mathbb{B}(x_s)/f(x_s)$ maps the solution to the constant 1 whatever its scale, which is what lets a source enter Equation (4.3). The scale is fixed after preparation (§ 4.1).

A zero row is exact only at a point where the solution's derivative of that order vanishes. The paper places its inhomogeneous examples' invariant constraints at zeros and extrema of the known solution rounded to two decimal places ($f'(-0.2) = 0$ on R4a, for an extremum at $x \approx -0.196$), so the row demands a zero the solution lacks, least squares trades it against the equation, and the error floors: 3.1×10^{-3} on R4a for every n from 4 to 10, against 3.8×10^{-14} at $n = 4$ at the exact point. The standard form takes the exact point, the root of the solution's derivative nearest the paper's value. That root is set-up that depends on the answer: we compute it from the reference solution, outside the method, so a new problem must supply its

constraint points with its data. We keep the invariant row, rather than impose the problem's stated data, so that the residual keeps the paper's rows and only the defect changes. The stated data could enter without the answer: $f'(0) = -\frac{1}{8}$ on R4a, for instance, is a point datum row of Definition 10, which no run here uses. On R4b to R4d the rounded point and the truncated source each floor the error on their own, and the error converges only with both replaced (Appendix K.2).

For an ordinary equation of order k whose leading coefficient has no zero on $[-1, 1]$, the invariant constraints fix the solution up to its scale, which the regular datum then fixes, exactly when they have rank $k - 1$ on the equation's k-dimensional solution space (Proposition 12). The rank is that on the solutions, not of the functionals, so $k - 1$ constraints chosen without regard to the equation can fail. Where the count does not apply, for an inhomogeneous equation or a vanishing leading coefficient, and for the gap at a given n, the computed kernel decides (§ 7.2).

A partial differential equation has an infinite-dimensional solution space and no such count. The paper's invariant sets (§ 3.2) omit the function that fixes the solution: on Laplace's equation (R5a), every mode $\sin \frac{\ell\pi(x+1)}{2} \sinh \frac{\ell\pi(y+1)}{2}$ satisfies the equation and its three homogeneous sides, and only $\ell = 1$ matches the omitted top side (Appendix C.5). Truncation hides the other modes at the printed resolution, where the printed set leaves a trusted one-dimensional kernel, and admits them as n grows: at $n = (5, 5)$ the computed near-kernel has dimension 8. The sets of the heat equation (R5b) and the wave equation (R5c), read off the known solutions, fail in the same way. The standard form imposes the initial or boundary function g itself, on its slice $x_a = x^*$, by the *datum-slice* rows

$$\mathbb{S}_a(x^*) - \frac{\hat{g}\ \langle \tau(x_s)|}{f(x_s)}, \tag{4.4}$$

with $\mathbb{S}_a(x^*)$ the zero row $\mathbb{B}(x^*)$ on axis a and the identity on the others, $\hat{g}$ the coefficients of g broadcast along axis a, and x_s a regular point. They vanish on the solution and its multiples and leave the scale free (Proposition 10), are built from the problem's statement, serve both forms, and on R5a leave a one-dimensional kernel that is trusted at every resolution computed from $n = (3, 3)$ to $(6, 6)$, with gap ratio 1.7×10^9 at the last. The heat problem is posed forward from $t = -1$, with the paper's solution evaluated there as its initial function, because the paper's $t = 0$ lies inside $[-1, 1]$ and would make $[-1, 0]$ a backward heat problem, ill posed (Appendix C.3); like R4's constraint points, these data are set up from the known solution.

The paper's associated Legendre examples R3c and R3d (its Figs 4c and 4d, $l = 5$ and 6 with $m = 1$) multiply the equation through by $1 - x^2$, which leaves a leading coefficient $(1 - x^2)^2$ with a double zero at each end and a solution $P_l^1 = -\sqrt{1 - x^2}\, P_l'$ with a branch point there, which a Chebyshev series approximates only algebraically: at the printed $n = 5$ neither problem has a trusted ground state, and the error on R3c is 8.7×10^{-1}. The standard form solves for $v = f/\sqrt{1 - x^2}$ instead, whose equation has a leading coefficient with simple zeros and the polynomial solution $v = -P_l'$, represented exactly once $N > l - 1$ (Proposition 11): the error on R3c is 4.9×10^{-15} at $n = 3$ (Appendix C.4). It is also the form the descriptor can run

(§ 5.6).

Two smaller points complete the data. The paper prints the solution of its Laplace example as $\operatorname{csch}\pi\,\cosh(\pi x/2)\sinh(\pi(y+1)/2)$, which does not satisfy its boundary conditions; the solution, and the reference used throughout, is $\cos(\pi x/2)\sinh(\pi(y+1)/2)/\sinh\pi$. Vector data, such as the initial values of a coupled system, enter as ratio rows, defined with systems in § 6.2.

4.4 The Standard Form

The equation block and the constraint rows are stacked into one residual,

$$R_{\text{std}} = \begin{pmatrix} A_{\text{std}} \\ C_1 \\ \vdots \\ C_L \end{pmatrix}, \qquad H_{\text{std}} = R_{\text{std}}^{\dagger} R_{\text{std}} = A_{\text{std}}^{\dagger} A_{\text{std}} + \sum_i C_i^{\dagger} C_i, \tag{4.5}$$

every block at weight one, so that H_{std} is a sum of Gram matrices of the paper's form, Equation (3.4), with the datum slices among the C_i. The ground state of H_{std} is the smallest right singular vector of R_{std}. The classical reference computes it from R_{std} by a singular value decomposition, or by Lanczos iteration beyond its reach (§ 7.2), and never forms H_{std}, which would square the residual's condition number: on R2a at $n = 8$ the relative gap of H_{std} is 2.3×10^{-17}, below double precision. The circuit likewise encodes R_{std}, and H_{std} is never formed (§ 7.3).

Written this way, H_{std} is positive semi-definite, and in the very case the method aims at, a solution the truncation represents exactly, it is not definite, contrary to the paper's description (§ 3.2). When the solution is a polynomial of degree below N, any source is represented exactly and every constraint holds exactly at its point, $\mathbb{G}$ differentiates the solution and the lifts multiply it exactly, so its coefficients are annihilated by R_{std} and give H_{std} an exact zero eigenvalue; Legendre's equation $(1-x^2)f'' - 2xf' + 20f = 0$ (R3a, the paper's Fig. 4a), whose solution is P_4, has one from $n = 3$. Otherwise the truncated solution leaves a residual, $\lambda_0 > 0$, and the ground state minimises the total squared residual: the construction is least squares. Nor need the bottom of the spectrum be distinct, as the paper's partial differential sets show (§ 4.3). The thesis relies on neither property; it relies on an isolated ground state, measured as a trusted kernel (§ 7.2).

Definition 10 (The standard form): For a linear problem on one or more Chebyshev axes, the *standard form* is the stacked residual R_{std} of Equation (4.5), whose equation block is Equation (4.3), lifted on every axis where a coefficient or a source needs the exact product, and whose constraint rows, each at weight one, are built from four kinds of row: zero rows $\mathbb{B}(x^*)\mathbb{G}^j$, at a point or on a slice, for data that vanish; the collapse row $\mathbb{D}^{(0)}(x_s)$, which carries a non-zero point datum $f^{(j)}(x^*) = y^*$ as the row $\mathbb{B}(x^*)\mathbb{G}^j - y^*\mathbb{D}^{(0)}(x_s)$, as it carries a source into the equation block; ratio rows for vector data (§ 6.2); and datum slices (Equation (4.4)) for initial and boundary functions. Its Hamiltonian is $H_{\text{std}} = R_{\text{std}}^{\dagger} R_{\text{std}}$, and its solution is the

smallest right singular vector of R_{std}, scaled by the regular datum $f(x_s) = y_s$.

Table 6 lists the four kinds as model inputs, and a benchmark label such as R2a names the problem with the data of this section. Table 2 collects every departure from the printed method: those argued above, and those of the pipeline, argued where they are built. The departures in the residual change the data it carries, never $\mathbb{G}$, the lifts or the Gram structure, so the standard form's Hamiltonian has the paper's norm to leading order, the constraint rows growing more slowly than the equation block (Appendix E.4). The one exception is R3c and R3d, whose substitution changes the equation; their cost is compared in the form the standard form runs. The departures in the pipeline do change the cost, and in one direction: the structured encoding of § 4.5 is far cheaper than the paper's. A comparison of the descriptor against the standard form is therefore conservative, since its baseline has the paper's Hamiltonian, the cheapest exact encoding built, and better accuracy.

The baseline is validated against the paper itself. With every printed choice restored, the rounded points, the Maclaurin sources and the hand-picked invariant sets, the same construction reproduces the matrices printed in the paper's supplementary material at $n = 2$ entry by entry, which checks the operators. Its basis gives, for the exact solution, the printed scale factor of each of the paper's eleven ordinary examples to within one unit of the last printed digit, which checks the normalisation: for Fig. 3b, as printed at $n = 3$, the exact solution gives $\eta_e = 1.29009$ against the printed 1.29. The printed scale factors of its three partial differential and two nonlinear examples are not reproduced, by the exact solution's η_e or by any ground state we computed, and the cause is not settled. The record of every example, the triage of the printed choices and the verdict on each of the paper's claims are given in Appendix K.

4.5 Encoding the Derivative

Every term of the standard form is a product of lifts, rank-one rows and powers of $\mathbb{G}$. The rank-one rows have exact circuits whose subnormalisation is fixed by their own norm, so the term whose encoding decides the encoder's cost is $\mathbb{G}$.

4.5.1 The Paper's Encoding

The paper's circuit for $\mathbb{G}$ (its Fig. 9; § 3.3) selects each entry (k, p) by a register subtraction over a superposition of offsets and loads the column value $2p$ as an amplitude through a ladder of small controlled rotations. Its claims about gates and entry errors hold, the first as a bound and the second to leading order (Appendix D.1). The claim that decides its cost is the prefactor of Equation (3.5), which is valid but not compared with $\|\mathbb{G}\|_2$. Since $\|\mathbb{G}\|_S = \|\mathbb{G}^\top\mathbb{G}\|_1$ grows as $N^4/3$, the prefactor grows as $N^5/2$, against $\|\mathbb{G}\|_2 \sim N^2/(2j_{-3/4,1})$, so its looseness tends to $j_{-3/4,1}N^3$ with $j_{-3/4,1} \approx 1.06$; at $n = 7$ it is 2.2×10^6 (exact classical computation, Appendix E.5). A circuit with polynomially many gates is cheap to apply once, but the filter applies it a number of times set by its subnormalisation (Equation (2.6)): the encoding is efficient in gates and inefficient in subnormalisation. To cost it beside the others we rebuild its

Table 2: **Every departure from the printed method removes a stated defect or adds a stated capability.** The method of Wu et al. [1] against the standard form and pipeline of this thesis. Rows above the rule change the residual, and none of them changes the differentiation matrix $\mathbb{G}$, the lifts or the Gram structure of the Hamiltonian except the substitution for R3c and R3d, which changes the equation. Rows below the rule change how the residual is encoded, filtered, solved or read.

Aspect	Wu et al.	This thesis	Defect removed or property gained	Where
Constraint points	zeros and extrema of the known solution, rounded (-0.2, -0.73, 0.91, -0.75)	the problem's point, exact (set up from the reference solution)	a zero the solution lacks contradicts the equation; the error floor is removed	§ 4.3
Sources	Maclaurin series to order $\bar{p} = 3, 5, 7$	Chebyshev coefficients on the lifted register	a truncated source poses another equation; spectral accuracy	§ 4.2
PDE data	invariant sets chosen by hand; the top side and the initial functions omitted	datum slices for initial and boundary functions	no degeneracy as n grows; the heat problem posed forward	§ 4.3
Double zero (R3c, R3d)	the equation times $1 - x^2$	solved for $v = f/\sqrt{1-x^2}$	simple zeros and a polynomial solution, represented exactly	§ 4.3
Axes	two	any number (two run)	generality	Appendix C.2
Gram sum	$\sum \mathcal{T}(\cdot)$	the stacked residual R_{std}, H_{std} never formed	the same H_{std}; R_{std} solved and encoded without squaring its conditioning	§ 4.4
Fields	one real scalar field	several fields; ratio rows for vector data; complex fields by realification	coupled systems, and the Carleman lift	§ 6
Nonlinearity	the doubled space	the doubled space, and a Carleman lift	a preparation, not a projection, for polynomial systems; the lift's convergence measured separately	§§ 6.4 and 6.5
Encoding of $\mathbb{G}$	rotation-loaded, approximate; α looser than $\|\mathbb{G}\|_2$ by about N^3	an exact factorisation; α within a factor 2.12 of $\|\mathbb{G}\|_2$	the encoder's cost held to a constant for constant coefficients	§ 4.5
Filter	imaginary time on $H/\|H\|_F$ by mixed-parity QSVT, phases fitted, time by Equation (3.6)	minimax edge filter on $H_{\mathrm{std}}/\alpha_R^2$ by generalised quantum signal processing (GQSP), phases by Weiss's complement and the inverse nonlinear Fourier transform (NLFT), degree a priori	the block encoding's own subnormalisation; no parity split; suppression to ε by construction, which the time rule does not give	§§ 7.3 and 7.4
Initial state	not specified	geometric product state on the field block; its overlap reported	answer-independent	§ 7.4
Classical solve	the ground state as an eigenvector of H	the smallest right singular vector of R_{std}	resolves a gap below double precision on H_{std}	§ 7.2
Readout	interferometric, point by point	the field decoded classically from the prepared state	the protocol built at small n only, with a correction for $p_G < 1$	§ 7.6

mechanism. The rebuilt circuit's own subnormalisation is a third of the printed prefactor, so every cost quoted for the paper's encoding is the kinder reading; its entries are approximate, with an error of order N^{-5} (Proposition 14), so it enters resource estimates only, composed into the same stacked residual as the exact encoding below. The rebuild, its subnormalisation, its error bound and the verdict on each of the paper's circuit claims are given in Appendix D.1.

4.5.2 An Exact Factorisation

The paper's circuit is approximate, and its prefactor large, because it loads the column value of each entry as a rotation amplitude. The structure of $\mathbb{G}$ allows a different route. Every non-zero entry of Equation (4.2) is a product of three factors, each depending on one index alone: a row weight, equal to $1/\sqrt{2}$ in row 0 because of the normalisation of T_0 and to one elsewhere; a selector depending only on the offset $p-k$, non-zero when $p>k$ and $p-k$ is odd; and the column value $2p$, the weight with which differentiation passes T_p to the lower polynomials. Each factor has an exact block encoding with no amplitude loaded by rotation.

Proposition 1 (Exact factorisation of the differentiation matrix)**.** *For every $n \geq 1$ the differentiation matrix on the normalised Chebyshev coefficients factorises as*

$$\begin{aligned} &\mathbb{G} = R_0 \left(2U_{\text{odd}}\right) \Lambda, \\ R_0 = I - \left(1 - \tfrac{1}{\sqrt{2}}\right) |0\rangle \langle 0| \,, \qquad &\left(U_{\text{odd}}\right)_{kp} = \left[p > k,\ p-k \ \textit{odd}\right], \qquad \Lambda = \operatorname{diag}(p), \end{aligned} \tag{4.6}$$

where U_{odd} is the sum of the $N/2$ truncated shifts by odd offsets.

The proof compares the two sides entry by entry with Equation (4.2) and is given in Appendix E.1. As a check on our implementation, not on the proposition, the block of the composed circuit matches the independently assembled $\mathbb{G}$ entrywise to 3.4×10^{-13} (absolute) for $n \leq 7$.

4.5.3 Circuits and Cost

Each factor is encoded as a linear combination of unitaries [9] whose terms are unitary already, so no part of the encoding is approximate. The column index Λ is a sum of $n+1$ Pauli-Z strings, encoded at $\alpha_\Lambda = N - 1 = \|\Lambda\|_2$, and the row weight R_0 is a positive combination of the identity and the reflection about $|0\rangle$, encoded at $\alpha_{R_0} = 1 = \|R_0\|_2$ (Appendix E.1). The sum of shifts is the one factor that needs arithmetic (Figure 1). Its PREPARE puts an offset register in the uniform superposition over the $N/2$ odd offsets, and its SELECT is the in-place subtraction $p \leftarrow p - d$ of a ripple-carry circuit [10], which sets a borrow flag when $p < d$. Post-selecting the flag on zero removes exactly the terms that would wrap around, so the block is $\frac{2}{N}\left[p > k,\ p-k \text{ odd}\right]$: an exact encoding of U_{odd} at $\alpha = N/2$ with $O(n)$ Toffoli gates, in which one PREPARE serves all $N/2$ terms because SELECT is a permutation (Appendix E.2).

The three encodings multiply, and so do their subnormalisations. Their product is an exact block encoding of $\mathbb{G}$ at $\alpha_{\mathbb{G}} = 1 \cdot N \cdot (N-1)$, with the following cost.

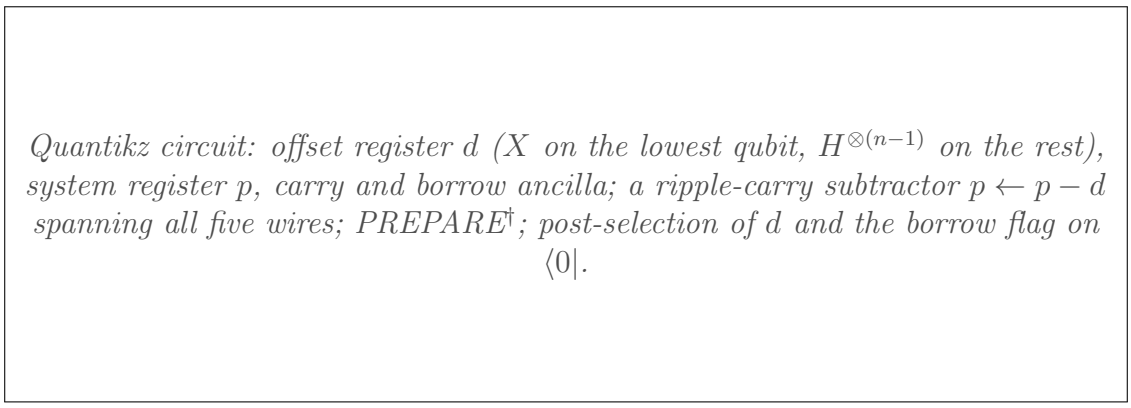


Figure 1: **The odd-offset shifts are one subtraction.** PREPARE (left) puts the offset register d in a uniform superposition over the $N/2$ odd offsets, by an X on its lowest qubit and Hadamards on the rest. SELECT is a ripple-carry subtractor [10] that replaces the system register p by $p - d$ and sets the borrow flag when $p < d$. PREPARE† (right), with post-selection of d and the borrow flag on $|0\rangle$, keeps $\frac{2}{N}\,|k\rangle\langle p|$ for every odd $d = p - k \geq 1$, and the wrapped terms vanish exactly. The block is $U_{\text{odd}}/(N/2)$, so $\alpha = N/2$, on n system and $n + 2$ ancilla qubits.

Proposition 2 (Cost of the structured encoding)**.** *The factorisation of Proposition 1 compiles to an exact block encoding of* $\mathbb{G}$ *on* n *system qubits with*

$$\alpha_{\mathbb{G}} = N(N-1), \qquad \frac{\alpha_{\mathbb{G}}}{\|\mathbb{G}\|_2} \longrightarrow 2j_{-3/4,1} \approx 2.117, \qquad O(n^2)\ \textit{gates}, \qquad n+4\ \textit{ancilla}, \tag{4.7}$$

of which $n - 1$ *ancilla are the selector of the* $N/2$*-term combination* U_{odd}*.*

The limit is that of $\|\mathbb{G}\|_2/N^2$ (§ 4.2). The ancilla are the $n + 2$ of U_{odd}'s offset register, carry and borrow, and one flag for each of the two products, the other factors' selectors reusing those qubits; the count is exact in the circuit for $n = 2$ to 8. The gate count assumes an ancilla-free synthesis of multi-controlled gates and falls to $O(n \log n)$ with a linear-cost synthesis that borrows idle qubits [11]; the transpiled two-qubit gate counts are available only at small n (109 at $n = 2$ and 471 at $n = 5$), consistent with the bound but too few to fit it. The proofs are given in Appendix E.3. The limit is asymptotic, and the sizes we reach are measured: the ratio $\alpha_{\mathbb{G}}/\|\mathbb{G}\|_2$ is 1.62 at $n = 2$ and rises to 2.11 at $n = 10$ (Table 3), against a ratio for the paper's encoding that grows as N^3. R_0 and Λ are encoded at their own norms, so the limiting factor is $\pi/2$ from U_{odd} and $4j_{-3/4,1}/\pi \approx 1.35$ from the product (Lemma 5).

The $n - 1$ qubits that hold the offsets' superposition are the price of writing U_{odd} as one combination of its $N/2$ shifts, since an exact block encoding built from L terms needs $\Omega(\log L)$ selector qubits [12]; an encoding of U_{odd} that is not such a combination is not excluded. The descriptor form of § 5.7 never forms the sum of shifts, and its ancilla grow as $\log n$.

The remaining terms of the residual are built from the same pieces: a polynomial coefficient through its lift, a combination of truncated shifts built by the same post-selected subtraction (Appendix C.7); the rank-one rows by the paper's exact circuits; a source by one rotation tree over its Chebyshev coefficients; and a power $\mathbb{G}^k$ as a product of encodings that share one

Table 3: **The structured encoding stays within a constant of the norm; the paper's prefactor does not.** Subnormalisation of the two encodings of the differentiation matrix, relative to $\|\mathbb{G}\|_2$. The structured encoding of Proposition 2 rises towards its limit $2j_{-3/4,1} \approx 2.117$; the paper's prefactor (Equation (3.5)) grows as N^3. The full table to $n = 10$ is Table 15. Exact classical computation of the norms and subnormalisations.

n	$\|\mathbb{G}\|_2$	Structured (Proposition 2)	The paper's (Equation (3.5))
2	7.39	1.623	49.9
4	121	1.986	4.05×10^3
7	7739	2.100	2.20×10^6
10	495 308	2.115	1.14×10^9

ancilla register with one flag per product, at subnormalisation $\alpha_{\mathbb{G}}^k$ while $\|\mathbb{G}^k\|_2 < \|\mathbb{G}\|_2^k$. The stacked residual is assembled into one reflection (§ 7.3) and verified against the independently assembled Hamiltonian (§ 7.5).

4.6 What Remains: The Standard-Form Hamiltonian

For constant coefficients the structured encoding holds the encoder's cost of the whole residual, as well as that of $\mathbb{G}$, to a constant. On R2a at $n = 8$ the stacked residual's encoding has $\alpha_R = 4.26 \times 10^9$ against $\|R_{\text{std}}\|_2 = 5.51 \times 10^8$, both computed exactly from the built circuit and the assembled residual, a ratio of 7.74. From 4.72 at $n = 2$ the ratio rises to 7.57 at $n = 6$, levelling off towards the looseness of the residual's dominant term, $\mathbb{G}^2$, whose ratio $\alpha_{\mathbb{G}}^2/\|\mathbb{G}^2\|_2$ tends to a constant (Appendix E.4); no limit is proved for the whole residual. Since the degree is linear in α_R, this ratio is the whole of the encoder's cost in the degree; in $\alpha_H = \alpha_R^2$ it is squared, 59.9 at $n = 8$. With the paper's encoding the same ratio is 9.4×10^{11} at $n = 7$ and 6.1×10^{13} at $n = 8$: it grows as N^{3k}, a factor N^3 for each of the k factors of $\mathbb{G}$ in the leading term.

No encoder can change the norm it encodes. For an equation of order k with constant coefficients and $a_k \neq 0$, $\|\mathbb{G}^k\|_2/N^{2k}$ tends to a positive constant, so $\|R_{\text{std}}\|_2 = \Theta(N^{2k})$ and $\|H_{\text{std}}\| = \|R_{\text{std}}\|_2^2 = \Theta(N^{4k})$, the constraint rows growing more slowly (Appendix E.4). The gap is measured; no proof is given. On R2a it is 7.04 at $n = 3$ and 7.04 at $n = 10$, and on $f'' - 2f' - 3f = 0$ (R2b, the paper's Fig. 3c) it is 0.415 at $n = 4$ and 0.415 at $n = 10$, unchanged to the figures shown at every n between, while $\|H_{\text{std}}\|$ on R2a grows from 2.8×10^5 to 2.0×10^{22}. Since $\alpha_H \geq \|H_{\text{std}}\|$ for every block encoding, Equation (2.6) gives

$$\|H_{\text{std}}\| = \Theta\big(N^{4k}\big), \quad \Delta = \Theta(1) \text{ (measured)} \quad \Longrightarrow \quad d = \tilde{\Omega}\big(N^{2k}\big) \text{ for every block encoding,} \tag{4.8}$$

and $d = \tilde{\Theta}(N^{2k})$ for the structured encoding, whose $\alpha_R/\|R_{\text{std}}\|_2$ stays bounded. At second order, the order of the paper's examples, $\|H_{\text{std}}\| = \Theta(N^8)$ and no block encoding brings the degree below $\tilde{\Omega}(N^4)$. This is the Hamiltonian's cost. The degree at the ideal-encoding bound $\alpha = \|R_{\text{std}}\|_2$, which contains no encoding, grows as N^{2k} on the constant-coefficient one-axis

problems measured, from $n = 7$ to 8: as $N^{2.00}$ on the first-order R1 and as $N^{4.00}$ on R2a, as Equation (4.8) predicts with a flat gap. The floor is stated for the minimax filter that prepares every state in this thesis; for polynomial filters in general it is what the optimality of Lin and Tong's construction supports [13], and we claim no more. At $n = 8$ the estimated degree for R2a is 1.2×10^{10} as built and 1.5×10^{9} at the ideal-encoding bound, both resource estimates far beyond what exact emulation reaches; their ratio is the encoder's cost of the previous paragraph.

The polynomial gate count that Wu et al. report for their operators [1], and that the rebuilt circuits bear out (Appendix D.1), is the cost of one query. The filter makes $\tilde{\Omega}(N^{2k})$ queries, exponentially many in n, however few gates each takes. The paper's claim of efficiency therefore holds for the circuit and says nothing about the method, consistent with its leaving the gap to future work.

The growth of the degree between $n = 7$ and $n = 8$, again from resource estimates, shows the two costs apart. With the paper's encoding the degree grows as $N^{10.0}$, with the structured encoding as $N^{4.01}$, and at the ideal bound as $N^{4.00}$. The difference between the first two, N^{3k} at $k = 2$, is the encoder's cost, whose growth the structured encoding removes; the common exponent is the Hamiltonian's. That exponent is set by the order of the derivatives. The degree grows as the square root of α_H/Δ, and for a ground value near zero $\Delta \approx \sigma_1^2$, so $\sqrt{\|H_{\text{std}}\|/\Delta} \approx \|R_{\text{std}}\|_2/\sigma_1$: the filter pays the conditioning of the residual, and the squared conditioning of $H_{\text{std}} = R_{\text{std}}^\dagger R_{\text{std}}$ is paid only by an eigensolver acting on H_{std} (§ 4.4). What makes $\|R_{\text{std}}\|_2$ grow as N^{2k} is that each of the k derivatives is eliminated through $\mathbb{G}$, at a factor N^2 each. A residual whose blocks each carry one derivative would grow only as the norm of one derivative operator, whatever the order, and less again if that operator were banded. These are the classical remedies: first-order-system least squares carries the derivatives as unknowns so that no block is of high order [8], and the ultraspherical method makes differentiation banded [4] (§§ 2.3.2 and 2.3.3).

The conclusion has a limited scope. The structured encoding bounds $\alpha_H/\|H_{\text{std}}\|$ for constant coefficients only, and is loose for a variable coefficient that is lifted. On R3a the circuit pays for the product $\mathbb{M}_a\mathbb{G}^2$ as $\|\mathbb{M}_a\|\,\|\mathbb{G}^2\| \gg \|\mathbb{M}_a\mathbb{G}^2\|$, because the coefficient $1 - x^2$ vanishes at the ends of the interval, where the derivatives of Chebyshev polynomials are largest. For the same reason the residual's norm is smaller than the constant-coefficient law gives: $\|H_{\text{std}}\|$ is 6.04×10^{8} at $n = 7$ and 9.70×10^{9} at $n = 8$, growing as N^4 where the constant-coefficient law gives $N^{4k} = N^8$. The encoder's cost reappears: the as-built degree grows as $N^{4.01}$, while at the ideal bound it grows as $N^{2.00}$, the exponent the descriptor's degree also has there ($N^{2.00}$ at the ideal bound). At $n = 8$ the as-built degree is 1.7×10^{10} against 1.1×10^{5} at the ideal bound, both resource estimates. For this reason § 8.3 quotes the ideal-encoding bound beside every as-built cost, and where the comparison between the forms reverses at that bound, as it does on R3, reports the reversal as a finding.

This answers the first question. For constant-coefficient problems the cost has two parts: the

encoder's, a factor growing as N^{3k} in the subnormalisation of the residual under the paper's encoding, which the structured encoding holds to a constant; and the Hamiltonian's, $\tilde{\Omega}(N^{2k})$ in the filter's degree, which no encoding removes because it is the residual's norm, a factor N^2 for each derivative eliminated through $\mathbb{G}$. § 8.3 tests both on the benchmark problems. What remains open to change is the residual itself, and § 5 changes it by carrying the derivatives as unknowns instead of eliminating them into the dense $\mathbb{G}$.

5 The Descriptor Reformulation

Can the residual be reformulated so that its Hamiltonian is banded and well conditioned without changing its solution, and at what cost? This is the second question of § 3.5. The descriptor form is the reformulation, and its kernel is the standard form's (§§ 5.1 to 5.4). It erodes the spectral gap, which an a priori rescaling restores on one axis (§§ 5.5 and 5.6); §§ 5.7 to 5.9 give its block encoding, its cost and its effect on the filter's degree.

5.1 The Idea

In the standard form of § 4 the state holds the field alone, and every derivative is eliminated through the dense differentiation matrix $\mathbb{G}$, so that a term $f^{(j)}$ acts as $\mathbb{G}^j$ on the field's coefficients. § 4.6 located the cost of that form in its Hamiltonian: $\|\mathbb{G}^k\|_2 = \Theta(N^{2k})$, so $\|H_{\text{std}}\| = \Theta(N^{4k})$ for constant coefficients, whatever the encoder. The density is a property of the representation: it arises because the derivative is required to be expanded in the same basis as the function. If the derivative may instead live in a neighbouring polynomial basis, differentiation is banded, and $\mathbb{G}$ factors into two banded matrices whose product is dense only because one of them is inverted. We therefore change the form in which the equation is written and leave the equation, its data and its solution as they were.

The descriptor form keeps the factors instead of the product. Where the standard form substitutes $\mathbb{G}^j$ for each derivative, the descriptor form carries $f', \ldots, f^{(k)}$ as unknowns beside f, ties consecutive derivatives by banded recurrences, and writes the equation over all of them. A first-order equation or system needs no extra unknowns, because the equation itself gives the derivative: multiplying the standard rows through by one banded factor is enough. We call the first layout the *chain* and the second the *mass form* (§ 5.3).

Neither ingredient is new to numerical analysis. Banded differentiation is the ultraspherical spectral method of Olver and Townsend [4], of which the factors of § 5.2 are the discrete shadow (§ 2.3.2). Carrying derivatives as unknowns so as not to square the condition number of a least-squares functional is the idea of first-order-system least squares [8] (§ 2.3.3). The physics-informed Hamiltonian is a least-squares functional, so both apply to it. The contribution of this thesis is their transfer to a block-encoded physics-informed Hamiltonian: a proof that the kernel is unchanged, an encoding of the residual as one flat linear combination of unitaries, an a priori rescaling that restores the gap the new unknowns erode, and a measurement of what

all of this does to ground-state preparation.

Figure 2 shows the effect on the running example $f''+4f'+4f = 0$ (R2a). At $n = 7$ the standard form's stacked residual holds 8384 non-zero entries on 128 unknowns, and the descriptor's holds 1402 on 512: four times as many unknowns, because of the block register that § 5.3 counts, and six times fewer entries. Apart from the rank-one constraint rows, which both forms share, the descriptor's residual is banded. That trade, a larger state for a sparser residual, is the subject of the chapter. Carrying the intermediate derivatives instead of eliminating them turns the residual from dense to banded with its kernel unchanged, so that the Hamiltonian's norm falls from $\Theta(N^{4k})$, which is $\Theta(N^8)$ at second order, to $\Theta(N^2)$ at fixed block weights; the price is a block register and a gap that the derivative blocks erode, which an a priori rescaling restores on one axis at no gate cost.

Two sparsity (spy) plots side by side for $f'' + 4f' + 4f = 0$ (R2a) at $n = 4$: left, R_{std} with the dense upper triangle of $\mathbb{G}^2 + 4\mathbb{G} + 4I$ and its constraint rows; right, R_{desc} with its block-banded pattern (recurrence blocks, equation row, padding) and the same constraint rows.

Figure 2: **Dense against banded: the two residuals of the same equation.** Sparsity of the stacked residuals of $f'' + 4f' + 4f = 0$ (R2a) at $n = 4$, constraint rows included. Left, the standard form's R_{std}, whose equation block $A_{\text{std}} = \mathbb{G}^2 + 4\mathbb{G} + 4I$ has a full upper triangle. Right, the descriptor's R_{desc}, whose recurrence blocks, equation row and padding are banded; the only dense rows are the rank-one constraint rows, which both forms share. The descriptor's residual has four times as many columns and, at $n = 7$, six times fewer non-zero entries (§ 5.1).

5.2 The Banded Factorisation

The Chebyshev coefficients of a function and of its derivative are tied by a three-term recurrence, each row of which ties one coefficient of the function to two of the derivative; the recurrence and its derivation are given in Appendix B. Truncated to N coefficients and written in the normalised basis of § 4.1, with $\hat{f}^{(0)}$ and $\hat{f}^{(1)}$ the coefficient vectors of f and f', it is the matrix identity

$$\tilde{\mathbb{B}}\,\hat{f}^{(1)} = \tilde{\mathbb{D}}\,\hat{f}^{(0)}. \tag{5.1}$$

$\tilde{\mathbb{D}}$ has a single non-zero diagonal, the first superdiagonal, with entries $(2\sqrt{2}, 4, 6, \dots, 2(N-1))$; $\tilde{\mathbb{B}}$ has the diagonal $(2, 1, 1, \dots)$ and the second superdiagonal $(-\sqrt{2}, -1, -1, \dots)$, the irregular first entries coming from the normalisation of T_0. Both hold $\mathcal{O}(N)$ non-zero entries against

the $N^2/4$ of $\mathbb{G}$. Two properties are used throughout: $\tilde{\mathbb{B}}$ is upper triangular with a non-zero diagonal, so it is invertible; and since the singular values of a single-diagonal matrix are its entries' magnitudes, $\|\tilde{\mathbb{D}}\|_2 = 2(N-1)$, while $\|\tilde{\mathbb{B}}\|_2 \le 2+\sqrt{2}$.

The dense differentiation matrix is therefore the quotient

$$\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}, \tag{5.2}$$

exactly at every N (Appendix B). Away from its first row, $\tilde{\mathbb{B}}^{-1}$ is a triangle of ones on each parity class, and the density of $\mathbb{G}$ is that inverse; Equation (5.1) is the same linear map before the inverse is taken. This is the discrete form of the ultraspherical spectral method: up to the basis normalisation, $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ are twice that method's operators for conversion from first-kind to second-kind Chebyshev polynomials and for differentiation into the second kind [4] (Appendix B.5). The ultraspherical method uses such operators to eliminate the derivatives, raising the basis once per derivative, and solves a banded system in f alone. We instead keep Equation (5.1) as a row of the system and each derivative as an unknown, which lets one pair of factors serve every order, every axis and every coupled field. The method's own reduction of an equation of order $k \ge 2$ to a banded operator on the N coefficients of f would also keep the standard form's kernel; it is not built here, and what it would change is stated in Appendix B.5.

5.3 Two Layouts and Kernel Equivalence

Consider the linear ODE of order k

$$\sum_{j=0}^{k} a_j(x)\, f^{(j)}(x) = r(x), \qquad x \in [-1,1], \tag{5.3}$$

with the data of § 4.3, every function in it represented by its first N coefficients in the normalised Chebyshev basis. Which layout carries it depends on its order.

The chain ($k \ge 2$). Let $\hat{f}^{(j)} \in \mathbb{C}^N$ be the coefficient vector of $f^{(j)}$ for $j = 0, \dots, k$, stacked into the state

$$w = (\hat{f}^{(0)};\ \hat{f}^{(1)};\ \dots;\ \hat{f}^{(k)}). \tag{5.4}$$

Two kinds of row couple the blocks. The *recurrence rows*

$$-\tilde{\mathbb{D}}\,\hat{f}^{(j)} + \tilde{\mathbb{B}}\,\hat{f}^{(j+1)} = 0, \qquad j = 0, \dots, k-1, \tag{5.5}$$

apply Equation (5.1) block by block. They contain no derivative operator, and on the kernel they are equivalent to $\hat{f}^{(j+1)} = \mathbb{G}\hat{f}^{(j)}$. The *equation row* is the equation itself, $\sum_j \mathbb{M}_{a_j}\hat{f}^{(j)} - \mathbb{M}_r\mathbb{D}^{(0)}(x_s)\hat{f}^{(0)} = 0$, with $\mathbb{M}_{a_j}$ the multiplication operator of § 4.2, lifted exactly where the standard form lifts it, and the source entering through the collapse row as in Equation (4.3). The equation row therefore applies the standard form's operators to the blocks that carry

the derivatives. The derivatives are unknowns of w and are never replaced by $\mathbb{G}^j \hat{f}^{(0)}$, so the dense quotient Equation (5.2) never appears. The name is borrowed from descriptor systems in control theory, which likewise keep intermediate quantities as unknowns beside the state and tie them by algebraic rows.

Definition 11 (Descriptor residual operator): Collecting the recurrence rows and the equation row, the descriptor residual operator A_{desc} is the block matrix

$$A_{\mathrm{desc}} = \begin{pmatrix} -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} & & & \\ & -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} & & \\ & & \ddots & \ddots & \\ & & & -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} \\ \mathbb{M}_{a_0} - \mathbb{M}_r \mathbb{D}^{(0)} & \mathbb{M}_{a_1} & \cdots & \cdots & \mathbb{M}_{a_k} \end{pmatrix} \tag{5.6}$$

with $(k+1)N$ columns before padding, so that $A_{\mathrm{desc}}\, w = 0$ is the equation. Its first k block rows are the recurrences and its last is the equation row.

The padded block register. On a quantum register the block index is held by $n_f = \lceil \log_2(k+1) \rceil$ qubits, which address 2^{n_f} blocks rather than $k+1$. The state is therefore padded to $2^{n_f} N$ amplitudes by blocks $b > k$, each pinned to zero by its own row $c_{\mathrm{pad}} \hat{f}^{(b)} = 0$ with $c_{\mathrm{pad}} = 1$. No other row reads a padding block, so the padding decouples: the Hamiltonian is block diagonal, and the spectrum of the physical blocks is unaffected. The padding is inert but not free. The chain's state has $2^{n_f} N$ amplitudes against the standard form's N, which is n_f more qubits and up to twice the unpadded $(k+1)N$ when $k+1$ is not a power of two: a second-order equation carries f, f' and f'' in four blocks. The constraint rows of § 5.4 are added to these block rows, and every row block, with the several-axis case and the padded count, is given in full in Appendix G.1.

The mass form ($k = 1$). A first-order system $z' = \mathbb{A} z$ with C components and a constant coupling matrix $\mathbb{A}$, a single first-order equation being the case $C = 1$, has in the standard form the rows $I_C \otimes \mathbb{G} - \mathbb{A} \otimes I_N$, one block per component. The equation already determines each derivative, so a derivative block would carry nothing new; the mass form instead multiplies these rows on the left by $I_C \otimes \tilde{\mathbb{B}}$ and uses $\tilde{\mathbb{B}}\mathbb{G} = \tilde{\mathbb{D}}$,

$$(I_C \otimes \tilde{\mathbb{B}})\big(I_C \otimes \mathbb{G} - \mathbb{A} \otimes I_N\big) = I_C \otimes \tilde{\mathbb{D}} - \mathbb{A} \otimes \tilde{\mathbb{B}}. \tag{5.7}$$

Both factors are banded and neither contains $\mathbb{G}$. The name is taken from the ODE literature, $\tilde{\mathbb{B}}$ playing the mass matrix of $\tilde{\mathbb{B}}\dot{z} = \ldots$. For $f' + 2f = 0$ with $f(-1) = 1$ (R1) the mass form is $\tilde{\mathbb{D}} + 2\tilde{\mathbb{B}}$ on N unknowns, where a chain would need $2N$ unknowns and elimination through $\mathbb{G}$ gives the dense $\mathbb{G} + 2I$; all three are written out at $N = 4$ in Appendix B. For a scalar equation with a constant coefficient the mass form is the ultraspherical method's own first-order operator. It has the standard form's state, initial state and readout, and differs from it only by a banded

invertible factor on the equation rows. It adds no qubits and has no derivative block to rescale (§ 5.6). Coupled fields and the Carleman lifts of § 6.5 are first-order systems with a constant coupling matrix, so in the descriptor form they take the mass form, which § 6.1 applies.

Both layouts rest on one fact.

Theorem 1 (Kernel equivalence)**.** *Let R_{std} be the standard form's stacked residual (Definition 10) of a linear problem on one axis. Let R_{desc} stack the descriptor's recurrence, equation and padding rows with the same constraint rows, each reading the block of the derivative it constrains (§ 5.4), the equation row applying the standard form's multiplication operators. Let R_{mass} be the mass form's stacked residual of a first-order system with a constant coupling matrix. Then*

$$\ker R_{\text{desc}} = \left\{ (p,\ \mathbb{G}p,\ \ldots,\ \mathbb{G}^k p,\ 0) \ :\ p \in \ker R_{\text{std}} \right\}, \qquad \ker R_{\text{mass}} = \ker R_{\text{std}}. \tag{5.8}$$

Proof. Write $Ep = (p; \mathbb{G}p; \ldots; \mathbb{G}^k p; 0)$. Chain: the padding rows force the padding blocks of a kernel vector w to zero, and since $\tilde{\mathbb{B}}$ is invertible each recurrence row gives $w_{j+1} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}\, w_j = \mathbb{G}w_j$, so $w = Ew_0$. The recurrence rows vanish on Ep for every p, because $(\tilde{\mathbb{B}}\mathbb{G} - \tilde{\mathbb{D}})\mathbb{G}^j p = 0$, and every other row of R_{desc} acts on Ep as the standard form's row of the same name acts on p: the equation row gives $\sum_j \mathbb{M}_{a_j}\mathbb{G}^j p - \mathbb{M}_r \mathbb{D}^{(0)}(x_s)p$, the standard form's equation block, and a constraint row reads the block that holds $\mathbb{G}^j p$. Hence $R_{\text{desc}}Ep$ vanishes exactly when $R_{\text{std}}p$ does, and E is injective because its first block is the identity. Mass form: R_{mass} is R_{std} with its equation rows multiplied on the left by the invertible $I_C \otimes \tilde{\mathbb{B}}$ (Equation (5.7)) and its constraint rows unchanged. □

The theorem holds on several axes, with E carrying $\mathbb{G}_a^j$ along each axis's chain, and for the block-rescaled residual of § 5.6, whose kernel is the image of this one under a diagonal map; both extensions are proved in Appendix G.3.

The theorem concerns kernels, and it is exact. An exact kernel exists only where the solution lies in the span of the basis and meets every constraint exactly, as Legendre's P_4 does on R3a; otherwise both residuals have full rank, and their ground states are the smallest right singular vectors of different matrices. The descriptor measures a state's residual against a norm that carries its derivatives (§ 5.5), and the mass form weights the equation rows by $\tilde{\mathbb{B}}$. We therefore claim equal kernels only, and observe a common limit. Both decoded fields converge to the solution, at rates we measure but do not prove. On R2a the descriptor's error against the analytic solution is 3.3×10^{-15} at $n = 5$ and 3.7×10^{-13} at $n = 10$, and the standard form's 3.2×10^{-13} and 1.4×10^{-6}: past $n = 5$ the standard form's error grows by orders of magnitude while the descriptor's grows far more slowly. This is consistent with the rounding error of the computed singular vector, which grows with the residual's norm relative to its gap, $\Theta(N^{2k})$ in the standard form against $\Theta(N)$ here (§ 5.9); we do not test the explanation further.

5.4 Constraints and Partial Differential Equations

A data constraint sits on the block that holds the quantity it constrains. A Dirichlet condition $f(x_z) = 0$ reads block 0 with the zero row $\mathbb{B}(x_z)$, as in the standard form. A Neumann condition $f'(x_m) = 0$ reads block 1 with the same row, $\mathbb{B}(x_m)\hat{f}^{(1)} = 0$, where the standard form must write $\mathbb{B}(x_m)\mathbb{G}$. The regular datum reads block 0, so the field is decoded and scaled exactly as in the standard form (§ 4.1). A variable coefficient and a source enter the equation row as they enter the standard form's equation block, so the two forms differ only in how the derivatives are carried.

On several axes the field is block 0, and each axis a owns a chain $\partial_a f, \dots, \partial_a^{k_a} f$ up to the highest order that the equation or a constraint takes along it, tied by that axis's factors $\tilde{\mathbb{B}}_a$ and $\tilde{\mathbb{D}}_a$ tensored with the identity on the other axes. A mixed derivative would need a block of its own; the construction does not build one, which limits it to equations without mixed derivatives. Initial and boundary functions enter both forms as the datum slices of Equation (4.4), read on the field block. Neither form doubles the state of a linear PDE, so the descriptor's advantage on these problems is not one of state size; it pays the larger state. The heat equation ($k_t = 1$, $k_x = 2$) carries four blocks, and Laplace's and the wave equation ($k = 2$ on each axis) five, padded to eight, so their states are four and eight times the standard form's, two and three more qubits.

5.5 The Gap of a Carried Derivative

Carrying the derivatives costs the descriptor something the standard form does not pay. With E the map of Theorem 1, the descriptor's Rayleigh quotient on the states that satisfy the recurrences is

$$\frac{\|R_{\text{desc}} E p\|^2}{\|Ep\|^2} = \frac{\|R_{\text{std}}\, p\|^2}{\sum_{m=0}^{k} \|\mathbb{G}^m p\|^2}, \tag{5.9}$$

because the recurrence and padding rows vanish on Ep and the remaining rows act as the standard form's. By the min–max principle, restricting the trial states to the range of E can only raise the quotient, so the descriptor's singular values are bounded above by the generalised singular values of the pair (R_{std}, E). The denominator carries every derivative, so the gap can be flat in N only where the equation controls each carried derivative: where no state p independent of the solution has a residual small against its largest carried derivative. The bound is an upper one in two senses. It does not prove flatness, and states that break the recurrences can lie below it, since the least singular value of $\tilde{\mathbb{B}}$ falls with N, its vector at the ends of the interval (Appendix G.4). The gap is therefore measured. The standard form measures the same states by $\|p\|$, against which their residuals are large, so its gap stays flat; its difficulty is the other side of the quotient, the norm of Equation (4.8).

Try to prove that the gap is flat for a regular

On a one-axis equation whose leading coefficient has no zero on $[-1, 1]$, the continuum a priori estimate $\|f^{(k)}\| \le (\|\sum_j a_j f^{(j)}\| + \|\sum_{j<k} a_j f^{(j)}\|)/\min|a_k|$ suggests that the quotient stays bounded away from zero on the complement of the kernel, and the measured gap, without

rescaling, is flat: on R2a it is 0.458 at $n = 3$ and 0.458 at $n = 10$. Flat is not large. The norm weighs block m by the size of $f^{(m)}$, so where the solution's derivatives are large the derivative blocks dominate it, and the ground state's weight on the field block, $w_f = \|\hat{f}^{(0)}\|^2/\|w\|^2$, is small; for an exact kernel it is $\|b\|^2/\sum_m \|\mathbb{G}^m b\|^2$, with b the solution's coefficients. An initial state prepared on the field block then overlaps the ground state by $\gamma \le w_f^{1/2}$, and amplification costs $1/\gamma$. On the stiff equation $f'' + 5f' + 400f = 0$ (R2c, the paper's Fig. 3d), whose solution oscillates at an angular frequency near 20, the gap is 7.7×10^{-6} at $n = 5$ and 7.5×10^{-6} at $n = 10$, flat and small, and w_f is 5.4×10^{-6}, against 4.3×10^{-3} on R2a.

The gap falls with N in two cases.

1. *A leading coefficient that vanishes.* A polynomial of degree N can concentrate within $\mathcal{O}(N^{-2})$ of an endpoint, where the Chebyshev points cluster, and each of its derivatives grows by a factor $\mathcal{O}(N^2)$ there. Where a_k vanishes at that endpoint, the leading term $a_k p^{(k)}$ of such a state is small, the lower-order terms are small as well, and the quotient is small against the top block. The measured gap falls as about N^{-4}: on Legendre's equation $(1 - x^2)f'' - 2xf' + 20f = 0$ (R3a), whose leading coefficient vanishes at both ends, it is 7.6×10^{-5} at $n = 5$ and 1.1×10^{-10} at $n = 10$, and on $(x - 1)f'' - xf' + f = (x - 1)^2$ (R4a, the paper's Fig. 5a), whose leading coefficient vanishes at $x = 1$, 4.9×10^{-5} and 4.3×10^{-11}. At an interior zero, where the Chebyshev resolution is $\mathcal{O}(N^{-1})$, the same argument gives N^{-2}; no benchmark problem has one, and the rate is measured on a controlled equation in Appendix G.4.

2. *Two axes.* The equation constrains only a sum of top derivatives, $f_{xx} + f_{yy}$ or $f_t - \kappa f_{xx}$. A high-frequency state with $\mathbb{G}_x^2 p \approx -\mathbb{G}_y^2 p$, or with its weight in the block the equation weights least, has a small residual against large blocks, and only the datum rows penalise it. On the heat equation $\partial_t f - \frac{1}{25}\partial_x^2 f = 0$ (R5b, the paper's Fig. 6d) the gap is 3.2×10^{-5} at $n = 3$ per axis and 2.5×10^{-7} at $n = 5$, where the standard form's is 0.264 and 0.433.

The collapse is not an artefact of the lift, the padding, the arithmetic precision, the constraints, the basis or the classical reference: each was tested and ruled out, and a descriptor built in the ultraspherical basis collapses at the same rates where the leading coefficient vanishes. The tests and the full mechanism are given in Appendix G.4.

5.6 Block Rescaling

The remedy is to rescale the carried derivatives so that no block outweighs the equation's own scale. The block of order m is carried divided by ζ^m,

$$\hat{f}^{(m)} \longmapsto \zeta^{-m}\,\hat{f}^{(m)}, \qquad m = 0, \dots, k, \tag{5.10}$$

which replaces R_{desc} by $R_{\text{desc}}S$, with S diagonal, equal to ζ^m on block m and to one on the padding blocks. Because S is invertible, the kernel becomes $S^{-1} \ker R_{\text{desc}}$ and Theorem 1 survives.

The field block has weight one, so the readout is unchanged, and the padding keeps its weight and still decouples. What changes is the Hamiltonian, $SR_{\text{desc}}^{\dagger}R_{\text{desc}}S$. This is a congruence, so the spectrum moves: by Equation (5.9) the denominator becomes $\sum_m \zeta^{-2m}\|\mathbb{G}^m p\|^2$, and ζ sets how much each carried derivative weighs against the residual. From here on R_{desc} and H_{desc} denote the rescaled residual and its Hamiltonian, and the residual of § 5.5 is called *unscaled*.

In the circuit the rescaling adds nothing. Scaling the columns of block m by ζ^m multiplies the coefficient of every term of the block encoding of § 5.7 by ζ raised to the order of its column block, which adds no gate and no ancilla and changes only the subnormalisation. A general diagonal equilibration, with a different weight for each amplitude, would need a diagonal block encoding of its own at $\Theta(\dim w)$ gates; that is why the weights are constant on each block, and general equilibration appears in no circuit in this thesis.

The a priori rule. ζ is computed from the equation's coefficients and the resolution, never from a solution or a spectrum. For a one-axis equation whose leading coefficient a_k has no zero on $[-1, 1]$,

$$\zeta = 2\left(\bar{a}_j/\bar{a}_k\right)^{1/(k-j)}, \tag{5.11}$$

with j the lowest order present and $\bar{a}$ the mean of $|a|$ over the interval. For constant coefficients and $j = 0$, $(\bar{a}_0/\bar{a}_k)^{1/k}$ is the geometric mean of the magnitudes of the characteristic roots: the scale at which a solution $e^{\lambda x}$ has $\|f^{(m)}\| \approx |\lambda|^m\|f\|$, so that dividing block m by its m-th power balances the blocks at the equation's own scale and returns the field block's weight to order one. Where a_k has a zero on the interval, $\zeta = N/2$, which raises the quotient of the concentrated states of § 5.5 to order one. On several axes $\zeta_a = N_a$, weighted on an axis of second order so that the top-order terms balance across axes: the wave equation $f_{tt} = c^2 f_{xx}$ takes (cN, N). Whether a_k vanishes is read from the equation.

The rule's two constants, the 2 in Equation (5.11) and the $\frac{1}{2}$ in $N/2$, were set once from a sweep of ζ over the benchmark problems and are never refitted to a problem. The rule was then checked against the cheapest ζ on a grid for the same problems, so the check shows that one rule serves the whole benchmark set and says nothing about a class of equation the set does not contain. The filter's cost stays within twice its least value over a window of ζ two to eight times wide, so ζ need only be right to about a factor of two, and the rule's cost is within 1.2 times the grid's best on 27 of 30 problem resolutions; at the other three, R2c at $n = 5$ and R3c and R3d at $n = 7$, it is within 2.2 times, $N/2$ overshooting a best nearer $N/4$. The sweep, the window and the check are tabulated in Appendix G.5.

What it restores. On every one-axis problem the descriptor runs, the rescaled gap no longer falls with N, and the field block regains its weight. The gap meant here is that of the physical subspace, the one the filter meets, since the padding alone would cap the gap of the whole spectrum at one. On R2c the gap rises from 7.5×10^{-6} to 0.715 and w_f from 5.4×10^{-6} to 0.774 at $n = 7$. On R3a the rescaled gap is 2.27 at $n = 5$ and 3.65 at $n = 10$, against 1.1×10^{-10}

unscaled: it rises with n and levels off.

What it does not remove. Three limits remain, each a finding.

1. *The norm grows with ζ.* Block m carries ζ^m, so $\|R_{\text{desc}}\| \gtrsim \zeta^k \bar{a}_k$ and $\|H_{\text{desc}}\| \gtrsim \zeta^{2k}$. With ζ constant this costs nothing. With $\zeta \propto N$, which a simple zero requires, $\|H_{\text{desc}}\| = \Theta(N^{2k})$, and at a flat gap the degree at second order is about $\tilde{\Theta}(N^2)$. This is no advantage over the standard form on such a problem, because a vanishing leading coefficient lowers the standard form's norm too: on R3a $\|H_{\text{std}}\|$ grows as N^4 where N^8 is the constant-coefficient law (§ 4.6). At the ideal-encoding bound both degrees grow as about N^2, and the standard form's is the lower, 1.1×10^5 against 2.8×10^5 on R3a at $n = 8$; as built the order reverses, 1.7×10^{10} against 5.7×10^5, because the structured encoding pays for the lifted multiplier in full. All four are resource estimates, and the reversal is a finding (§ 8.3).

2. *Two axes.* The rescaled gap still falls with N, though far more slowly than unscaled: on R5b it is 0.253 at $n = 3$ per axis and 0.0942 at $n = 5$, and on Laplace's equation (R5a) 0.372 and 1.1×10^{-2}, against 1.1×10^{-4} and 1.1×10^{-8} unscaled.

3. *A double zero is beyond any block rescaling.* If a_k vanishes to second order at an endpoint, the concentrated states have quotient $\mathcal{O}(N^{-8}\zeta^{2k})$, which is the unscaled N^{-8} at $\zeta = 1$, so balancing them needs $\zeta \propto N^{4/k}$. At $k = 2$ that is $\zeta \propto N^2$, at which $\|H_{\text{desc}}\| = \Theta(N^8)$: the norm of a second-order standard form with constant coefficients, and more than the standard form's own norm on such a problem, which the vanishing coefficient lowers. The paper's associated Legendre examples R3c and R3d (its Figs 4c and 4d) multiply the operator by $1 - x^2$ and so have exactly this double zero. Both forms instead run the equivalent problem for $v = f/\sqrt{1 - x^2}$ of § 4.3, whose leading coefficient has simple zeros, and we make no claim for the double zero beyond the measurement on the paper's form in Appendix G.5.

Block rescaling is the descriptor's counterpart of the diagonal preconditioner of the ultraspherical method [4], which bounds the condition number of its banded system. Here the diagonal scales the state's columns and is constant on each block, which is why it costs nothing in the circuit.

5.7 The Block Encoding: One Flat Linear Combination

Every block of A_{desc} is a short sum of *bands*, each a diagonal followed by a truncated shift. Write $\Lambda = \text{diag}(0, 1, \ldots, N - 1)$ for the index diagonal of Proposition 1. The diagonal of $\tilde{\mathbb{D}}$ is 2Λ except at column 1, where the normalisation of T_0 makes it $2\sqrt{2}$; the diagonals of $\tilde{\mathbb{B}}$ and of a constant multiplication are constant except at their ends; and a variable coefficient's multiplication and its truncation at the edge of the register contribute a few further exceptions. An affine diagonal $aI + b\Lambda$ is a sum of $n + 1$ Pauli-Z terms, and each exception is one reflection

Q_m:

$$\Lambda = \frac{N-1}{2}\, I - \sum_{j=0}^{n-1} 2^{j-1}\, Z_j, \qquad Q_m = I - 2\,|m\rangle\,\langle m|\,. \tag{5.12}$$

A_{desc} is therefore one flat LCU of $\mathcal{O}(n)$ terms,

$$A_{\text{desc}} = \sum_{t=1}^{L} c_t\, U_t, \qquad U_t = (\text{field word}) \otimes \text{SHIFT}^{\ell_t}\, P_t, \qquad \alpha_A = \sum_{t=1}^{L} |c_t|, \tag{5.13}$$

where the field word, a Pauli word on the block register, places the term's row and column blocks, SHIFT^{ℓ_t} is the truncated shift by ℓ_t, and P_t is the identity, a Z_j or a reflection. An affine diagonal whose end values a and $a + b(N-1)$ share a sign is encoded tightly, at $\alpha = \max(|a|, |a + b(N-1)|) = \|aI + b\Lambda\|_2$, so the dominant term $\tilde{\mathbb{D}}$ is encoded at its own norm $2(N-1)$ and its exception costs $\mathcal{O}(1)$. A truncated shift of the N-dimensional register is not unitary; it is applied on the register widened by one overflow qubit, so that rows falling outside the interval land in the half that the final projection discards. That qubit is the shift's only ancilla.

Algorithm 1 compiles the terms into the circuit of Figure 3.

Quantikz circuit: selector registers (PREPARE$_R$ loads $\text{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ over their joint index), field register (n_f qubits), system register (n qubits) widened by one overflow qubit; SELECT applies each term's field word, diagonal (Z-string or reflection) and truncated shift (a ladder of multi-controlled-X gates); PREPARE$_L^\dagger$ unloading $\sqrt{|c_t|/\alpha_A}$; post-selection on the selectors, the overflow and the flag.

Figure 3: **The whole descriptor residual is one preparation, one SELECT and one unpreparation.** PREPARE$_R$ loads the signed amplitudes $\text{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ over the joint index of the factored selector registers, and PREPARE$_L^\dagger$ unloads the unsigned $\sqrt{|c_t|/\alpha_A}$, so that the signs of the terms survive. SELECT applies each term's field word on the block register (n_f qubits), then on the system register (n qubits, widened by one overflow qubit) its diagonal, a Z-string or a reflection, and its truncated shift. Post-selecting the selectors, the overflow qubit and the flag on $|0\rangle$ leaves the system rows of A_{desc} divided by α_A, with the block rescaling carried entirely in the coefficients c_t.

The terms (lines 1–4). Lines 1 and 2 are the decomposition above. Line 3 gathers terms whose axis factors agree across the rows and columns they join, so that the block register carries their pattern once, as a small matrix expanded in Pauli words; the recurrences' common $\tilde{\mathbb{D}}$ is paid for once and not k times. Line 4 applies the block rescaling.

Input: the blocks of A_{desc} as bands, the block weights ζ
Output: PREPARE_R, SELECT and $\text{PREPARE}_L^\dagger$; the subnormalisation α_A
Write every block as a sum of bands: a diagonal and a truncated shift;
Fit each diagonal as affine in the column index plus a few exceptions, Equation (5.12);
Merge terms with equal axis factors across blocks; expand the resulting block-register pattern in Pauli words;
Multiply the coefficient of each term by ζ^m for its column block m;
$\alpha_A \leftarrow \sum_t |c_t|$; on the joint selector index of term t, PREPARE_R loads $\text{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ and PREPARE_L loads $\sqrt{|c_t|/\alpha_A}$;
SELECT: controlled on each selector register, apply the field word, then on each axis the diagonal and the shift;
return *PREPARE$_R$, SELECT, PREPARE$_L^\dagger$, α_A*;
Algorithm 1: Compiling the rows of A_{desc} into two preparations and one SELECT.

The preparations and SELECT (lines 5–6). The two preparations differ only in the signs, so that $\langle 0|\,\text{PREPARE}_L^\dagger\,\text{SELECT}\,\text{PREPARE}_R\,|0\rangle = \sum_t c_t U_t/\alpha_A$; one preparation and its own inverse would give $\sum_t |c_t| U_t/\alpha_A$ instead. The selectors are factored: one register for the field operation and, on each axis, one for the diagonal and one for the shift, each only as wide as the number of distinct values its terms use. SELECT then applies each distinct operation once, controlled on its own register, rather than once per term, which is what keeps the ancilla at $\mathcal{O}(\log n)$ (§ 5.8). The rank-one rows, the sources and the data constraints, are encoded through the feature map, with a rotation tree loading a source's coefficients, so no dense state preparation appears anywhere. The terms, their compilation and the check of the terms against the assembled rows are given in Appendix G.6.

5.8 Cost of the Descriptor Encoding

Proposition 3 (Cost of the descriptor encoding)**.** *For a descriptor residual of a fixed problem, the flat LCU of Equation* (5.13) *is an exact block encoding of the system rows (the recurrence, equation and padding rows) at $\alpha_A = \sum_t |c_t|$, with $\mathcal{O}(\log n)$ ancilla and $\mathcal{O}(n^2)$ gates. For the second-order family R2,*

$$\lceil \log_2(n+4) \rceil + 8 \text{ ancilla}, \qquad \mathcal{O}(n^2) \text{ gates}. \tag{5.14}$$

For the mass form of a scalar first-order equation, and for a second-order chain with polynomial coefficients whose leading coefficient has no zero on $[-1,1]$, at a fixed $\zeta \geq 1$, $\alpha_A/\|A_{\text{desc}}\|_2 = 1 + \mathcal{O}(N^{-1})$ without a source and $1 + \mathcal{O}(N^{-1/2})$ with one.

The ancilla are the selector registers, the overflow qubit and a flag. Only the diagonal selector's width depends on n: for R2 it indexes the identity, the n gates Z_j and three reflections ($\tilde{\mathbb{B}}$'s diagonal and second superdiagonal and the first entry of $\tilde{\mathbb{D}}$), so it has $\lceil \log_2(n+4) \rceil$ qubits, while the shift selector holds the offsets 0, 1 and 2 on two qubits and the field selector has four. The gate count follows because each distinct operation is applied once and the shift's multi-controlled ladder costs $\mathcal{O}(n^2)$. The subnormalisation tends to the norm because the dominant

terms, the recurrences' $\zeta^j\tilde{\mathbb{D}}$, share one pattern on the block register that its Pauli expansion encodes at its largest entry, so their contribution to α_A is $2\zeta^{k-1}(N-1)$, which is also a lower bound on $\|A_{\mathrm{desc}}\|_2$; every other term is $\mathcal{O}(1)$ in N at fixed ζ, except a source's rank-one term, whose coefficient vector grows as $N^{1/2}$. The proofs are given in Appendix G.7, and the ancilla count is verified for R2's system rows at $n = 2$, 4, 5 and 12.

The ancilla count is what expanding the diagonals in Pauli strings forces, and no more. A combination whose SELECT indexes L distinct operations needs $\lceil \log_2 L \rceil$ selector qubits [12], and the index diagonal Λ has exactly $n+1$ non-zero Walsh coefficients, so an LCU that expands it in Pauli-Z strings has at least $n+1$ terms and $\Omega(\log n)$ selector qubits. The count is therefore of the least order among exact LCU encodings that expand their diagonals in Pauli strings; it is not claimed against encodings built otherwise, by arithmetic on the index for instance. The structured encoding of the standard form needs $n-1$ selector qubits for $\mathbb{G}$ alone, because it writes U_{odd} as one combination of its $N/2$ shifts (§ 4.5.3), so its ancilla grow as n where the descriptor's grow as $\log n$; the descriptor never forms that sum, though an encoding of U_{odd} that is not such a combination is not excluded. The better scaling is not yet visible at the sizes that circuits reach. For the whole reflection on R2a the descriptor needs 12 ancilla at $n = 2$ and 13 at $n = 5$, against 10 and 13 for the standard form.

The ratio $\alpha/\|\cdot\|$ is the factor by which an encoding exceeds the least possible subnormalisation, and the filter's degree is linear in it. The proposition concerns the system rows; for the whole stacked residual, constraints included, the ratio is computed exactly from the built circuit's α_R and the assembled residual's norm. It approaches one on regular one-axis problems, $\alpha_R/\|R_{\mathrm{desc}}\|_2 = 1.04$ on R2a at $n = 8$. It stays near two where the leading coefficient vanishes, 2.12 on R3a at $n = 7$, because $\zeta\tilde{\mathbb{D}}$, $\zeta^2\tilde{\mathbb{B}}$ and $\zeta^2 a_2$ are then comparable and comparable terms add in ℓ^1 even when each is tight. On two axes it is 2.29 on R5b at $n = 4$ per axis. The limit of one therefore holds for regular one-axis problems only. For comparison, the structured standard-form encoding is at 7.74 on R2a and 3.7×10^4 on R3a, the latter the lifted multiplier's looseness of § 4.6.

The gates per query go the other way on the constant-coefficient problems. On R2a, at the sizes reached, one query of the descriptor costs more than one of the structured standard form, having more terms and a block register: the transpiled two-qubit count of the reflection is 25 914 against 8094 at $n = 2$ and 34 098 against 22 856 at $n = 5$. The ratio falls with n, but both counts are $\mathcal{O}(n^2)$, so the data do not show that it crosses one. On R3a, whose lifted multiplier the structured standard form pays for in full, the descriptor's query is the cheaper from $n = 4$: 38 864 against 43 108 at $n = 4$ and 48 542 against 52 038 at $n = 5$. On the constant-coefficient problems the descriptor's advantage is therefore the number of queries, the filter's degree, and not the gates per query, and its chain state is larger by the block register.

5.9 Consequences for the Spectrum and the Filter

Every block of the rescaled descriptor is $\mathcal{O}(N)$ at fixed ζ, the recurrences' $\zeta^j\tilde{\mathbb{D}}$ being the largest, and the block $\zeta^{k-1}\tilde{\mathbb{D}}$ alone has norm $2\zeta^{k-1}(N-1)$, so $\|R_{\text{desc}}\|_2 = \Theta(N)$ and $\|H_{\text{desc}}\| = \Theta(N^2)$ whatever the order k: the derivative is banded, and its norm is that of $\tilde{\mathbb{D}}$ rather than of $\mathbb{G}^k$. Since the degree is $d = \Theta(\sqrt{\alpha_H/\Delta}\log 1/\varepsilon)$ with $\alpha_H \geq \|H\|$ for every block encoding (Equation (2.6)), the rescaled gap, which is flat on a one-axis ODE whose leading coefficient does not vanish, gives for that class of problems

$$\|H_{\text{desc}}\| = \Theta(N^2), \qquad \Delta = \Theta(1) \text{ (measured)} \quad \Longrightarrow \quad d = \tilde{\Theta}(N), \tag{5.15}$$

against $\tilde{\Theta}(N^{2k})$ for the standard form with the best exact encoding (Equation (4.8)). Where the leading coefficient vanishes, or on two axes, the rescaling's weights raise the norm, and the degree is about $\tilde{\Theta}(N^2)$ or grows more steeply; for a double zero no claim is made (Table 4). Neither form is polylogarithmic in N: $\tilde{\Theta}(N)$ is polynomial in N and so exponential in n, and the claim is a lower exponent, not an exponential speed-up. The degree is that of the minimax edge filter, which prepares every comparison in this thesis; for polynomial filters in general the floor is no stronger than Lin and Tong's optimality supports [13].

Table 4: **The descriptor lowers the degree's exponent where the equation controls every carried derivative.** Norm, gap and filter degree of each form by class of problem, with the block weight ζ set by the a priori rule of § 5.6. Norms follow from the block weights and are proved (Appendix G.7); gaps are measured. A vanishing leading coefficient lowers the standard form's norm too, so the third row is no advantage over the standard form on the same problem (§ 5.6).

Case	ζ	$\|H\|$	Gap	Degree
Standard, order k, constant coefficients	–	$\Theta(N^{4k})$	flat	$\tilde{\Theta}(N^{2k})$
Descriptor, one axis, $a_k \neq 0$ on $[-1,1]$	fixed	$\Theta(N^2)$	flat	$\tilde{\Theta}(N)$
Descriptor, one axis, simple zero, $k=2$	$N/2$	$\Theta(N^4)$	flat	$\tilde{\Theta}(N^2)$
Descriptor, two axes	N_a per axis	$\Theta(N^4)$	falls	steeper than $\tilde{\Theta}(N^2)$
Descriptor, double zero	$\propto N^2$ to balance	$\Theta(N^8)$	–	not claimed

The encoding sits at that norm. In the terms of § 4, the encoder's cost $\alpha_H/\|H\|$ on R2a at $n=8$ is 1.09 for the descriptor against 59.9 for the structured standard form, so the descriptor lowers the Hamiltonian's cost and keeps the encoder's near one. At $n=8$ the estimated degree on R2a is 9.1×10^3 for the descriptor as built and 8.7×10^3 at the ideal-encoding bound $\alpha = \|R\|_2$, against 1.2×10^{10} and 1.5×10^9 for the standard form, all four resource estimates; the measured growth with N is reported in § 8.3. Every comparison is quoted both as built and at $\alpha = \|R\|_2$, which contains neither form's encoding, and where the order reverses, as it does on R3 at the ideal bound (§ 5.6) and on the heat equation at small n, the reversal is reported as a finding (§§ 8.3 and 8.4).

Two further facts bound what the comparison says. The first is the relative gap $\Delta/\|H\|$, which is $\Theta(N^{-2})$ here and $\Theta(N^{-4k})$ in the standard form: on R2a at $n=8$ it is 6.9×10^{-7} for the

descriptor and 2.3×10^{-17} for the standard form, below double precision. That is why the classical reference never forms the Hamiltonian (§ 4.4); the consequence of the small relative gap is the degree; the ground state stays resolvable. The second concerns a single query: one post-selected application of a block encoding of R to a Haar-random state of dimension D succeeds with mean probability $\|R\|_F^2/(D\alpha_R^2)$, which is $\Theta(1)$ for the descriptor and $\Theta(N^{-1})$ per axis for the structured standard form, whose dense $\mathbb{G}^k$ dominates the Frobenius norm (Appendix G.8). This compares the encodings only; it is not a cost of the preparation, which post-selects no single query.

5.10 Summary

The residual can be reformulated so that its Hamiltonian is banded without changing its solution, and the reformulation has a price.

1. Keeping the banded factors $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ of $\mathbb{G}$ and carrying the derivatives as unknowns, as a chain for equations of order two and above or in mass form for first-order equations and systems, makes the residual banded, apart from the rank-one constraint rows both forms share, with its kernel exactly the standard form's (Theorem 1).

2. Its block encoding is one flat LCU with $\lceil \log_2(n+4) \rceil + 8$ ancilla for R2's system rows, the least order for an exact LCU that expands its diagonals in Pauli strings, and its subnormalisation tends to the norm on regular one-axis problems (Proposition 3). Its price is a larger state, by the chain's block register, and, on the constant-coefficient problems, more gates per query at the sizes reached.

3. The Hamiltonian's norm is $\Theta(N^2)$, and for a one-axis equation whose leading coefficient does not vanish the filter's degree is $\tilde{\Theta}(N)$, against the standard form's $\tilde{\Theta}(N^{2k})$ (Equation (5.15)). Neither is polylogarithmic.

4. *The limit.* The carried derivatives shrink the gap where the solution's derivatives are large, and make it fall with N where the leading coefficient vanishes and on two axes. An a priori block rescaling restores it on one axis at no gate cost, at the price of a larger norm where the leading coefficient vanishes, where the standard form is then no worse at the ideal-encoding bound; on two axes the gap still falls, and a double zero is beyond any block rescaling.

§§ 8.3 and 8.4 test the cost and the gap on the benchmark problems. Put briefly: the descriptor replaces a dense residual by a banded one with the same kernel, lowering the Hamiltonian's norm from $\Theta(N^{4k})$ to $\Theta(N^2)$, and pays for it with a block register and a gap that its derivative blocks erode and an a priori rescaling restores on one axis, without a gate.

6 Beyond a Single Real Field: Systems, Complex Fields and Nonlinearity

The physics-informed construction of § 3 solves for one real scalar field. Its authors chose the Chebyshev basis partly to keep it so: with a Fourier basis the differentiation matrix is imaginary, and "due to the complex-valued quantum state, the resulting total energy function is of a complicated form, rendering it difficult to find the ground state" [1]. Their route to nonlinearity, the doubled space of § 3.4, multiplies the register instead of adding to it. This chapter takes the construction to several coupled fields, to complex-valued linear equations and to polynomial nonlinearity, and on the way answers the third question of § 3.5: *can the solution of a nonlinear equation be prepared as the unique ground state of a Hamiltonian, rather than selected by projection from a degenerate kernel?*

Each extension that prepares is carried as a coupled linear system on one state, with one block per component: several fields directly, a complex field by realification as two real ones, and a polynomial nonlinearity by the Carleman lift, whose components are the field's monomials. The route that is not such a system, the doubled space, only projects. We state the system and its data once (§§ 6.1 and 6.2), carry complex-valued equations on it (§ 6.3), show why the doubled space fails and the lift succeeds (§§ 6.4 and 6.5), put the couplings and products on a circuit (§ 6.6), and specify the fluids that test the lift (§ 6.7). § 6.8 collects the answer. The constructions and proofs in full are in Appendix H.

6.1 Several Fields on One State

Consider a first-order system of C fields with a constant coupling matrix $\mathbb{A}$, $z_i' = \sum_j \mathbb{A}_{ij} z_j$, each field holding its own first derivative. A time interval $t \in [0, T]$ is carried on the Chebyshev axis $\xi = 2t/T - 1$, so $dz/d\xi = \frac{T}{2}\mathbb{A}z$. The state holds one block of N Chebyshev coefficients per field, the field index on a register of $\lceil \log_2 C \rceil$ qubits as the most significant index, padded where C is not a power of two with blocks pinned to zero by rows that no other row reads, so the padding decouples as the chain's does (§ 5.3). In the standard form (Definition 10) the state holds the fields alone and every derivative is eliminated through the dense differentiation matrix $\mathbb{G}$; the descriptor carries a system in the mass form of Equation (5.7). The equation rows are

$$\underbrace{I_C \otimes \mathbb{G}_\xi - \tfrac{T}{2}\,\mathbb{A} \otimes I_N}_{\text{standard form}}, \qquad \underbrace{I_C \otimes \tilde{\mathbb{D}}_\xi - \tfrac{T}{2}\,\mathbb{A} \otimes \tilde{\mathbb{B}}_\xi}_{\text{descriptor (mass form)}} = \big(I_C \otimes \tilde{\mathbb{B}}_\xi\big)\Big(I_C \otimes \mathbb{G}_\xi - \tfrac{T}{2}\,\mathbb{A} \otimes I_N\Big), \tag{6.1}$$

and by Theorem 1 their kernels coincide. With the data rows of § 6.2, an initial-value problem has a one-dimensional near-kernel in either form, for the reason given in § 6.5. R7's rows are written out in Appendix H.1.

The descriptor's chain could instead carry each field with its derivative as a block, as it carries a field of higher order. The equation already gives each derivative as $\mathbb{A}z$, so such a block holds

nothing the rows do not determine; it doubles the state and adds a block for the rescaling of § 5.6 to balance. It also costs gap: on the harmonic oscillator $u' = v$, $v' = -\pi^2 u$ (R7) at $n = 8$, the mass form's gap is 0.132 and the block-rescaled chain's 0.0628, a ratio of about 2.1 that holds from $n = 3$ upwards. The chain is kept only as a labelled ablation.

The two forms of a system therefore share the state, the data rows, the initial state and the readout, and differ only by the banded, invertible factor $I_C \otimes \tilde{\mathbb{B}}_\xi$ on the equation rows. Every system and lift of §§ 8.5 and 8.6 is compared on this footing: the same unknowns, data and prepared object, and Hamiltonians that differ by one banded factor along the time axis. A system of higher order enters by the usual reduction to first order, as R7 is $u'' = -\pi^2 u$ written as two fields; none of higher order than R7's was run.

6.2 Data for Several Fields: Ratio Rows

A single field's initial or boundary value is imposed by a homogeneous zero row, which leaves the field's scale free, and a regular datum, which fixes it (§ 4.3). Vector data $y = (y_1, \dots, y_C)$ at a point x^* need the same separation. With an anchor component a, $y_a \neq 0$, and $y_\perp = \sum_{c \neq a} y_c \, |c\rangle$ the rest of the data, the *ratio row* on the field register is

$$\mathbb{F} = y_a\big(I - |a\rangle\langle a|\big) - y_\perp \langle a| \,, \qquad (\mathbb{F}v)_c = y_a v_c - y_c v_a \;\; (c \neq a), \quad (\mathbb{F}v)_a = 0, \tag{6.2}$$

and the data rows are $(\mathbb{F}/y_a) \otimes \mathbb{B}(x^*)$. The kernel of $\mathbb{F}$ is exactly the line through y, so the rows impose $z_c(x^*) = (y_c/y_a)\, z_a(x^*)$: the direction of the data, with their scale free. The regular datum on the anchor fixes the scale, a component whose datum is zero is held to zero exactly, and the rows' singular values, $\|y\|$, $|y_a|$ and 0, weigh the data against the equation at the data's own scale (Appendix H.1, with their three-term circuit).

The initial state of a system is the geometric product state of § 7.4 on each field's block, weighted by the problem's stated data $w_0/\|w_0\|$: the problem's statement, never its solution. Equal weights would not do, since they cancel exactly on the Taylor–Green vortex of § 6.7, whose two modes carry data of opposite sign.

6.3 Complex-Valued Fields

The paper's remark locates the difficulty in the energy function, but the construction does not meet it. If an equation has complex coefficients, data or sources, its stacked residual R is complex, yet $H = R^\dagger R$ is still Hermitian and positive semi-definite, its ground state is still the smallest right singular vector of R, and the stacked reflection and the filter of § 7 act on $H' = 2H/\alpha_R^2 - I$ exactly as before. The non-Hermitian operator is R, which the filter never sees, so no non-Hermitian filter is needed. This is a statement about the formulation: no complex residual was prepared in this thesis.

The route that was run is realification. Writing $z = u + \mathrm{i}v$, a complex linear equation becomes

a real system of twice as many fields; for one mode $z' = cz$ with a complex coefficient c,

$$\begin{pmatrix} u \\ v \end{pmatrix}' = \begin{pmatrix} \operatorname{Re} c & -\operatorname{Im} c \\ \operatorname{Im} c & \operatorname{Re} c \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}, \tag{6.3}$$

a system of § 6.1 with one more field-register qubit. The map $z \mapsto (\operatorname{Re} z, \operatorname{Im} z)$, a real-linear isomorphism, carries the complex problem's solutions onto the real system's, and real ratio rows on the realified data fix the solution's phase as well as its direction, so an initial-value problem keeps a one-dimensional near-kernel (§ H.1.2). R7 is an instance: with $z = \pi u + \mathrm{i}v$, its equations are the real and imaginary parts of $z' = -\mathrm{i}\pi z$, a single-mode Schrödinger-type equation, its state being the realified one with the first block divided by π. The fluids of § 6.7 meet the paper's remark directly, since they are Fourier–Galerkin systems; they are carried on real bases instead. Burgers' solution from odd data stays odd and lives on sine modes, so it needs no realification, and the vorticity is carried on $\cos(\mathbf{k}\cdot\mathbf{x})$ and $\sin(\mathbf{k}\cdot\mathbf{x})$, an isometric real embedding of the complex Fourier variables (§ H.1.2).

One encoding is natively complex: the tensor layout of the Carleman coupling (§ 6.6) acts on complex Fourier variables, and its LCU weights carry the phases $\mathrm{i}^{\sum_d a_d}$ of the Fourier symbols $\prod_d k_d^{a_d}$ it applies. It is verified, exactly at the smallest size and by IR to field scale, and prepares no state. Native complex preparation is not built. The problem specification types coefficients and data as real, and the circuit loaders for sources, datum slices and the collapse row's value, like the term IR, refuse complex data; no complex case is among R1–R10, and none was run in the experiments of § 8. This was a limit of the implementation, not a result about the method. The classical reference and the GQSP and multiplication components already carry complex amplitudes, so lifting the real-data restrictions should need little modification, but that path is untested. The research proposal's "complex-valued, non-Hermitian equations" are therefore answered at the level of the formulation, with realification as the route run (Table 32); native complex preparation is future work (§ 10.2).

6.4 The Doubled Space: Exact Encoding, Degenerate Kernel

A polynomial nonlinearity has two treatments here, the paper's doubled space and the Carleman lift. The first is not a linear system on one state, and that is where it fails; we run it in the standard form only, since the failure is a property of its kernel, which neither form changes. Its encoding is exact: with $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$, each product is folded to one Chebyshev series by the nodal fold of § 6.6 and each linear term is carried by a copy of the collapse row (Appendix H.2). Its kernel is large. The equation block maps the N^2 amplitudes of Ψ to the $2N$ coefficients of the folded register, so it has rank at most $2N$, and each of the c invariant constraints adds rank at most one:

$$\dim\ker H_{\otimes} \;\geq\; N^2 - 2N - c, \tag{6.4}$$

above the paper's count of $2^{2n-1} = N^2/2$ from $n = 3$ for $c \le 2$. On $4f'' + 2(f')^2 + f = 0$ (R6a) at $n = 5$ the kernel has dimension 962 of 1024.

Proposition 4 (A filter projects onto the doubled kernel)**.** *Let $H_\otimes \succeq 0$ act on $\mathbb{R}^{N^2}$, $N \ge 2$, with Π the projector onto its kernel.*

1. *The symmetric product states $\{u \otimes u\}$ do not form a linear subspace, and a kernel does. A kernel that contains two independent product states therefore contains states that are not products, and no Hamiltonian has the product states as its kernel.*
2. *Let P_d be filters with $P_d(0) = 1$ whose largest magnitude on the rest of the spectrum tends to zero, as for the minimax filter as $d \to \infty$ or for $e^{-\beta H_\otimes}$ as $\beta \to \infty$. For any ψ_{init} with $\Pi\psi_{\text{init}} \neq 0$,*
$$\lim_{d\to\infty} \frac{P_d(H_\otimes)\,\psi_{\text{init}}}{\left\|P_d(H_\otimes)\,\psi_{\text{init}}\right\|} = \frac{\Pi\,\psi_{\text{init}}}{\left\|\Pi\,\psi_{\text{init}}\right\|}. \tag{6.5}$$
3. *A penalty $P \succeq 0$ with $\ker(H_\otimes + P) = \operatorname{span}\{\psi_* \otimes \psi_*\}$ must vanish on $\psi_* \otimes \psi_*$ and on no other vector of $\ker H_\otimes$: a condition stated in terms of the solution ψ_*.*

The proof is given in Appendix H.3. A filter returns the initial state's projection onto a kernel that holds the solution's product state among exponentially many others, chosen by ψ_{init}, not by the equation. A penalty that singles out the solution must be built from it and is excluded as answer-dependent; the paper's selection of "a degenerate state that closely matches the analytical solution" is such a choice. We report the doubled space as a projection, never as a preparation.

6.5 The Carleman Lift in Space–Time

The Carleman lift of a polynomial system is a linear system on the system's monomials [14], [15] (§ 2.2.2), so everything in §§ 6.1 and 6.2 carries over: the components become monomials, the ratio rows carry the data's moments, and $\mathbb{A}$ becomes the lifted generator $\mathbb{A}_K$. This section adds only the lift. For $u \in \mathbb{R}^M$,

$$\frac{du}{dt} = \sum_{p=1}^{P} F_p\, u^{\otimes p} \quad \longrightarrow \quad \frac{dz}{dt} = \mathbb{A}_K\, z, \qquad z = \big(u_\alpha\big)_{1\le|\alpha|\le K}, \quad u_\alpha = \textstyle\prod_i u_i^{\alpha_i}, \tag{6.6}$$

where the state holds every monomial of degree 1 to K once, level by level (the symmetric layout), and each row of $\mathbb{A}_K$ comes from differentiating a monomial and substituting u', the terms beyond degree K being dropped. A damped linear part gives a damped lift, since the eigenvalues of $\mathbb{A}_K$ are sums of those of F_1. The lift is carried in the scaled variable u/s, $s = \max(\|u_0\|, 1)$, a diagonal similarity of $\mathbb{A}_K$ that leaves the truncated solution unchanged and keeps the data's higher moments from growing as $\|u_0\|^k$ and pulling the ground state's weight into the top level. Both facts are proved in Appendix H.4.

How fast the lift converges is a property of the lift, not of the residual. With λ_1 the eigenvalue

of F_1 of largest real part and $\|F_2\|$ taken on symmetric tensors,

$$\rho = \frac{\|u_0\|\,\|F_2\|}{\left|\operatorname{Re}\lambda_1(F_1)\right|} \tag{6.7}$$

is the ratio of Liu et al., who prove for a dissipative quadratic system that $\rho < 1$ suffices for the truncation error at order K to fall as $\mathcal{O}(\rho^K)$, uniformly in time [14]. The condition is sufficient; nothing in the bound says the lift diverges at $\rho \geq 1$. It is a worst case over all data of norm $\|u_0\|$, since $\|F_2\|$ is a supremum over every pair of inputs and $|\operatorname{Re}\lambda_1|$ the slowest decay of any mode. And it says nothing about the residual. For $y' = -y + y^2$ from $y(0) = \frac{1}{2}$ (R8), $\rho = \frac{1}{2}$; on the fluids it exceeds one on most builds. The convergence is therefore measured on every build, order by order against the exact solution of the truncated lift (§ 8.5).

What the residual guarantees is uniqueness. On the time axis the lifted system is Equation (6.1) with $\mathbb{A} \to \mathbb{A}_K$, in either form, with its data as ratio rows at $t = 0$ on the scaled moments of u_0. The truncated lift is a linear initial-value problem (IVP) whose initial value is fixed up to scale, so its solutions form a one-dimensional space. The residual, restricted to polynomials of degree below N_t, then has a one-dimensional near-kernel whose singular value falls spectrally in N_t, in both forms and for every $\mathbb{A}_K$, including a lift whose truncation diverges (Appendix H.5). The uniqueness is the initial-value problem's, which the lift carries; neither form contributes it. The decoded field's error splits accordingly,

$$\|\hat{u} - u\|_\infty \;\leq\; \underbrace{\left\|z_1^{(K)} - u\right\|_\infty}_{\text{the lift's}} + \underbrace{\left\|\hat{u} - z_1^{(K)}\right\|_\infty}_{\text{the time axis's}}, \tag{6.8}$$

with $z_1^{(K)}$ the first level of the truncated lift's exact solution. The lift's part does not depend on n_t and the time axis's falls spectrally to rounding, so the field's error falls with n_t until it meets the lift's floor and then holds; every build records both parts.

6.6 Couplings and Products on a Circuit

Everything in a system's or a lift's rows except its coupling is built from the bands and rank-one rows of § 5.7; the ratio rows are three LCU terms (Appendix H.1). This section adds the coupling and the products.

The coupling as a sum of matchings. A system's coupling $\mathbb{A}$, or a lift's $\mathbb{A}_K$ in the symmetric layout, is sparse and irregular, and its expansion in Pauli words has many terms even when it has few entries. We split it instead into *matchings*, each with at most one entry in every row and column: a permutation of the field register's basis states followed by a diagonal, whose entries are rotations on one flag qubit (§ H.1.1). Each matching is encoded at its largest entry, so the coupling's subnormalisation is the sum of the matchings' largest entries. The split is an edge colouring of $\mathbb{A}$'s bipartite graph of rows and columns: a greedy colouring uses at most $2\Delta - 1$ matchings, Δ the largest number of entries in a row or column, Δ is the least

possible by König's theorem [16], and on every lift built the greedy colouring reached Δ. The field-register factor is the same in both forms, so they pay the same for the coupling. On Burgers' lift on sixteen modes at $K = 3$, $\mathbb{A}_K$ has 968 components and splits into 31 matchings at $\alpha/\|\mathbb{A}_K\| = 1.34$; on the Navier–Stokes lifts the ratio is larger, 7.34 on eighty modes at $K = 2$ (Table 20).

The tensor layout. The matchings grow with $\mathbb{A}_K$'s entries, which grow with the number of modes. For a Fourier–Galerkin equation, whose linear operators are Fourier multipliers and whose products are convolutions, a second layout's term count does not: level k is carried as the whole tensor $u^{\otimes k}$ in k slots of wavevectors, and the coupling is a sum of terms "F_p on slots $j, \ldots, j + p - 1$", $2K - 1$ of them for a quadratic field whatever the number of modes (Proposition 23). Its multipliers vanish off the truncation, so it maps physical states to physical states with no leakage, and its weights carry the complex phases of § 6.3. Per query it becomes the cheaper of the two as the lift grows: on Burgers on sixteen modes at $K = 3$ it takes 17 218 CNOTs against the matchings' 525 436. Its subnormalisation is looser in two dimensions, where the fold's α is N. It is verified exactly on Burgers' smallest lift and by IR to 316 Navier–Stokes modes, and it encodes the coupling alone: it prepares no state.

Products. A product is pointwise on a grid, so a fold transforms to node values, keeps the diagonal and transforms back. The doubled space's nodal fold does this with the Chebyshev–Gauss transform $\mathbb{U}$: it is exact and block-encoded at $\alpha = 2$, within 1.21 of its norm at every n computed, so tight to a constant but not optimal (Proposition 26). $\mathbb{U}$ is the discrete cosine transform, budgeted as a cited primitive [17] and not built gate by gate. The tensor layout's Fourier fold uses the quantum Fourier transform (QFT), which is built, at $\alpha = N^{d/2}$ on d axes, the convolution's own norm.

6.7 Fourier–Galerkin Fluids: Burgers and Navier–Stokes

The lift is tested on two periodic flows whose quadratic terms are convolutions. Truncated to finitely many Fourier modes on a real basis (§ 6.3) they are polynomial systems of the form of Equation (6.6). A build's error has three parts, each measured on its own: the truncation's, against the equation itself; the lift's, the truncated lift's exact solution against the Galerkin system's IVP; and the time axis's (Equation (6.8)). The systems, their couplings and their parameters are given in full in Appendix H.7.

Viscous Burgers (R9). On $x \in [0, 2\pi)$,

$$u_t + u\,u_x = \nu\,u_{xx}, \qquad u(x, 0) = a \sin x, \tag{6.9}$$

whose solution stays odd and lives on the sine modes $\sin kx$, $k \le M$, with $M = 4$, 8 or 16. The Galerkin system has $F_1 = \mathrm{diag}(-\nu k^2)$ and F_2 the projection of $-\frac{1}{2}(u^2)_x$. The least damped

mode is $k = 1$, so $\rho = a\|F_2\|/\nu$, and $\|F_2\|$ grows with M. The strength runs from $a/\nu = 0.03$ to 10, across the lift's convergence boundary, and the Cole–Hopf solution [18], [19] measures the truncation's error.

Two-dimensional Navier–Stokes (R10). In vorticity form on the periodic square,

$$\omega_t + J(\psi, \omega) = \nu \nabla^2 \omega, \qquad \nabla^2 \psi = -\omega \ \text{ on } [0, 2\pi)^2, \qquad J(\psi, \omega) = \psi_x \omega_y - \psi_y \omega_x, \tag{6.10}$$

with $\nu = 1$, the vorticity on the real modes $\cos(\mathbf{k} \cdot \mathbf{x})$ and $\sin(\mathbf{k} \cdot \mathbf{x})$ of the half-plane with $0 < |\mathbf{k}|^2 \leq k_c^2$ (12 to 80 modes), the stream function eliminated as $\hat{\psi}_{\mathbf{k}} = \hat{\omega}_{\mathbf{k}}/|\mathbf{k}|^2$, and every coupling rational and exact. Without viscosity the truncation conserves the energy and the enstrophy, the two invariants that make two-dimensional flow what it is, and modes of equal $|\mathbf{k}|$ do not interact (Appendix H.7); a pseudo-spectral direct numerical simulation (DNS) measures the truncation's error. The Taylor–Green vortex $\omega_0 = 2 \sin x \sin y$ [20] occupies two modes of equal $|\mathbf{k}|$, so $\omega = \omega_0 e^{-2\nu t}$ exactly and its lift is exact at every order: R10a validates the whole chain at a known answer and does not test the lift's convergence. A counter-rotating vortex pair, scaled to $\|w_0\| = a$ and run at $a/\nu = 3$, 10 and 30, spans several $|\mathbf{k}|$, so its triads are active and its lift is a real test. The tensor layout's coupling alone is built further, to 148 and 316 modes.

6.8 Preparation Versus Projection

Table 5 places the two nonlinear routes: the doubled space's kernel is a subspace that the solution's product structure is not, so a filter projects; the Carleman lift is a linear system on one state with its data fixed up to scale, so a filter prepares it, in either form.

Table 5: **The Carleman lift prepares; the doubled space projects.** For each nonlinear route: the forms it is run in, its kernel, what a filter returns from an initial state ψ_{init}, and what a trusted kernel certifies. $\rho < 1$ is sufficient for the lift's truncation error to fall as ρ^K; the convergence is measured on every build, not inferred from ρ (§ 6.5).

Route	Forms run	Kernel	A filter returns	A trusted kernel certifies
Doubled space (Wu et al.; R6)	standard only	degenerate, dimension $\geq N^2 - 2N - c$	the projection of ψ_{init} onto the kernel	nothing: the kernel is never one-dimensional, so never trusted
Carleman lift (R8–R10)	both	one-dimensional near-kernel, from the lifted IVP's uniqueness	the truncated lift's solution	the lifted *linear* problem, not the lift's convergence

The caveat that matters is in the last column. A unique, trusted kernel certifies that the residual singles out the truncated lift's solution, not that the truncated lift is close to the nonlinear solution, which is the first term of Equation (6.8), measured separately. The two can

part company: past the convergence boundary of Burgers' lift the kernel stays one-dimensional and trusted while the lift diverges (§ 8.5).

What the descriptor adds is the time axis. At one Carleman order the two forms differ only by $I \otimes \tilde{\mathbb{B}}_\xi$ (Equation (6.1)), the banded left factor of § 5.1 applied in time, so the time axis contributes $\|\tilde{\mathbb{D}}_\xi\| = 2(N_t - 1)$ to the mass form's norm against $\|\mathbb{G}_\xi\| = \Theta(N_t^2)$ to the standard form's. Where the gap is flat, as on R7, the mass form's degree therefore grows more slowly with N_t; the coupling's part, $\frac{T}{2}\|\mathbb{A}_K\|$, is common to both and grows with the modes, so at fixed n_t the advantage narrows as the truncation widens. The descriptor's degree is lower in every Navier–Stokes pair compared, and on R8 and R9 from $n_t = 4$ (§§ 8.5 and 8.6).

The answer to Q(iii) is therefore yes, for polynomial nonlinearities and in either form: the Carleman lift turns the equation into a linear initial-value problem on one state, whose residual has a one-dimensional near-kernel that a filter prepares within ε (by exact emulation throughout, and by circuit simulation at the smallest sizes), where the doubled space can only project. What is prepared is the truncated lift's solution, and how close it lies to the nonlinear solution is a property of the lift, measured, not certified, by the construction. §§ 8.5 and 8.6 give the evidence, from the scalar R8 to two-dimensional Navier–Stokes.

remove once previous section complete

7 Preparing and Verifying the Ground State

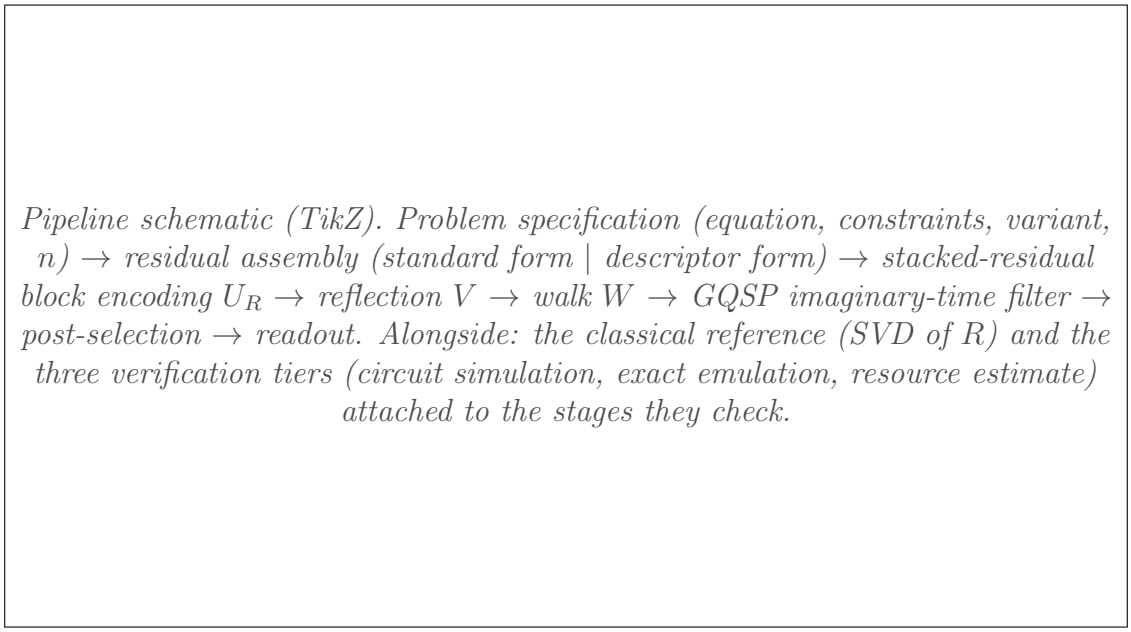


Figure 4: Model configuration: from problem specification to decoded field.

7.1 Problems and Model Inputs

Table 6: The shared constraint layer: every constraint row, in either form, is one of four kinds.

Kind	Operator	Use
Zero (invariant)	$\mathbb{B}(x)$ on the chosen field or derivative block	$f(x_z) = 0$, $f'(x_m) = 0$
Collapse (regular)	$\mathbb{D}^{(0)}(x_s)$	constants, sources; lifting linear terms in the doubled space
Ratio (vector data)	$\mathbb{F} = y_a\big(I - \lvert a\rangle\langle a\rvert\big) - y_\perp \langle a\rvert$, $y_\perp = \sum_{c\neq a} y_c \lvert c\rangle$	multi-component initial data (Carleman); coupled fields
Datum slice	$\big(\mathbb{S}_a(x^*) - \hat{g}\,\tau(x_s)^\top/v\big)\,\Pi_f$	inhomogeneous boundary or initial functions, without doubling

7.2 The Classical Reference

$$\lvert\psi_{\text{ref}}\rangle = \arg\min_{\lVert\psi\rVert=1} \lVert R\psi\rVert_2, \qquad \kappa\big(R^\dagger R\big) = \kappa(R)^2. \tag{7.1}$$

Table 7: The benchmark ladder. Each problem adds one capability to those below it; R2–R6 are the paper's Figs 3–7.

	Problem	Reference	New capability
R1	$f' + 2f = 0$, $f(-1) = 1$	$e^{-2(x+1)}$	smallest end-to-end system
R2	Fig. 3: $\{1, 4, 4\}$, $\{1, -2, -3\}$, $\{1, 5, 400\}$	closed form	order 2; Dirichlet and Neumann invariants; stiffness
R3	Fig. 4: Legendre, $m = 0$ ($l = 4, 5$) and $m = 1$ ($l = 5, 6$)	P_l, P_l^1	variable coefficients and lifts; an exact kernel ($m = 0$)
R4	Fig. 5: four inhomogeneous equations	closed form	source injection via $\mathbb{D}^{(0)}$; Maclaurin sources
R5	Fig. 6: Laplace (2-D), heat and wave (1+1-D)	closed form	multi-axis tensors; invariants versus datum slices
R6	Fig. 7: $4f'' + 2f'^2 + f = 0$; $f'' - 2f^2 + x = 0$	$1 - x^2/8$; boundary-value solve	doubled space (kernel degeneracy)
R7	$u' = v$, $v' = -\omega^2 u$, $\omega = \pi$	sin, cos	two fields; ratio rows
R8	$y' = -y + y^2$, $y(0) = \frac{1}{2}$	$1/(1 + e^t)$	Carleman lift, both forms in time
R9	viscous Burgers, periodic, Fourier–Galerkin	Galerkin IVP; Cole–Hopf	field-scale Carleman; convolution F_2; QFT fold
R10	2-D vorticity Navier–Stokes, periodic	Galerkin IVP; Taylor–Green; DNS	the culmination

7.3 From Stacked Residual to Quantum Walk

$$R = \begin{pmatrix} R_1 \\ \vdots \\ R_L \end{pmatrix}, \qquad \big(\langle j| \otimes \langle 0_a|\big)\, U_R \,\big(|0\rangle \otimes |0_a\rangle\big) = \frac{R_j}{\alpha_R}, \qquad \alpha_R = \Big(\sum_{j=1}^{L} \alpha_j^2\Big)^{1/2}, \qquad H = R^\dagger R. \tag{7.2}$$

Proposition 5 (Stacked-residual reflection)**.**

$$\begin{aligned} V &= U_R^\dagger \big(2\Pi_{a=0} - I\big)\, U_R, \qquad V^\dagger = V, \quad V^2 = I, \\ \Pi_{\text{in}}\, V\, \Pi_{\text{in}} &= \frac{2H}{\alpha_R^2} - I \equiv H', \qquad C(V) = U_R^\dagger\, C\big(2\Pi_{a=0} - I\big)\, U_R. \end{aligned} \tag{7.3}$$

$$W = \big(2\Pi_{\text{in}} - I\big)\, V, \qquad \Pi_{\text{in}}\, W^k\, \Pi_{\text{in}} = T_k(H'). \tag{7.4}$$

7.4 The Filter and Its Parameters

$$\Pi_{\text{in}}\, P(W)\, \Pi_{\text{in}} = \sum_{k=0}^{d} q_k\, T_k(H'), \qquad P(z) = \sum_{k=0}^{d} q_k z^k, \qquad \sup_{|z|=1} |P(z)| \le 1. \tag{7.5}$$

$$f_\beta(x) = e^{-\beta(1+x)}, \qquad f_\beta(H') = e^{-2\beta H/\alpha_R^2}, \qquad q_k = \big(2 - \delta_{k0}\big)(-1)^k e^{-\beta} I_k(\beta). \tag{7.6}$$

Proposition 6 (Normalisation of the imaginary-time filter)**.**

$$\sum_{k=0}^{\infty} |q_k| = 1. \tag{7.7}$$

$$\beta = \frac{\alpha_R^2}{2\Delta}\,\ln\frac{1}{\varepsilon_f}, \qquad d \approx \sqrt{2\beta\,\ln(2/\varepsilon_t)}, \tag{7.8}$$

$$|\psi_{\text{init}}\rangle \propto \bigotimes_{j=0}^{n-1}\Big(|0\rangle + r^{2^j}\,|1\rangle\Big), \qquad \gamma = \big|\langle\psi_{\text{ref}}|\psi_{\text{init}}\rangle\big|. \tag{7.9}$$

$$p_G = \big\|f_\beta(H')\,\psi_{\text{init}}\big\|^2 \;\to\; \gamma^2, \qquad \text{gates} \approx \frac{\pi}{4\gamma}\Big[\,2d\,g(U_R) + d\,g(\text{refl})\Big]. \tag{7.10}$$

Optional quantikz circuit: one GQSP ancilla; $R_0 \prod_{k=1}^{d}\big[C_0W\,R_k\big]$ with C_0W the controlled walk built from U_R, $U_R^\dagger$ and the two reflections, only the reflections controlled (Proposition 5); post-selection on the ancilla and the block-encoding register. The R_k phases realise whichever filter is in force (minimax, primary; imaginary time, the paper's).

Figure 5: The GQSP filter circuit.

7.5 Verification and the Error Budget

Table 8: Operator-verification modes: what each compares, up to what size, and to what tolerance.

Mode	Comparison	Size	Tolerance
Exact	statevector per input column; $\alpha\,\Pi_{\text{out}}U\Pi_{\text{in}}$ against the independently assembled operator	$\lesssim 20$ qubits	max-abs $\le 10^{-10}\|X\|$
Probe	16–32 Haar-random and structured inputs against $X\psi/\alpha$ by sparse matrix–vector products	$\lesssim$ 28–30 qubits	relative $\le 10^{-9}$
IR	compiled term IR against the residual assembly at every size; each term type exactly at small n	any n	$\le 10^{-12}$
Approximation	published $\mathbb{G}$ only: entry error $\epsilon_{\mathbb{G}}(n)$ against the small-angle bound	exact sizes	reported, not gated

Table 9: Acceptance criteria for every state-preparation run.

Level	Metric	Acceptance
State	infidelity $1 - \lvert \langle \psi_{\mathrm{ref}} \vert \psi \rangle \rvert^2$ in the form's own space	$\le \varepsilon = 10^{-6}$, and $\le 10\times$ the predicted bound
Probability	measured p_G against $\lVert P(H')\psi_{\mathrm{init}} \rVert^2$	relative difference $\le 10^{-8}$
Field	decoded f_Q against the analytic solution (L^∞, relative L^2)	$\le \varepsilon_{\mathrm{disc}} + \kappa_{\mathrm{read}}\sqrt{\varepsilon_{\mathrm{prep}}}$
T1 against T2	state difference	$\le 10^{-10}$
Phases	realised against target polynomial	$\le 10^{-12}$

$$\lVert f_Q - f \rVert \le \underbrace{\lVert f_{\mathrm{ref}} - f \rVert}_{\varepsilon_{\mathrm{disc}}} + \kappa_{\mathrm{read}}\big(\varepsilon_{\mathrm{filter}} + \varepsilon_{\mathrm{trunc}} + \varepsilon_{\mathrm{phase}} + d\,\varepsilon_{\mathrm{BE}}\big) + \varepsilon_{\mathrm{sample}}. \tag{7.11}$$

7.6 Reading Out the Solution

$$f^*(x) = 4P_C(x) - P_G(x) - \frac{1}{2N} = \sqrt{\frac{p_G}{N}}\,\tau(x)\cdot\psi, \tag{7.12}$$

$$f_Q(x) = f(x_s)\,\frac{f^*(x)}{f^*(x_s)}, \qquad \eta_e = \frac{f(x_s)^2\,p_G}{N\,f^*(x_s)^2}. \tag{7.13}$$

7.7 Design of the Numerical Experiment

7.8 Software and Computation

remove once previous section complete

8 Numerical Study

8.1 Reproducing Wu et al.

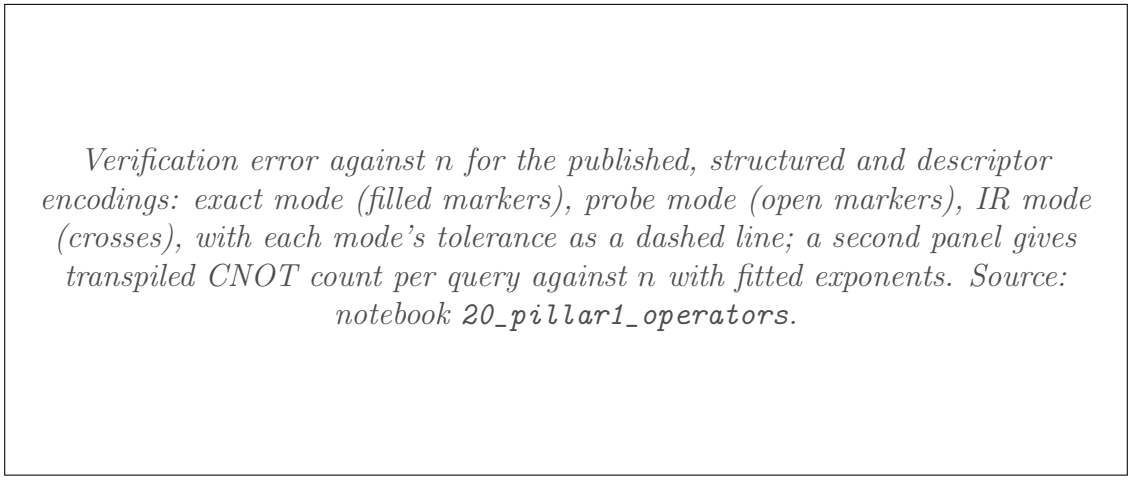
The closed form of the ideal scale factor is Equation (4.1), derived in § 4.1.

Table 10: The paper's printed scale factors against the exact solution's $\|b\|^2$ (Equation (4.1)) and the paper-faithful ground state at the printed n. †: the ground state is not trusted (a degenerate kernel, or a gap ratio below 20). Classical computation.

Fig.	Problem	n	Printed η_e	Ideal $\|b\|^2$	Ground state	Agreement
3b	R2a	3	1.29	1.29009	1.28731	✓
3c	R2b	3	32.47	32.47741	32.22636†	✓(one unit low)
3d	R2c	5	451.71	451.70962	451.52312	✓
4a	R3a	3	1.75	1.74512	1.74512	✓
4b	R3b	3	1.49	1.48779	1.48779	✓
4c	R3c	5	108.51	108.51142	109.70433†	✓
4d	R3d	5	135.38	135.38360	137.46159†	✓
5a	R4a	3	1.35	1.34649	1.34478	✓
5b	R4b	3	14.74	14.74797	14.72531	✓(one unit low)
5c	R4c	4	0.82	0.82116	0.82521	✓
5d	R4d	4	77.51	77.51167	76.93737	✓
6b	R5a	3	5.21559	5.21653	5.24407	×
6d	R5b	4	603.863	598.86606	598.87966	×
6f	R5c	5	216.469	239.82712	239.82712	×
7b	R6a	2	3.75367	3.52344	3.52344†	×
7c	R6b	2	0.07984	0.04383	0.04182†	×

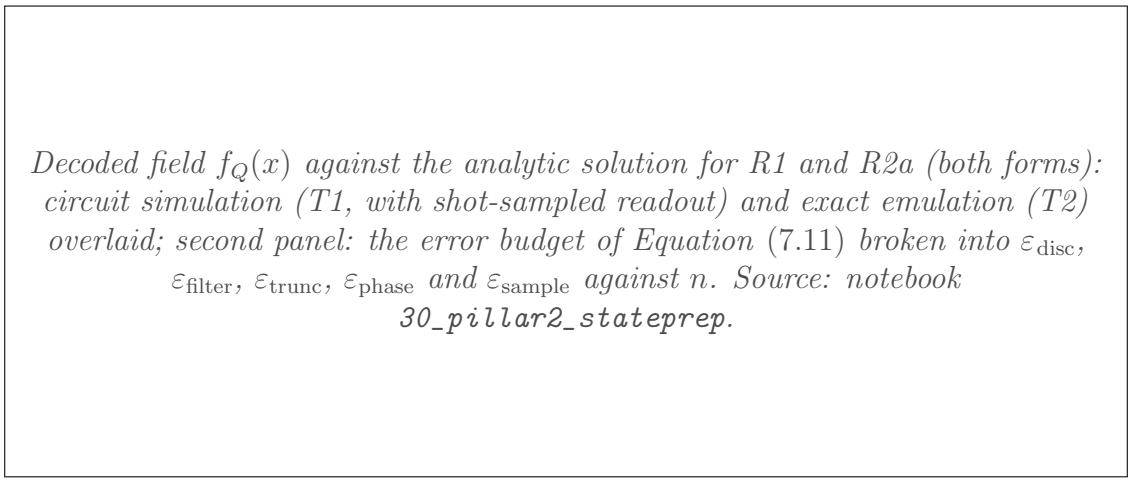
Four panels, the paper's Figs 5a–5d (R4a–R4d): the analytic solution, the paper-faithful ground state (rounded constraint points, Maclaurin sources) and R4a–R4d solved with exact roots and Chebyshev sources, at the printed n; inset or second row: L^∞ error against n for both variants. Source: notebook `10_paper_reproduction`. Classical computation.

Figure 6: Printed choices distort the paper's solutions at the printed resolution.



Verification error against n for the published, structured and descriptor encodings: exact mode (filled markers), probe mode (open markers), IR mode (crosses), with each mode's tolerance as a dashed line; a second panel gives transpiled CNOT count per query against n with fitted exponents. Source: notebook `20_pillar1_operators`.

Figure 7: Every encoding is verified to its mode's tolerance at every size tested.



Decoded field $f_Q(x)$ against the analytic solution for R1 and R2a (both forms): circuit simulation (T1, with shot-sampled readout) and exact emulation (T2) overlaid; second panel: the error budget of Equation (7.11) broken into $\varepsilon_{\text{disc}}$, $\varepsilon_{\text{filter}}$, $\varepsilon_{\text{trunc}}$, $\varepsilon_{\text{phase}}$ and $\varepsilon_{\text{sample}}$ against n. Source: notebook `30_pillar2_stateprep`.

Figure 8: The prepared state reproduces the analytic solution to within the predicted error budget.

Table 11: Where the cost lies: the standard form under the paper's encoding and under the structured exact encoding, against the descriptor form, for Fig. 3b ($\{1, 4, 4\}$, $f(-1) = 0$). Tier: T2 where measured, T3 beyond.

Quantity	Standard, published	Standard, structured	Descriptor
$\|H\|$	$\Theta(N^8)$	$\Theta(N^8)$	$\Theta(N^2)$
Gap Δ at $n = 8$	7.04 (flat)	7.04 (flat)	2.87 (flat)
$\alpha_{\mathbb{G}}/\|\mathbb{G}\|_2$	$\sim N^3/2$	≤ 2.12	no $\mathbb{G}$
$\alpha_H/\|H\|$	$\sim N^{12}$	$22 \to 60$	$5.3 \to 1.05$
Gates per query	$O(n^3)$, approximate	$O(n^2)$, exact	$O(n^2)$, exact
Ancilla	$2n + 3$ ($\mathbb{G}$ alone)	$n + O(\log n)$ per $\mathbb{G}$	$\lceil \log_2(n + 4) \rceil + 8$
Degree at $n = 8$ ($\varepsilon = 10^{-6}$)	unknown	1.2×10^{10}	9.1×10^3
Degree scaling	$\sim N^{10}$	$\tilde{\Theta}(N^4)$	$\tilde{\Theta}(N)$

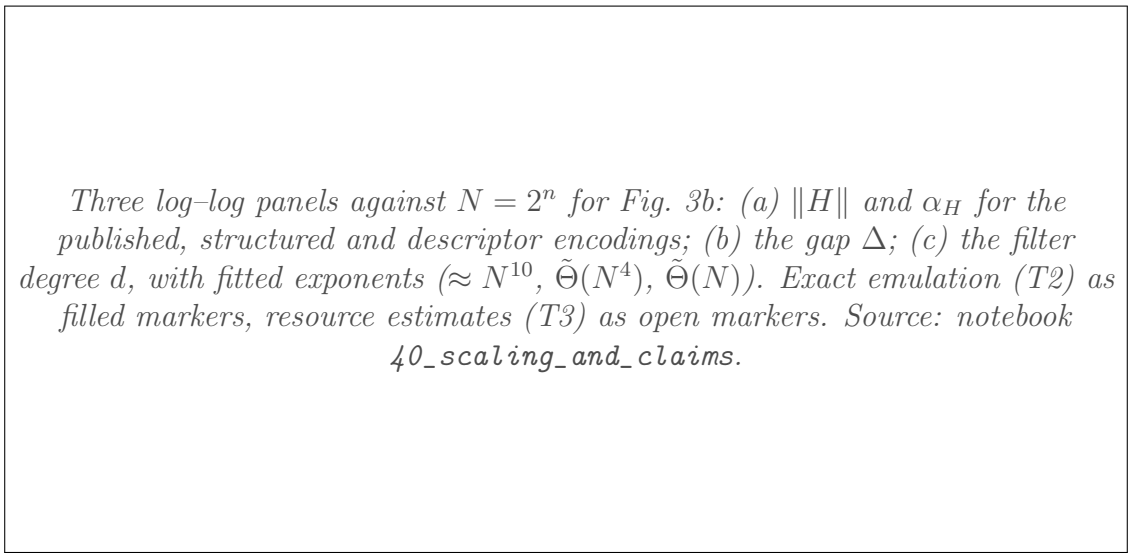


Figure 9: The encoder wall and the Hamiltonian wall are separate gaps.

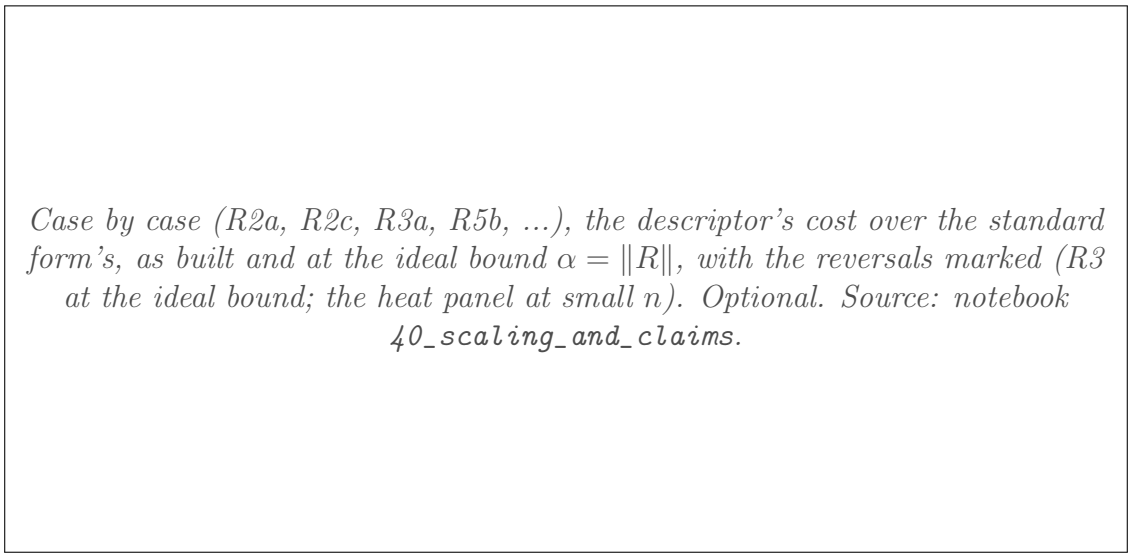


Figure 10: Three representative problems: where the comparison, quoted as built, reverses at the ideal encoding bound.

8.2 The Pipeline Is Faithful

8.3 Where the Cost Lies

8.4 The Descriptor's Gap and Its Rescaling

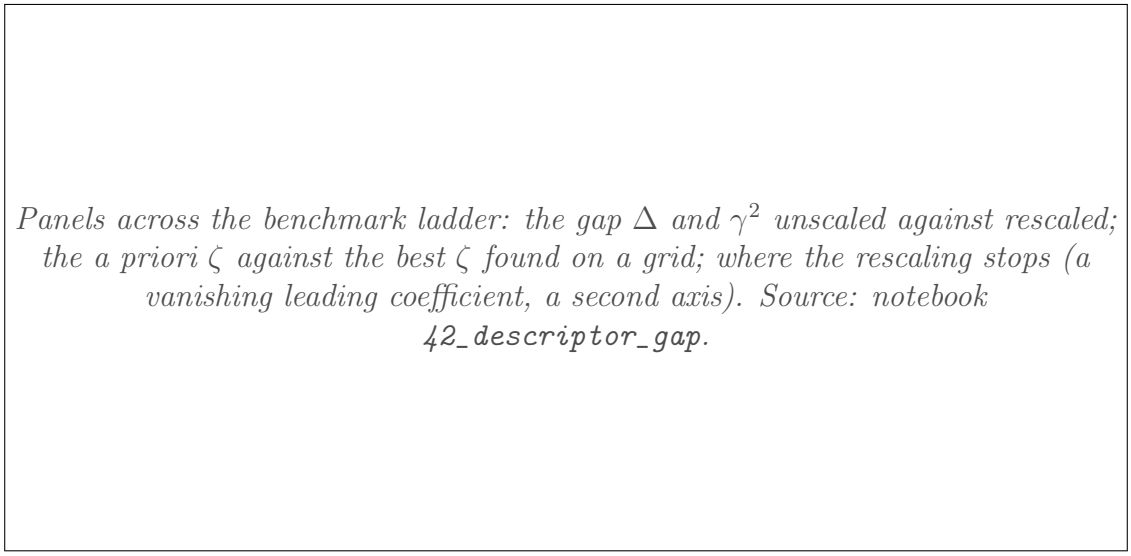


Figure 11: Block rescaling restores an O(1) gap on one axis, and stops at a double zero or on two axes.

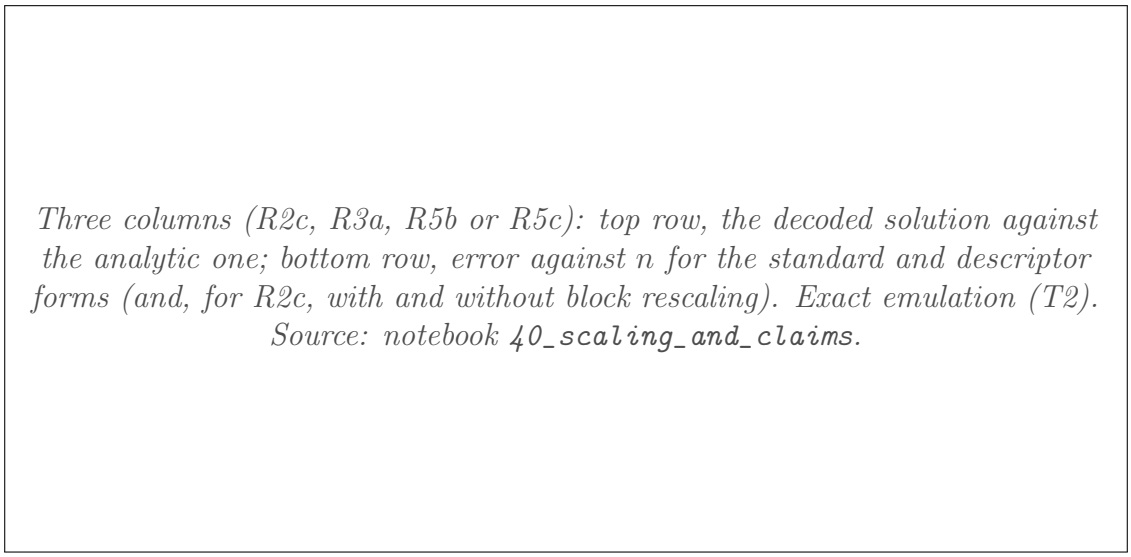
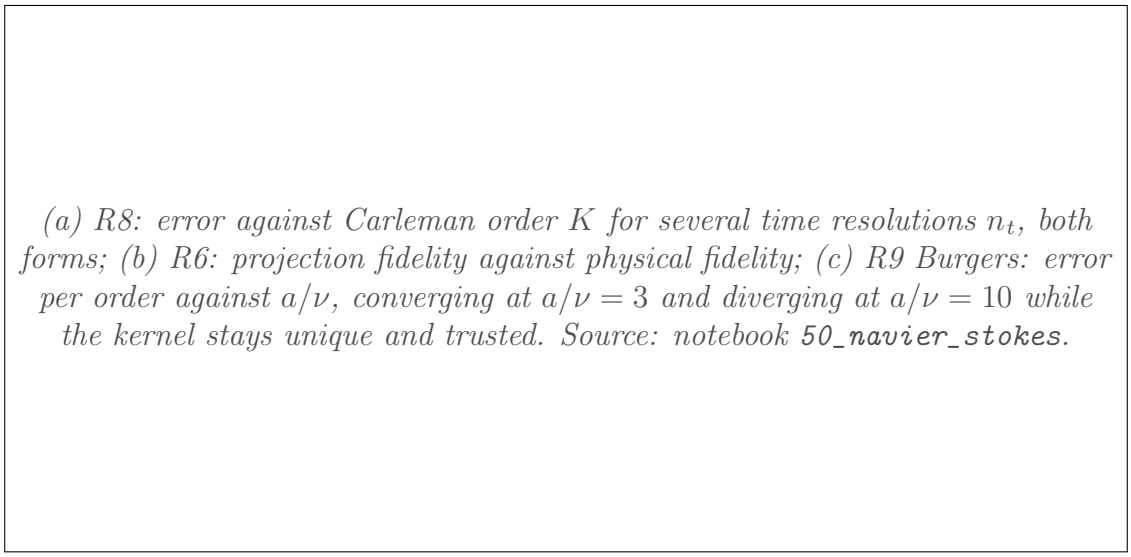


Figure 12: Three representative problems, solved in both forms.

8.5 Systems and Nonlinear Problems

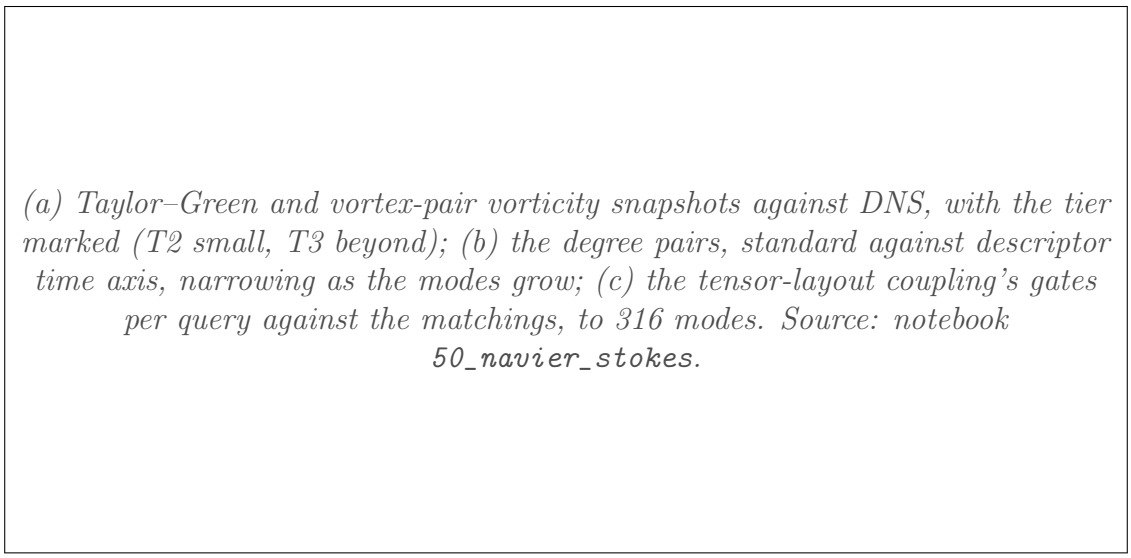
8.6 Case Study: Two-Dimensional Navier–Stokes

8.7 Summary of Findings



(a) R8: error against Carleman order K for several time resolutions n_t, both forms; (b) R6: projection fidelity against physical fidelity; (c) R9 Burgers: error per order against a/ν, converging at $a/\nu = 3$ and diverging at $a/\nu = 10$ while the kernel stays unique and trusted. Source: notebook `50_navier_stokes`.

Figure 13: Carleman truncation converges geometrically once time is resolved; the doubled space only projects; on Burgers the error tracks a/ν, not ρ.



(a) Taylor–Green and vortex-pair vorticity snapshots against DNS, with the tier marked (T2 small, T3 beyond); (b) the degree pairs, standard against descriptor time axis, narrowing as the modes grow; (c) the tensor-layout coupling's gates per query against the matchings, to 316 modes. Source: notebook `50_navier_stokes`.

Figure 14: Navier–Stokes is reached by exact emulation and a resource estimate, not by circuit simulation.

remove once previous section complete

9 Discussion

9.1 Why the Descriptor Form Wins, and What It Costs

9.2 End-to-End Complexity

9.3 What the Reproduction Says About the Published Method

9.4 Where the Construction Applies

9.5 Feasibility on Fault-Tolerant Hardware

9.6 Limitations

remove once previous section complete

10 Conclusions and Future Work

10.1 Conclusions

10.2 Future Work

remove once previous section complete

11 References

[1] H.-Y. Wu, A. E. Paine, E. Philip, A. A. Gentile and O. Kyriienko, *Quantum algorithm for solving nonlinear differential equations based on physics-informed effective Hamiltonians*, 2025. arXiv: 2504.13174 [quant-ph].

[2] O. Kyriienko, A. E. Paine and V. E. Elfving, 'Solving nonlinear differential equations with differentiable quantum circuits,' *Physical Review A*, vol. 103, p. 052 416, 2021. arXiv: 2011.10395 [quant-ph].

[3] A. E. Paine, V. E. Elfving and O. Kyriienko, *Physics-Informed Quantum Machine Learning: Solving nonlinear differential equations in latent spaces without costly grid evaluations*, 2023. arXiv: 2308.01827 [quant-ph].

[4] S. Olver and A. Townsend, 'A Fast and Well-Conditioned Spectral Method,' *SIAM Review*, vol. 55, no. 3, pp. 462–489, 2013. DOI: 10.1137/120865458 arXiv: 1202.1347.

[5] O. Kyriienko, 'Quantum inverse iteration algorithm for programmable quantum simulators,' *npj Quantum Information*, vol. 6, p. 7, 2020. arXiv: 1901.09988 [quant-ph].

[6] S. Scali, C. Umeano and O. Kyriienko, *Quantum topological data analysis via the estimation of the density of states*, 2023. arXiv: 2312.07115 [quant-ph].

[7] A. E. Paine, *Effective-Hamiltonian Quantum Solvers for Differential Equations: Alternative Constructions and Function Encodings*, 2026. arXiv: 2609.26330 [quant-ph].

[8] Z. Cai, R. Lazarov, T. A. Manteuffel and S. F. McCormick, 'First-Order System Least Squares for Second-Order Partial Differential Equations: Part I,' *SIAM Journal on Numerical Analysis*, vol. 31, no. 6, pp. 1785–1799, 1994. DOI: 10.1137/0731091

[9] A. M. Childs and N. Wiebe, 'Hamiltonian simulation using linear combinations of unitary operations,' *Quantum Information and Computation*, vol. 12, pp. 901–924, 2012. arXiv: 1202.5822 [quant-ph].

[10] S. A. Cuccaro, T. G. Draper, S. A. Kutin and D. P. Moulton, *A new quantum ripple-carry addition circuit*, 2004. arXiv: quant-ph/0410184 [quant-ph].

[11] A. Barenco et al., 'Elementary gates for quantum computation,' *Physical Review A*, vol. 52, p. 3457, 1995.

[12] S. Chakraborty, R. Hazra, X. Li, C. Shao, Q. Wang and Z. Zhang, *Quantum singular value transformation without block encodings: near-optimal complexity with minimal ancilla*, 2025. arXiv: 2504.02385 [quant-ph].

[13] L. Lin and Y. Tong, 'Near-optimal ground state preparation,' *Quantum*, vol. 4, p. 372, 2020. arXiv: 2002.12508 [quant-ph].

[14] J.-P. Liu, H. Ø. Kolden, H. K. Krovi, N. F. Loureiro, K. Trivisa and A. M. Childs, 'Efficient quantum algorithm for dissipative nonlinear differential equations,' *Proceedings of the National Academy of Sciences*, vol. 118, no. 35, e2026805118, 2021. arXiv: 2011.03185 [quant-ph].

[15] H. Krovi, 'Improved quantum algorithms for linear and nonlinear differential equations,' *Quantum*, vol. 7, p. 913, 2023. arXiv: 2202.01054 [quant-ph].

[16] D. König, 'Über Graphen und ihre Anwendung auf Determinantentheorie und Mengenlehre,' *Mathematische Annalen*, vol. 77, no. 4, pp. 453–465, 1916. DOI: 10.1007/BF01456961

[17] A. Klappenecker and M. Rötteler, 'Discrete cosine transforms on quantum computers,' *Proc. SPIE 4740, Image and Signal Processing and Analysis (ISPA 2001)*, 2001. arXiv: quant-ph/0111038.

[18] E. Hopf, 'The partial differential equation $u_t + uu_x = \mu u_{xx}$,' *Communications on Pure and Applied Mathematics*, vol. 3, no. 3, pp. 201–230, 1950. DOI: 10.1002/cpa.3160030302

[19] J. D. Cole, 'On a quasi-linear parabolic equation occurring in aerodynamics,' *Quarterly of Applied Mathematics*, vol. 9, no. 3, pp. 225–236, 1951. DOI: 10.1090/qam/42889

[20] G. I. Taylor and A. E. Green, 'Mechanism of the Production of Small Eddies from Large Ones,' *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, vol. 158, no. 895, pp. 499–521, 1937. DOI: 10.1098/rspa.1937.0036

[21] A. W. Harrow, A. Hassidim and S. Lloyd, 'Quantum Algorithm for Linear Systems of Equations,' *Physical Review Letters*, vol. 103, p. 150 502, 2009. DOI: 10.1103/PhysRevLett.103.150502

[22] D. W. Berry, A. M. Childs, A. Ostrander and G. Wang, 'Quantum Algorithm for Linear Differential Equations with Exponentially Improved Dependence on Precision,' *Communications in Mathematical Physics*, vol. 356, pp. 1057–1081, 2017. arXiv: 1701.03684 [quant-ph].

[23] A. M. Childs and J.-P. Liu, 'Quantum Spectral Methods for Differential Equations,' *Communications in Mathematical Physics*, vol. 375, pp. 1427–1457, 2020. DOI: 10.1007/s00220-020-03699-z

[24] P. C. S. Costa, D. An, Y. R. Sanders, Y. Su, R. Babbush and D. W. Berry, 'Optimal Scaling Quantum Linear-Systems Solver via Discrete Adiabatic Theorem,' *PRX Quantum*, vol. 3, p. 040 303, 2022. DOI: 10.1103/PRXQuantum.3.040303

[25] D. An, J.-P. Liu and L. Lin, 'Linear Combination of Hamiltonian Simulation for Nonunitary Dynamics with Optimal State Preparation Cost,' *Physical Review Letters*, vol. 131, p. 150 603, 2023. DOI: 10.1103/PhysRevLett.131.150603

[26] S. Jin, N. Liu and Y. Yu, 'Quantum simulation of partial differential equations: Applications and detailed analysis,' *Physical Review A*, vol. 108, p. 032 603, 2023. arXiv: 2212.14703 [quant-ph].

[27] D. Jennings, M. Lostaglio, R. B. Lowrie, S. Pallister and A. T. Sornborger, *The cost of solving linear differential equations on a quantum computer: fast-forwarding to explicit resource counts*, 2023. arXiv: 2309.07881 [quant-ph].

[28] M. Lubasch, J. Joo, P. Moinier, M. Kiffner and D. Jaksch, 'Variational quantum algorithms for nonlinear problems,' *Physical Review A*, vol. 101, p. 010 301, 2020. DOI: 10.1103/PhysRevA.101.010301

[29] R. D. Somma, Y. Subasi and D. Orsucci, 'Quantum algorithms for systems of linear equations inspired by adiabatic quantum computing,' *Physical Review Letters*, vol. 122, p. 060 504, 2019. DOI: 10.1103/PhysRevLett.122.060504

[30] D. An and L. Lin, 'Quantum Linear System Solver Based on Time-Optimal Adiabatic Quantum Computing and Quantum Approximate Optimization Algorithm,' *ACM Transactions on Quantum Computing*, vol. 3, no. 2, p. 5, 2022. DOI: 10.1145/3498331

[31] L. Lin and Y. Tong, 'Optimal Polynomial Based Quantum Eigenstate Filtering with Application to Solving Quantum Linear Systems,' *Quantum*, vol. 4, p. 361, 2020. DOI: 10.22331/q-2020-11-11-361

[32] L. Lin, *Lecture notes on quantum algorithms for scientific computation*, https://arxiv.org/abs/2201.08 Jan. 2022.

[33] C. A. Williams, S. Scali, A. A. Gentile, D. Berger and O. Kyriienko, *Addressing the Readout Problem in Quantum Differential Equation Algorithms with Quantum Scientific Machine Learning*, 2024. arXiv: 2411.14259 [quant-ph].

[34] G. Brassard, P. Høyer, M. Mosca and A. Tapp, *Quantum Amplitude Amplification and Estimation*, 2000. arXiv: quant-ph/0005055.

[35] D. W. Berry, A. M. Childs, R. Cleve, R. Kothari and R. D. Somma, 'Exponential improvement in precision for simulating sparse Hamiltonians,' in *Proceedings of the 46th Annual ACM Symposium on Theory of Computing (STOC 2014)*, 2014, pp. 283–292. arXiv: 1312.1414 [quant-ph].

[36] Y. Suzuki, S. Uno, R. Raymond, T. Tanaka, T. Onodera and N. Yamamoto, 'Amplitude estimation without phase estimation,' *Quantum Information Processing*, vol. 19, p. 75, 2020. arXiv: 1904.10246 [quant-ph].

[37] D. Grinko, J. Gacon, C. Zoufal and S. Woerner, 'Iterative Quantum Amplitude Estimation,' *npj Quantum Information*, vol. 7, p. 52, 2021. arXiv: 1912.05559 [quant-ph].

[38] H. Nishi, T. Kosugi, Y. Huang, K. Hirose, Y. Okayama and Y.-i. Matsushita, *Quantum State Readout via Overlap-Based Feature Extraction*, 2025. arXiv: 2505.08613 [quant-ph].

[39] D. Camps, R. Van Beeumen, L. Lin and C. Yang, 'Explicit Quantum Circuits for Block Encodings of Certain Sparse Matrices,' *SIAM Journal on Matrix Analysis and Applications*, vol. 45, no. 1, pp. 801–827, 2024. arXiv: 2203.10236 [quant-ph].

[40] C. Sünderhauf, E. Campbell and D. Camps, 'Block-encoding structured matrices for data input in quantum computing,' *Quantum*, vol. 8, p. 1226, 2024. arXiv: 2302.10949 [quant-ph].

[41] Z. Zhang and C. Shao, *Low-ancilla block encodings via Hamiltonian simulation*, 2026. arXiv: 2607.01843 [quant-ph].

[42] D. Motlagh and N. Wiebe, 'Generalized Quantum Signal Processing,' *PRX Quantum*, vol. 5, p. 020 368, 2024. arXiv: 2308.01501 [quant-ph].

[43] *QRTlib*, 2025. arXiv: 2510.16625 [quant-ph].

[44] *Dyadic-Order Quantum Fractional Transforms*, 2026. arXiv: 2604.09295 [quant-ph].

[45] D. Maslov, 'Advantages of using relative-phase Toffoli gates with an application to multiple control Toffoli optimization,' *Physical Review A*, vol. 93, p. 022 311, 2016. arXiv: 1508.03273 [quant-ph].

[46] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th Anniversary. Cambridge University Press, 2010.

[47] B. K. Berntson and C. Sünderhauf, 'Complementary polynomials in quantum signal processing,' *Communications in Mathematical Physics*, vol. 406, p. 161, 2025. arXiv: 2406.04246.

[48] L. Laneve, *Generalized Quantum Signal Processing and Non-Linear Fourier Transform are equivalent*, 2025. arXiv: 2503.03026 [quant-ph].

[49] H. Ni, R. Sarkar, L. Ying and L. Lin, *Inverse nonlinear fast Fourier transform on SU(2) with applications to quantum signal processing*, 2025. arXiv: 2505.12615 [quant-ph].

[50] A. Gilyén, Y. Su, G. H. Low and N. Wiebe, 'Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics,' in *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC 2019)*, 2019, pp. 193–204. arXiv: 1806.01838 [quant-ph].

[51] J. M. Martyn, Z. M. Rossi, A. K. Tan and I. L. Chuang, 'Grand Unification of Quantum Algorithms,' *PRX Quantum*, vol. 2, p. 040 203, 2021. arXiv: 2105.02859 [quant-ph].

[52] G. H. Low and I. L. Chuang, 'Hamiltonian Simulation by Qubitization,' *Quantum*, vol. 3, p. 163, 2019. arXiv: 1610.06546 [quant-ph].

[53] L. Grover and T. Rudolph, *Creating superpositions that correspond to efficiently integrable probability distributions*, 2002. arXiv: quant-ph/0208112.

[54] M. Möttönen, J. J. Vartiainen, V. Bergholm and M. M. Salomaa, *Transformation of quantum states using uniformly controlled rotations*, 2004. arXiv: quant-ph/0407010.

[55] L. N. Trefethen, *Spectral Methods in MATLAB*. SIAM, 2000.

[56] L. N. Trefethen, *Approximation Theory and Approximation Practice*. SIAM, 2013.

[57] R. J. LeVeque, *Finite Difference Methods for Ordinary and Partial Differential Equations: Steady-State and Time-Dependent Problems*. SIAM, 2007.

[58] J. P. Boyd, *Chebyshev and fourier spectral methods: Second revised edition*, 2nd. Courier Corporation, Dec. 2001.

[59] C. Canuto, M. Y. Hussaini, A. Quarteroni and T. A. Zang, *Spectral Methods: Fundamentals in Single Domains*. Springer, 2006.

[60] R. G. Voigt, D. Gottlieb and M. Y. Hussaini, *Spectral Methods for Partial Differential Equations*. SIAM, 1984.

[61] M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables* (NBS Applied Mathematics Series 55). National Bureau of Standards, 1964.

[62] P. B. Bochev and M. D. Gunzburger, *Least-Squares Finite Element Methods* (Applied Mathematical Sciences). Springer, 2009, vol. 166.

[63] P.-Å. Wedin, 'Perturbation Bounds in Connection with Singular Value Decomposition,' *BIT*, vol. 12, no. 1, pp. 99–111, 1972. DOI: 10.1007/BF01932678

[64] C. Davis and W. M. Kahan, 'The Rotation of Eigenvectors by a Perturbation. III,' *SIAM Journal on Numerical Analysis*, vol. 7, no. 1, pp. 1–46, 1970. DOI: 10.1137/0707001

[65] Qiskit contributors, *Qiskit: An Open-source Framework for Quantum Computing*, 2024. DOI: 10.5281/zenodo.2573505

[66] V. Bergholm et al., *PennyLane: Automatic differentiation of hybrid quantum-classical computations*, 2018. arXiv: 1811.04968.

Appendices

A Quantum Gate Reference **63**

B Chebyshev and Ultraspherical Identities **63**
B.1 The differentiation recurrence 63
B.2 The normalised basis 63
B.3 The scale factor and the feature vector's norm 64
B.4 The structure and norm of the derivative matrix 65
B.5 The Ultraspherical Connection 66
B.6 The $N = 4$ instance 67

C The Standard Form of Wu et al. in Full **68**
C.1 The faithful rebuild's conventions 69
C.2 The standard form on several axes 69
C.3 Datum-slice rows 70
C.4 The substitution for R3c and R3d 71
C.5 The uniqueness count for an ordinary equation 71
C.6 The paper's printed matrices at $n = 2$ 72
C.7 Lifts and folds 72

D The Published Circuits as Built **73**
D.1 The circuits as built 73
D.2 The small-angle ladder and its error 74
D.3 Measured resources and error 76

E The Structured Exact Encoding of the Differentiation Matrix **76**
E.1 Proof of Proposition 1, and the two diagonal factors 76
E.2 The arithmetic SELECT 76
E.3 Proof of Proposition 2 77
E.4 The norms of the derivative and its powers 78
E.5 Subnormalisation across n 80

F The Shared Pipeline: Identities and Formulas **80**
F.1 Proof of Proposition 5 . . . 80
F.2 Qubitisation: Chebyshev polynomials of the walk . . . 80
F.3 The controlled walk . . . 80
F.4 The imaginary-time filter . . . 80
F.5 GQSP phases . . . 80
F.6 The readout identity for probabilistic preparation . . . 80

G The Descriptor Construction in Full **80**
G.1 Block Rows in Full . . . 80
G.2 Variable Coefficients and the Padded Block Count . . . 81
G.3 Kernel Equivalence on Several Axes and Under Rescaling . . . 82
G.4 The Gap Mechanism and the Artefacts Ruled Out . . . 83
G.5 Block Rescaling . . . 85
G.6 The Term IR and Its Compilation . . . 87
G.7 Proof of the Cost Proposition, and the Norms . . . 88
G.8 One Query's Success . . . 90

H Systems and the Nonlinear Route in Full **90**
H.1 Systems in Full . . . 90
H.1.1 The coupling as a sum of matchings . . . 91
H.1.2 Realification and the real embedding . . . 92
H.2 The Doubled Space . . . 93
H.3 Proof of Proposition 4 . . . 94
H.4 The Carleman Lift . . . 94
H.5 The Time Axis, Uniqueness and the Error . . . 96
H.6 Product Folds . . . 97
H.7 Fourier–Galerkin Operators . . . 98

I Verification Protocol **100**
I.1 Modes, sizes and tolerances . . . 100
I.2 Structured probes . . . 100
I.3 Per-atom residuals . . . 100
I.4 The classical reference: eigh against SVD . . . 100
I.5 The circuit-simulation memory and time model . . . 102
I.6 Acceptance and the error budget . . . 102

J Benchmark Catalogue **102**
J.1 Linear problems . . . 102
J.2 Nonlinear and coupled problems . . . 102
J.3 Results by problem . . . 102

K Reproduction Record **102**
K.1 All panels . . . 102
K.2 The Fig. 5 triage . . . 102
K.3 Panels not reproduced, and printed errors . . . 102
K.4 The paper's claims . . . 102
K.5 The paper's filter and its parameters . . . 102
K.6 The tau-truncation ablation . . . 104

L Software and Reproducibility **105**
L.1 Architecture . . . 105
L.2 Records and provenance . . . 105
L.3 Campaigns and computational resources . . . 105
L.4 Tests . . . 105
L.5 Regenerating the results . . . 106
L.6 Environment . . . 106

M Research Proposal **106**

A Quantum Gate Reference

Table 12: The gates used in this thesis.

Gate	Circuit symbol	Matrix
Hadamard		
Pauli X, Y, Z		
Rotation $R_Y(\theta)$		
Phase $P(\phi)$		
CNOT		
Toffoli		
Multi-controlled X		
Zero-state reflection $2\left\|0\right\rangle\left\langle 0\right\| - I$		
Quantum Fourier transform		

B Chebyshev and Ultraspherical Identities

This appendix records the recurrence that defines the banded factors $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ of § 5.2, their entries, the exact factorisation $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$, its connection with the ultraspherical method, and the fully populated $N = 4$ instance of the first-order descriptor operator.

B.1 The differentiation recurrence

For $f = \sum_{\ell\geq 0} a_\ell T_\ell$ with derivative $f' = \sum_{\ell\geq 0} p_\ell T_\ell$ on $[-1, 1]$, the Chebyshev coefficients satisfy

$$c_{\ell-1}\, p_{\ell-1} - p_{\ell+1} = 2\ell\, a_\ell, \qquad \ell \geq 1, \tag{B.1}$$

with $c_0 = 2$ and $c_\ell = 1$ for $\ell \geq 1$; the $\ell = 1$ row therefore carries the factor $c_0 = 2$ on p_0. Truncating to $N = 2^n$ coefficients and dropping the p_N term, Equation (B.1) is the matrix identity $\tilde{\mathbb{B}}_{\text{raw}}\, p = \tilde{\mathbb{D}}_{\text{raw}}\, a$ with $\tilde{\mathbb{D}}_{\text{raw}}$ a single superdiagonal, $(\tilde{\mathbb{D}}_{\text{raw}})_{\ell,\ell+1} = 2(\ell+1)$, and $\tilde{\mathbb{B}}_{\text{raw}}$ two diagonals, $(\tilde{\mathbb{B}}_{\text{raw}})_{\ell,\ell} = c_\ell$, $(\tilde{\mathbb{B}}_{\text{raw}})_{\ell,\ell+2} = -1$.

The recurrence follows from two trigonometric identities. With $x = \cos\theta$, $T_k' = k \sin k\theta / \sin\theta$, and $\sin(k+1)\theta - \sin(k-1)\theta = 2\cos k\theta \sin\theta$ gives $T'_{k+1}/(k+1) - T'_{k-1}/(k-1) = 2T_k$ for $k \geq 2$, with $T_1 = T_2'/4$ and $T_0 = T_1'$. Integrating $f' = \sum_k p_k T_k$ term by term with these, the coefficient of T_ℓ in f collects $c_{\ell-1}p_{\ell-1}/(2\ell)$ from $T_{\ell-1}$ and $-p_{\ell+1}/(2\ell)$ from $T_{\ell+1}$, which is Equation (B.1); the constant of integration affects a_0 alone, which no row of the recurrence reads. The truncation is exact: the derivative of a polynomial of degree below N has degree at most $N-2$, so the dropped coefficients p_N and p_{N+1} vanish, and the last row, $p_{N-1} = 0$, holds.

B.2 The normalised basis

The construction of § 5.2 works in the normalised basis $\varphi_0 = T_0/2^{n/2}$, $\varphi_\ell = T_\ell/2^{(n-1)/2}$ $(\ell \geq 1)$, orthonormal under the discrete N-point Chebyshev inner product $\langle T_k, T_\ell\rangle_N = \frac{N}{2}\, c_\ell\, \delta_{k\ell}$ [55]:

since $N = 2^n$, this gives $\|\varphi_0\|_N = \|\varphi_\ell\|_N = 1$ exactly, so the coefficient vector of a unit-norm function is a unit vector. Writing $S = \operatorname{diag}(\sqrt{c_\ell})$, the normalised factors are $\tilde{\mathbb{B}} = S\,\tilde{\mathbb{B}}_{\text{raw}}\,S^{-1}$ and $\tilde{\mathbb{D}} = S\,\tilde{\mathbb{D}}_{\text{raw}}\,S^{-1}$, that is

$$\tilde{\mathbb{D}}_{\ell,\ell+1} = 2(\ell+1)\sqrt{c_\ell/c_{\ell+1}}, \qquad \tilde{\mathbb{B}}_{\ell,\ell} = c_\ell, \qquad \tilde{\mathbb{B}}_{\ell,\ell+2} = -\sqrt{c_\ell/c_{\ell+2}}, \tag{B.2}$$

so $\tilde{\mathbb{D}}$ has the single non-zero diagonal ($2\sqrt{2}$, 4, 6, 8, …) and $\tilde{\mathbb{B}}$ has main diagonal (2, 1, 1, …) and second superdiagonal ($-\sqrt{2}$, -1, -1, …). Both hold $\mathcal{O}(N)$ non-zero entries at any N. The dense Chebyshev derivative matrix is their exact quotient,

$$\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}, \tag{B.3}$$

an algebraic identity rather than a numerical approximation. $\tilde{\mathbb{B}}$ is upper triangular with the non-zero diagonal $(2, 1, \ldots, 1)$, so it is invertible. For any coefficients a the true derivative's coefficients satisfy the truncated rows of Equation (B.1), as shown above, and they are $\mathbb{G}a$; conjugating by S gives $\tilde{\mathbb{B}}\mathbb{G} = \tilde{\mathbb{D}}$, so the equality holds exactly at every N.

Proposition 7 (The inverse of the mass factor)**.** $\tilde{\mathbb{B}}^{-1}$ *is upper triangular and supported where the column index minus the row index is even. Its entries there are* 1 *in every row* $k \geq 1$*; in row* 0 *they are* $\frac{1}{2}$ *at column* 0 *and* $1/\sqrt{2}$ *at every even column* $q \geq 2$*.*

Proof. Away from row 0, $\tilde{\mathbb{B}} = I - P$ with P the shift by two, so on each parity class its inverse is $\sum_m P^m$, a triangle of ones. Row 0 of $\tilde{\mathbb{B}}x = e_q$ reads $2x_0 - \sqrt{2}x_2 = \delta_{0q}$, which gives $x_0 = \frac{1}{2}$ for $q = 0$ and, with $x_2 = 1$, $x_0 = 1/\sqrt{2}$ for even $q \geq 2$. □

Multiplying out $\tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$ with this inverse returns the entries of Equation (4.2): $\tilde{\mathbb{D}}$'s single diagonal forces the inner index to be $p-1$, the parity support of $\tilde{\mathbb{B}}^{-1}$ makes $p-k$ odd, and the entry is $2p$ times one, or times $1/\sqrt{2}$ in row 0. The density of $\mathbb{G}$ is the inverse of the banded $\tilde{\mathbb{B}}$, which is what the descriptor never forms. The triangle of ones has norm $\Theta(N)$ and $\tilde{\mathbb{D}}$ has norm $2(N-1)$; their product has the $\Theta(N^2)$ norm of $\mathbb{G}$.

B.3 The scale factor and the feature vector's norm

The basis functions $\varphi_0 = T_0/\sqrt{N}$ and $\varphi_k = T_k/\sqrt{N/2}$ $(1 \leq k < N)$ are orthonormal under the discrete inner product on the N Chebyshev nodes. Under the continuous Chebyshev weight $w(x) = (1-x^2)^{-1/2}$ they are orthogonal with a common norm: with $x = \cos\theta$, $\int_{-1}^{1} T_j T_k w\,dx = \int_0^\pi \cos j\theta \cos k\theta\,d\theta$, which is π for $j = k = 0$, $\pi/2$ for $j = k \geq 1$ and zero otherwise, so $\int \varphi_j\varphi_k w\,dx = (\pi/N)\,\delta_{jk}$ for every $j, k < N$. For $f_N = \sum_k b_k\varphi_k$ Parseval's identity is therefore $\int_{-1}^{1} f_N^2 w\,dx = (\pi/N)\|b\|^2$, which is Equation (4.1).

Proposition 8 (The feature vector's norm)**.** *With* $x = \cos\theta$*,*

$$\|\tau(x)\|^2 = 1 + \frac{1}{N}\Big(\frac{\sin(2N-1)\theta}{2\sin\theta} - \frac{1}{2}\Big), \qquad \|\tau(\pm 1)\|^2 = \frac{2N-1}{N}, \tag{B.4}$$

and $\|\tau(x)\|^2 \le (2N-1)/N < 2$ *on* $[-1,1]$.

Proof. Since $2T_k^2 = 1 + \cos 2k\theta$,

$$\|\tau(x)\|^2 = \frac{1}{N}\Big(T_0^2 + 2\sum_{k=1}^{N-1} T_k^2\Big) = \frac{1}{N}\Big(N + \sum_{k=1}^{N-1} \cos 2k\theta\Big).$$

The sum is the Dirichlet kernel $\sin((2N-1)\theta)/(2\sin\theta) - \frac{1}{2}$, at most $N-1$, which it attains at $\theta = 0$ and π. □

The norm oscillates about one and is largest at the ends, so a constraint row at an end of the interval weighs most. That weight is part of the paper's construction and is kept (§ 4.1). Normalising $\tau(x)$ point by point would also break the consistency of the collapse rows: two collapses $\mathbb{B}(x_s)/f(x_s)$ and $\mathbb{B}(x'_s)/f(x'_s)$ map the solution to the same constant, because $\mathbb{B}(x)b = f_N(x)\sqrt{N}\,|0\rangle$, whereas with $\tau(x)/\|\tau(x)\|$ they would map it to constants differing by the factor $\|\tau(x'_s)\|/\|\tau(x_s)\|$.

B.4 The structure and norm of the derivative matrix

Proposition 9 (Structure and norm of $\mathbb{G}$)**.** *For* $n \ge 2$*,* $\mathbb{G}$ *of Equation* (4.2) *is strictly upper triangular and nilpotent;* $\ker \mathbb{G}^j$ *is the space of polynomials of degree below* j*; it has* $N^2/4$ *non-zero entries; it is orthogonally similar to the matrix that maps the values of a polynomial of degree below* N *at the Chebyshev nodes to the values of its derivative there; and*

$$\frac{N^2}{4\sqrt{2}} \;\le\; \|\mathbb{G}\|_2 \;\le\; \frac{N^2}{\sqrt{2}}. \tag{B.5}$$

Proof. The entries are non-zero only for $p > k$, so $\mathbb{G}$ is strictly upper triangular and $\mathbb{G}^N = 0$. $\mathbb{G}$ represents differentiation exactly on polynomials of degree below N, so $\mathbb{G}^j b = 0$ exactly when the j-th derivative of the represented polynomial vanishes, that is when its degree is below j. Column p has a non-zero entry in each row $k < p$ of the other parity, $\lceil p/2 \rceil$ of them, and $\sum_{p<N} \lceil p/2 \rceil = N^2/4$.

For the similarity, let $\Phi_{jk} = \varphi_k(\bar{x}_j)$ at the nodes $\bar{x}_j$. The discrete orthonormality of the φ_k says that Φ is orthogonal, and Φb is the vector of nodal values of the polynomial with coefficients b, so $\Phi\mathbb{G}\Phi^\top$ maps nodal values to the nodal values of the derivative.

For the upper bound, $\|\mathbb{G}\|_2 \le \|\mathbb{G}\|_F$, and each of the $\lceil p/2 \rceil$ entries of column p is at most $2p$ in magnitude, so

$$\|\mathbb{G}\|_F^2 \le \sum_{p=0}^{N-1} 4p^2\Big(\frac{p}{2} + \frac{1}{2}\Big) = \frac{(N-1)^2N^2}{2} + \frac{(N-1)N(2N-1)}{3} = \frac{N^4}{2} - \frac{N^3}{3} - \frac{N^2}{2} + \frac{N}{3} \le \frac{N^4}{2}.$$

For the lower bound, let u be the unit vector uniform over the $N/4$ even rows $k < N/2$ and v

the unit vector uniform over the $N/4$ odd columns $p > N/2$. Every pair (k, p) has $p > k$ with $p - k$ odd, and its entry is $2p > N$ for $k \geq 2$ and $\sqrt{2}p > N/\sqrt{2}$ for $k = 0$, so

$$\|\mathbb{G}\|_2 \geq u^\top \mathbb{G} v \geq \frac{4}{N} \cdot \left(\frac{N}{4}\right)^2 \cdot \frac{N}{\sqrt{2}} = \frac{N^2}{4\sqrt{2}}. \qquad \square$$

The bounds give $\|\mathbb{G}\|_2 = \Theta(N^2)$. The exact limit, $\|\mathbb{G}\|_2/N^2 \to 1/(2j_{-3/4,1}) \approx 0.4724$, with $j_{-3/4,1}$ the first positive zero of the Bessel function $J_{-3/4}$, and the growth of every power, $\|\mathbb{G}^k\|_2 = \Theta(N^{2k})$, are proved in Appendix E.4.

B.5 The Ultraspherical Connection

The banded factorisation above is the first-derivative case of the ultraspherical spectral method of Olver and Townsend [4]. Stating the correspondence precisely is what fixes what this thesis adds to it.

Olver and Townsend represent a function's derivative not in the Chebyshev basis T_k but in the ultraspherical (Gegenbauer) basis $C_k^{(\lambda)}$, the family of polynomials orthogonal with respect to the weight $(1 - x^2)^{\lambda - 1/2}$; at $\lambda = 1$ these are the Chebyshev polynomials of the second kind, $C_k^{(1)} = U_k$. Differentiation of a Chebyshev series obeys

$$\frac{\mathrm{d}T_k}{\mathrm{d}x} = k\, C_{k-1}^{(1)}, \qquad k \geq 1 \tag{B.6}$$

[4, Eq. 2.5], so for $u = \sum_k u_k T_k$ the coefficients of u' in the $C^{(1)}$ basis are read off directly: $(\mathcal{D}_0 u)_k = (k+1)u_{k+1}$, a single superdiagonal, sparse at every truncation. Because u''s coefficients now live in a different basis to u's own, a second, purely algebraic operator is needed to convert a function's own Chebyshev coefficients into the $C^{(1)}$ basis without differentiating it: $\mathcal{S}_0$, with $(\mathcal{S}_0)_{k,k} = 1$ $(k = 0)$, $\frac{1}{2}$ $(k \geq 1)$, and $(\mathcal{S}_0)_{k,k+2} = -\frac{1}{2}$ [4, Eq. 2.8]. Comparing entries, $\tilde{\mathbb{D}}_{\text{raw}} = 2\mathcal{D}_0$ and $\tilde{\mathbb{B}}_{\text{raw}} = 2\mathcal{S}_0$ exactly: the recurrence $\tilde{\mathbb{B}}_{\text{raw}}\, p = \tilde{\mathbb{D}}_{\text{raw}}\, a$ of Equation (B.1) is literally $\mathcal{S}_0\, p = \mathcal{D}_0\, a$, doubled by the convention that folds the weight of T_0 into $c_0 = 2$ rather than into the polynomial itself. This is the discrete shadow the framing of § 5.2 claims: the descriptor recurrence does not compute a new identity, it asserts an existing one – that p's own Chebyshev coefficients, converted into the $C^{(1)}$ basis by $\mathcal{S}_0$, must equal the derivative computed directly by $\mathcal{D}_0$ – as one row of a linear system, rather than solving it for p via $\mathcal{S}_0^{-1}$.

The generalisation to order $k \geq 1$ in [4] raises the basis one level per derivative: a λ-th derivative needs coefficients in $C^{(\lambda)}$, obtained from the analogous pair $\mathrm{d}C_k^{(\lambda)}/\mathrm{d}x = 2\lambda\, C_{k-1}^{(\lambda+1)}$ ($k \geq 1$, giving $\mathcal{D}_\lambda$, still a single superdiagonal) and a conversion operator $\mathcal{S}_\lambda : C^{(\lambda)} \to C^{(\lambda+1)}$, still two diagonals [4, Eqs. 3.2–3.3]. To assemble one equation of order k, every term is raised to the common top basis $C^{(k)}$ by a chain $\mathcal{S}_{k-1} \cdots \mathcal{S}_{\lambda+1}$ applied after each term's own $\mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$, giving one almost-banded operator $\mathrm{L} = \mathcal{M}_k[a^k]\mathcal{D}_k + \sum_{\lambda=0}^{k-1} \mathcal{S}_{k-1} \cdots \mathcal{S}_\lambda \mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$ that is solved once for the coefficients of $u^{(k)}$ [4, §3.2] – so a k-th order equation needs k distinct conversion operators $\mathcal{S}_0, \ldots, \mathcal{S}_{k-1}$ chained together and the lower-order unknowns eliminated.

The descriptor construction of § 5.3 takes a different route to the same sparsity. Rather than raising the basis and eliminating, it keeps every derivative's coefficients in the *same* normalised Chebyshev basis, as its own block $\hat{f}^{(j)}$ of w, and reuses the single $\lambda = 0$ pair $(\tilde{\mathbb{B}}, \tilde{\mathbb{D}})$ between every consecutive pair of blocks (Equation (5.5)). No elimination is performed, and no operator beyond $\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$ and the multiplication operators of Appendix G.2 is required at any order k – the price is the larger state w that carries every intermediate derivative explicitly instead of eliminating it (Appendix G.2). What this thesis adds to [4] is not a new sparsification of differentiation – that is their result – but applying the *same* $\lambda = 0$ identity block-wise to build an unreduced, differential-algebraic residual for the physics-informed Hamiltonian of [1], whose block structure is exactly what § 5.7 block-encodes.

For a first-order equation the two coincide. The descriptor's mass form multiplies the standard rows by $\tilde{\mathbb{B}}$ (Equation (5.7)), and for $f' = af$ with a a number, $\tilde{\mathbb{B}}(\mathbb{G} - aI) = \tilde{\mathbb{D}} - a\tilde{\mathbb{B}}$, which in the raw basis is $2(\mathcal{D}_0 - a\mathcal{S}_0)$: the ultraspherical method's own operator in the basis $C^{(1)}$, on N unknowns with no derivative block.

A reduction the descriptor does not build. For $k \geq 2$ the same left factor exists. In the raw basis $\mathcal{S}_1\mathcal{S}_0\mathbb{G}^2 = \mathcal{D}_1\mathcal{D}_0$ exactly, so $\mathcal{S}_{k-1} \cdots \mathcal{S}_0 \, A_{\text{std}}$ is the ultraspherical method's operator for a constant-coefficient equation: banded, with a bandwidth independent of N, on the N coefficients of f, and, since the conversions are invertible, with the standard form's kernel, as in Theorem 1. It would remove the chain's block register and its recurrence and padding rows. It would also change three things this thesis measures: the residual's rows are weighted by the conversions, so its least-squares ground state is that of a differently weighted problem; the conversions' entries would have to be encoded; and the gap mechanism of § 5.5 does not carry over without a measurement. This reduction was not built into the pipeline, and the thesis's descriptor is not compared against it.

B.6 The $N = 4$ instance

At $n = 2$ ($N = 4$), Equation (B.2) gives

$$\tilde{\mathbb{B}} = \begin{pmatrix} 2 & 0 & -\sqrt{2} & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \tilde{\mathbb{D}} = \begin{pmatrix} 0 & 2\sqrt{2} & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \mathbb{G} = \begin{pmatrix} 0 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad \text{(B.7)}$$

The top-right entry $3\sqrt{2}$ of $\mathbb{G}$ is the fill-in: coefficient 0 of the derivative depends on coefficient 3 of the input. Row ℓ of $\mathbb{G}$ is non-zero at columns $\ell + 1, \ell + 3, \ldots$, so this is the extreme case – the lowest-order coefficient can depend on up to half the others, while the highest-order one (row 3 above) depends on none. The factorisation removes the fill-in entirely: no row of $\tilde{\mathbb{B}}$ or $\tilde{\mathbb{D}}$ touches more than two coefficients.

For the first-order equation $f' + 2f = 0$ (R1), the descriptor runs the mass form of Equation (5.7), on the field's own N coefficients and with no derivative block:

$$\tilde{\mathbb{D}} + 2\tilde{\mathbb{B}} = \begin{pmatrix} 4 & 2\sqrt{2} & -2\sqrt{2} & 0 \\ 0 & 2 & 4 & -2 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 2 \end{pmatrix} = \tilde{\mathbb{B}}\,(\mathbb{G} + 2I), \qquad \mathbb{G} + 2I = \begin{pmatrix} 2 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 2 \end{pmatrix}. \quad \text{(B.8)}$$

The standard form's equation block $\mathbb{G} + 2I$ carries the fill-in $3\sqrt{2}$; the mass form's has none, and every row touches at most three coefficients. Its kernel is that of $\mathbb{G} + 2I$, because $\tilde{\mathbb{B}}$ is invertible.

The same equation can be carried as a chain, with its derivative as a second block, $w = (\hat{f}^{(0)};\, \hat{f}^{(1)})$, the recurrence $-\tilde{\mathbb{D}}\,\hat{f}^{(0)} + \tilde{\mathbb{B}}\,\hat{f}^{(1)} = 0$ and the equation row $2\hat{f}^{(0)} + \hat{f}^{(1)} = 0$:

$$A_{\text{desc}} = \left[\begin{array}{cccc|cccc} 0 & -2\sqrt{2} & 0 & 0 & 2 & 0 & -\sqrt{2} & 0 \\ 0 & 0 & -4 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -6 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right]. \quad \text{(B.9)}$$

The top four rows are the recurrence and the bottom four the equation. This chain doubles the state for a block the equation already determines, which is why first-order equations take the mass form (§ 5.3); it is kept only as the mass form's ablation, whose kernel and decoded field agree with the mass form's.

C The Standard Form of Wu et al. in Full

This appendix gives the $N = 4$ instance of the standard-form residual of § 4.2 for a constant-coefficient equation, the entries of the lifts $\mathbb{M}_{x^p}$ that the same section uses for a variable coefficient, and the constructions and proofs that § 4.3 points to.

Why a condition for all x is a finite linear system. A condition required at every $x \in [-1, 1]$, such as the differential equation, reaches the form $\langle \tau(x)|\, C\, |\psi\rangle = 0$ for all x. The components of $\langle \tau(x)|$ are linearly independent functions of x, so $\langle \tau(x)|\, \mathbf{v}$ vanishes for all x only if $\mathbf{v} = 0$. Taking $\mathbf{v} = C\,|\psi\rangle$, the condition for all x is therefore equivalent to the finite system $C\,|\psi\rangle = 0$, which no longer depends on x. This is the step that §§ 3.2 and 4.2 take without proof.

The standard-form residual at $N = 4$. For Wu et al.'s stiff pilot equation [1], $f'' + 5f' + 400f = 0$, the equation block of Equation (4.3) is $A_{\text{std}} = \mathbb{G}^2 + 5\mathbb{G} + 400I$. At $N = 4$, with the entries of Equation (4.2),

$$A_{\text{std}} = \begin{pmatrix} 400 & 5\sqrt{2} & 4\sqrt{2} & 15\sqrt{2} \\ 0 & 400 & 20 & 24 \\ 0 & 0 & 400 & 30 \\ 0 & 0 & 0 & 400 \end{pmatrix}. \tag{C.1}$$

It is upper triangular, as every polynomial in the strictly upper-triangular $\mathbb{G}$ is. At this size the largest off-diagonal entry, 30, is about one order of magnitude below the diagonal; the entries of $\mathbb{G}^2$ grow as N^4, so at larger N the derivative terms dominate the constant one.

The lifting operator $\mathbb{M}_{x^p}$. $\mathbb{M}_{x^p}$ is exact and *rectangular*: multiplying a degree-$(N-1)$ model by x^p can raise its degree by as much as p, so the result is represented one register wider, $2N \times N$, with no truncation. At $n = 1$ ($N = 2$, so $p \in \{1, 2\}$),

$$\mathbb{M}_{x^1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}, \qquad \mathbb{M}_{x^2} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{3}{2\sqrt{2}} \\ \frac{1}{2} & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{pmatrix}, \tag{C.2}$$

which reproduce the paper's printed matrices. The general entry follows from the product-to-sum identity conjugated into the normalised bases of $|\tau(x)\rangle_n$ and $|\tau(x)\rangle_{n+1}$ (Appendix C.7).

C.1 The faithful rebuild's conventions

C.2 The standard form on several axes

Stub: write this subsection out in full (derivation of the register count, a worked three-variable

§ 3.2 writes the two-variable case. For ω variables $\mathbf{x} = (x_1, \ldots, x_\omega)$, each carried on its own n-qubit Chebyshev axis u, the state lives on ωn qubits and the feature vector is $|\tau(\mathbf{x})\rangle = |\tau(x_1)\rangle_n \otimes \cdots \otimes |\tau(x_\omega)\rangle_n$, so that $f_q(\mathbf{x}) = \langle \tau(\mathbf{x}) | \psi \rangle$. The derivative along axis u is $\mathbb{G}_u = I^{\otimes(u-1)} \otimes \mathbb{G} \otimes I^{\otimes(\omega-u)}$; these act on different factors and commute, and a derivative of multi-index $\boldsymbol{\chi} = (\chi_1, \ldots, \chi_\omega)$ is $\partial^{\boldsymbol{\chi}} \mapsto \mathbb{G}^{\boldsymbol{\chi}} = \prod_u \mathbb{G}_u^{\chi_u} = \bigotimes_u \mathbb{G}^{\chi_u}$.

Each term t of the equation has a constant c_t, a multi-index $\boldsymbol{\chi}_t$ and a coefficient m_t that is a polynomial in a single variable x_{u_t} (the class the code builds; a coefficient that depends on several variables is not built). An axis is *lifted* when it carries a non-constant coefficient or when the equation has a source, in which case every axis is lifted. The term acts on axis u as $\mathbb{M}_{m_t}\mathbb{G}^{\chi_{t,u}}$ if $u = u_t$, as $\mathbb{M}_1\mathbb{G}^{\chi_{t,u}}$ on a lifted axis without a coefficient, and as $\mathbb{G}^{\chi_{t,u}}$ on an unlifted axis, so that

$$A_{\text{std}} = \sum_t c_t \bigotimes_{u=1}^{\omega} \mathbb{M}_t^{(u)}\, \mathbb{G}^{\chi_{t,u}} \;-\; \mathbb{M}_r\, \mathbb{D}^{(0)}(\mathbf{x}_s), \tag{C.3}$$

with $\mathbb{M}_t^{(u)}$ as just described. The source term is the rank-one operator $\mathbb{M}_r\mathbb{D}^{(0)}(\mathbf{x}_s) = |\hat{r}\rangle \langle\tau(\mathbf{x}_s)| / y_s$, with $|\hat{r}\rangle$ the coefficients of r on the lifted register (the Kronecker product of the one-dimensional coefficient vectors where r is a product of single-variable polynomials). The equation block's output register has $\omega n + \omega_{\text{lift}}$ qubits, ω_{lift} being the number of lifted axes. For $\omega = 1$, Equation (C.3) is Equation (4.3).

An invariant constraint $\partial_u^j f = 0$ on the face $x_u = x_z$ is the row block $I^{\otimes(u-1)} \otimes \mathbb{B}_n(x_z)\mathbb{G}^j \otimes I^{\otimes(\omega-u)}$, or $\mathbb{B}_{n+1}(x_z)\mathbb{M}_1\mathbb{G}^j$ on its own axis if that axis is lifted. The identity factors are on n qubits whether or not their axes are lifted, so only the equation block, not each constraint, acts into the full lifted register. The regular datum at a point $\mathbf{x}_s$ is the rank-one row $\mathbb{D}^{(0)}(\mathbf{x}_s) = \bigotimes_u \mathbb{B}_n(x_{s,u})/y_s = 2^{\omega n/2} |0\rangle \langle\tau(\mathbf{x}_s)| / y_s$. Each $\mathbb{G}_u$ has the norm $\Theta(N^2)$ of $\mathbb{G}$.

C.3 Datum-slice rows

An inhomogeneous initial or boundary function g on the slice $x_a = x^*$ of a problem on several Chebyshev axes is imposed by the rows of Equation (4.4),

$$\mathbb{S}_a(x^*) - \frac{\hat{g}\ \langle\tau(x_s)|}{f(x_s)}, \qquad \mathbb{S}_a(x^*) = I \otimes \cdots \otimes \mathbb{B}(x^*) \otimes \cdots \otimes I \quad (\mathbb{B} \text{ on axis } a),$$

with $\hat{g}$ the coefficients of g on the other axes broadcast as a constant along axis a: its mode 0 on that axis, $\sqrt{N_a}$ times the coefficients of g, as $\mathbb{B}$ emits. $\langle\tau(x_s)|$ is the feature row at a regular point x_s of the full domain, where the value $f(x_s)$ is known.

Proposition 10 (Datum slices)**.** *On cf, with f the solution and c any scalar, the rows return $c\left(\hat{f}|_{x_a=x^*} - \hat{g}\right)$, with $\hat{f}|_{x_a=x^*}$ the broadcast slice of f. They are therefore homogeneous in the scale and vanish on the solution and its multiples.*

Proof. $\mathbb{S}_a(x^*)$ contracts axis a with $\tau(x^*)$, which gives the slice function's coefficients on the other axes times $\sqrt{N_a}\,|0\rangle$ on axis a: applied to cf, the broadcast slice $c\hat{f}|_{x_a=x^*}$. The second term applied to cf is $\hat{g}\, c\, f(x_s)/f(x_s) = c\,\hat{g}$. If $f|_{x_a=x^*} = g$ the difference is zero. □

The block is square on $\prod_a N_a$ coefficients but non-zero only on the $\prod_{b\neq a} N_b$ rows with mode 0 on axis a, which carry the slice; the rest are dropped from R without changing $R^\dagger R$. On one axis the slice is a point, and the rows reduce to the single row $\mathbb{B}(x^*) - y^*\mathbb{D}^{(0)}(x_s)$ of a point datum $f(x^*) = y^*$, or to $\mathbb{B}(x^*)\mathbb{G}^j - y^*\mathbb{D}^{(0)}(x_s)$ for a datum on a derivative (Definition 10). The rows are built from the problem's statement, the function g, the slice and one regular value, so they serve both forms and every resolution; whether the kernel is then one-dimensional is computed (§ 4.3).

Why the heat data sit at $t = -1$. The paper's heat problem, $\partial_t f = \frac{1}{25}\partial_x^2 f$ with $f(t, \pm 1) = 0$, states its initial function $\sin(2\pi x)$ at $t = 0$, inside the Chebyshev time domain $[-1, 1]$. Imposed there, it fixes the solution on $[-1, 0]$ by a backward heat problem: going back a time

s, a mode $\sin\frac{\ell\pi(x+1)}{2}$ of the data is multiplied by $\exp\bigl(\frac{1}{25}(\ell\pi/2)^2 s\bigr)$, a factor unbounded in ℓ, so the problem is ill posed in Hadamard's sense. The standard form states the problem forward from $t=-1$ instead, with the initial function $e^{4\pi^2/25}\sin(2\pi x)$, the paper's solution $e^{-4\pi^2 t/25}\sin(2\pi x)$ evaluated there. The solution is the same; the data are set up from it, as R4's constraint points are.

C.4 The substitution for R3c and R3d

The paper's R3c and R3d are the associated Legendre equation with $m=1$, multiplied through by $1-x^2$,

$$(1-x^2)^2 f'' - 2x(1-x^2)f' + \bigl(l(l+1)(1-x^2)-1\bigr)f = 0,$$

with $l=5$ and $l=6$. Its leading coefficient has a double zero at each end, and its solution $P_l^1 = -\sqrt{1-x^2}\,P_l'$ has a branch point there, so its Chebyshev coefficients decay only algebraically: $\sqrt{1-x^2}=|\sin\theta|$ has cosine coefficients of order k^{-2}.

Proposition 11 (The substitution)**.** *With $f=\sqrt{1-x^2}\,v$ the equation becomes*

$$(1-x^2)\,v'' - 4x\,v' + \bigl(l(l+1)-2\bigr)v = 0, \tag{C.4}$$

whose leading coefficient has simple zeros at ±1 and whose solution $v=-P_l'$ is a polynomial of degree $l-1$.

Proof. Write $s=\sqrt{1-x^2}$, so that $s'=-x/s$ and $s''=-1/s^3$. Then $f'=sv'-xv/s$ and $f''=sv''-2xv'/s-v/s^3$. Substituting, $(1-x^2)^2 f''=s^5v''-2xs^3v'-sv$, $-2x(1-x^2)f'=-2xs^3v'+2x^2sv$ and $\bigl(l(l+1)s^2-1\bigr)sv=l(l+1)s^3v-sv$. The sum is $s^5v''-4xs^3v'+\bigl(l(l+1)s^3-2s(1-x^2)\bigr)v = s^3\bigl[(1-x^2)v''-4xv'+(l(l+1)-2)v\bigr]$, and $s^3>0$ on $(-1,1)$. The solution is $v=f/s=-P_l'$. □

The data carry through: $f(\pm1)=0$ holds for every v and is dropped; R3c's $f'(0)=0$ is $v'(0)=0$, since $s(0)=1$ and $s'(0)=0$; R3d's $f(0)=0$ is $v(0)=0$; and the regular datum becomes $v(x_s)=f(x_s)/\sqrt{1-x_s^2}$, at $x_s=\frac{1}{2}$. The coefficients of Equation (C.4) are polynomials of degree at most two, inside the lift's range, and v is represented exactly once $N>l-1$, so the field error falls to rounding at the first such n (§ 4.3). The substitution removes this double zero because the branch point is a known factor of the solution; it is not a remedy for a double zero in general. The simple zeros it leaves are what the descriptor's block rescaling can treat (§ 5.6).

C.5 The uniqueness count for an ordinary equation

Proposition 12 (Uniqueness count)**.** *Let $\sum_{j=0}^{k} a_j f^{(j)} = 0$ on $[-1,1]$ with smooth coefficients and a_k nowhere zero, and let $c_1,\dots,c_m$ be invariant constraints (each sets a derivative of f at a point to zero). The solutions that satisfy all of them form a space of dimension $k-r$, where*

r is the rank of the c_i restricted to the k-dimensional solution space S. The ground state of the constraint energy is unique up to scale exactly when $r = k - 1$.

Proof. Since a_k has no zero, the equation is a regular linear system of order k and S has dimension k. The map $f \mapsto (c_i(f))_{i \le m}$ is linear on S with rank r, so its kernel, the solutions that satisfy every constraint, has dimension $k - r$. A ground state of zero energy is a nonzero element of that kernel, unique up to scale exactly when the kernel is one-dimensional. For $r = k$ the kernel is $\{0\}$ and no nonzero state has zero energy. For $r < k - 1$ it has dimension at least two. □

The rank is that of the constraints restricted to S, not of the functionals, so $k - 1$ constraints chosen without regard to the equation can fail. The regular datum fixes the scale of the selected solution and requires that it not vanish at the regular point. The proposition concerns the continuous problem: the assembled residual follows the count only as far as the truncation resolves S, and the dimension and gap of its kernel are computed, not inferred (§ 4.3). For an inhomogeneous equation, or one whose leading coefficient vanishes (R3a), no count is claimed here and uniqueness is judged by the computed kernel.

The Laplace invariant set. The modes $\sin \frac{\ell\pi(x+1)}{2} \sinh \frac{\ell\pi(y+1)}{2}$, $\ell \ge 1$, all satisfy the equation and its three homogeneous sides, and only $\ell = 1$ matches the top-side function, which the printed set does not impose. This is the continuous kernel that the truncation resolves as n grows.

C.6 The paper's printed matrices at $n = 2$

C.7 Lifts and folds

For $a(x) = \sum_m a_m T_m(x)$ the identity $T_m T_k = \frac{1}{2}\big(T_{m+k} + T_{|m-k|}\big)$ gives the Chebyshev coefficients of $a\,T_k$, $X_{jk} = \frac{1}{2}\sum_m a_m\big([\,j = m + k\,] + [\,j = |m - k|\,]\big)$. In the normalised bases of n input and $n + 1$ output qubits, with σ_k the normalisation of T_k on each register ($\varphi_k = T_k/\sigma_k$), the lift is $(\mathbb{M}_a)_{jk} = (\sigma_j^{\text{out}}/\sigma_k^{\text{in}})\, X_{jk}$; a monomial coefficient is converted to its Chebyshev series first.

Proposition 13 (The lift)**.** *$\mathbb{M}_a \in \mathbb{R}^{2N \times N}$ represents the product $a\,f_N$ without truncation for every f_N of degree below N if and only if* $\deg a \le N$*. A constant enters as $a_0\mathbb{M}_1 = a_0\sqrt{2}\,[\,I;\,0\,]$, and $\|\mathbb{M}_a\|_2 \le \sqrt{2}\,\sup_{[-1,1]}|a|$.*

Proof. The product has degree at most $N - 1 + \deg a$, and $2N$ coefficients hold degrees up to $2N - 1$, so $\deg a \le N$ suffices; if $\deg a > N$, a series of degree $N - 1$ gives a product of degree at least $2N$, which does not fit. Each normalisation σ_k on $n + 1$ qubits is $\sqrt{2}$ times that on n, since $\varphi_k = T_k/\sqrt{N/2}$ on n qubits and $T_k/\sqrt{N}$ on $n + 1$ for $k \ge 1$, and likewise for T_0, so the coefficients of a function are multiplied by $\sqrt{2}$ on the wider register, which gives $\mathbb{M}_1$. For the norm, Parseval's identity of Appendix B.3 on $n + 1$ qubits, where the product is exact, gives $\|\mathbb{M}_a b\|^2 = (2N/\pi)\int a^2 f_N^2 w\,dx \le 2\sup|a|^2\,(N/\pi)\int f_N^2 w\,dx = 2\sup|a|^2\,\|b\|^2$. □

The lift of x has the non-zero entries $(\mathbb{M}_x)_{10} = (\mathbb{M}_x)_{01} = 1$ and $(\mathbb{M}_x)_{k\pm1,k} = 1/\sqrt{2}$ otherwise, from $xT_0 = T_1$ and $xT_k = \frac{1}{2}(T_{k+1}+T_{k-1})$, which is Equation (C.2) at $n = 1$. On a circuit the lift of x is a combination of truncated shifts, built by the post-selected subtraction of Appendix E.2, and a polynomial coefficient is the sum of the powers of that block weighted by its monomial coefficients, so its subnormalisation is $\sum_p |a_p|\, \alpha_{\mathbb{M}_x}^p$. For a coefficient that vanishes where the derivatives are largest, such as the $(1-x^2)$ of R3a, this sum far exceeds what the product $\mathbb{M}_a\mathbb{G}^2$ needs, which is the looseness of the standard form's encoding on R3 (§ 4.6). The folds $\mathbb{n}_1$ and $\mathbb{n}_x$ of the doubled space follow from the same product-to-sum identity applied to a product of two series (Equation (3.7)); their circuits and norms are given in Appendix H.2.

D The Published Circuits as Built

§ 3.3 describes, at the level a reader needs to follow the argument, the circuit Wu et al. use to load $\mathbb{G}$'s column-linear entries as a rotation amplitude [1]. This appendix gives the mechanism in full, together with a rigorous bound on its entrywise error that this thesis derives and verifies – the source paper states only that the error is "negligible" and decreases "exponentially" with n, without a closed form.

D.1 The circuits as built

The constraint rows (Figs 2 and 8). $\mathbb{B}(x)$ and $\mathbb{D}^{(0)}(x_s)$ are block-encoded by the inverse feature map $U_\tau^\dagger(x)$ followed by $\hat{S}_{n+1} = 2\,|0\rangle\langle 0| - I$ controlled inside a Hadamard sandwich on one ancilla, a two-term LCU on $n+2$ qubits. The encoded block is $\mathbb{B}(x)/\sqrt{2^{n+1}}$, and $\mathbb{D}^{(0)}(x_s)/\big(\sqrt{2^{n+1}}/f(x_s)\big)$ for the collapse row. The global phase of -1 in $\hat{S}_{n+1}$ is a relative phase once the circuit is controlled, and cannot be dropped.

The derivative circuit (Fig. 9). The registers are, in order, a flag qubit acted on by $R_Y(\omega_c)$ with $\omega_c = 2\arccos\sqrt{1/3}$, an X ancilla, an offset register of $m = n$ qubits sandwiched between Hadamards, a value ancilla, and the n-qubit system. Two conditioned subroutines assign the entries on different bases. In $\hat{U}_R$ the offset $|i\rangle$ selects the $(2i+1)$th superdiagonal and the system $|j\rangle$ the jth column; it consists of n singly-controlled $R_Y(\theta_k)$ gates on the value ancilla, controlled by the bits of $|j\rangle$, followed by n cyclic shifts $\hat{R}_n$. In $\hat{U}_C$ the offset selects the $(i+1)$th row; it consists of n multi-controlled $R_Y(\varphi_k)$ gates, one shift $\hat{L}_n$ and $2n$ swaps. With $\mu = 2/\|\mathbb{G}\|_S$ the angles are $\theta_k = 2\arcsin(-2^{k-1}\mu)$ and $\varphi_k = 2\arcsin[-2^{k-1}\mu(1/\sqrt{2}-1)]$. The circuit uses $m+n+3 = 2n+3$ qubits, its gate count is linear in n before the multi-controlled gates are decomposed and $O(n^3)$ after an $O(n^2)$ decomposition of each, and $\mathbb{G}$ is the top-left $2^n \times 2^n$ block of the unitary multiplied by $(2^{n-1}+2^n)\|\mathbb{G}\|_S$. These are the paper's statements; their tests follow.

Costing the rebuild. The rebuild restricts the offsets to odd values. Every non-zero entry of $\mathbb{G}$ has $p-k$ odd, so an even offset, $d = 0$ above all, would load a value onto the diagonal and onto

entries that must vanish; the uniform superposition over every offset drawn in Fig. 9 does not exclude them. With the ladder coefficient $\mu = 2/\|\mathbb{G}\|_S$ and the weight $2/N$ that the odd-offset superposition gives each entry, the rebuilt circuit block-encodes $\mathbb{G}$ at $\alpha = N/\mu = N\|\mathbb{G}\|_S/2$.

Lemma 1 (The paper's norm of $\mathbb{G}$)**.** *For $n \geq 3$, $\|\mathbb{G}\|_S = \|\mathbb{G}^\top\mathbb{G}\|_1 = N(N-1)^2(N+1)/3$.*

Proof. $(\mathbb{G}^\top\mathbb{G})_{pq} = \sum_k \mathbb{G}_{kp}\mathbb{G}_{kq}$ is non-zero only for $p \equiv q \pmod 2$, the sum running over the $k < \min(p,q)$ of the other parity with weight $4pq/c_k$. For odd $p \leq q$ these k are $0, 2, \ldots, p-1$, and $\sum 1/c_k = \frac{1}{2} + \frac{p-1}{2} = p/2$; for even $p \leq q$ they are $1, 3, \ldots, p-1$, again $p/2$. So $(\mathbb{G}^\top\mathbb{G})_{pq} = 2\min(p,q)^2\max(p,q)$, non-negative and increasing in both indices, and the largest column sum is the last, $q = N-1$, over the odd $p \leq N-1$: $2(N-1)\sum_{i=1}^{N/2}(2i-1)^2 = N(N-1)^2(N+1)/3$. The column sums of $\mathbb{G}\mathbb{G}^\top$ are, to leading order, $N^4/4$, below this from $n = 3$ on. □

The rebuilt circuit's $\alpha = N\|\mathbb{G}\|_S/2$ is therefore a third of the printed prefactor $\frac{3N}{2}\|\mathbb{G}\|_S$ of Equation (3.5), so every cost quoted for the paper's encoding is on the kinder side of the paper's own. With $\|\mathbb{G}\|_S \sim N^4/3$ and $\|\mathbb{G}\|_2 \sim N^2/(2j_{-3/4,1})$ (Theorem 2), the printed prefactor's looseness is $\frac{3N}{2} \cdot \frac{N^4}{3} \cdot \frac{2j_{-3/4,1}}{N^2} = j_{-3/4,1}N^3$, approached from below.

The claims tested. Table 13 gives the verdict on each claim the paper makes for its circuits. The gate count holds as a bound, the error holds to leading order in μ (Proposition 14), and the rows of Figs 2 and 8 hold exactly. The qubit count of Fig. 9 is not testable: the rebuild uses registers of its own, so its $2n+5$ neither confirms nor refutes the printed $2n+3$. The prefactor is valid but exceeds $\|\mathbb{G}\|_2$ by a factor growing as N^3 (§ 4.5.1).

Table 13: **The paper's encodings hold their gate and error claims but not their efficiency.** The claims of Wu et al. for their block encodings [1], tested against the rebuilt circuits. Counts are for the rebuilt circuits; the error is to leading order in μ.

Circuit and claim	Printed	As rebuilt	Verdict
Fig. 2, $\mathbb{B}(x)$	$n+2$ qubits, $\sqrt{2^{n+1}}$	the same, exact	holds
Fig. 8, $\mathbb{D}^{(0)}(x_s)$	$\sqrt{2^{n+1}}/f(x_s)$	the same, exact	holds
Fig. 9, qubits	$2n+3$	$2n+5$ (own registers)	not testable
Fig. 9, gates	$O(n^3)$	$O(n^2)$	holds (as a bound)
Fig. 9, error	decreases exponentially	$O(N^{-5})$, bound derived	holds (leading order)
Fig. 9, prefactor	$(2^{n-1}+2^n)\|\mathbb{G}\|_S$	valid, $\sim j_{-3/4,1}N^3\|\mathbb{G}\|_2$	loose

D.2 The small-angle ladder and its error

The mechanism. A ladder of n singly-controlled R_y rotations acts on one flag qubit, initialised to $|1\rangle$: bit m of the (register-arithmetic-shifted) column index controls $R_y(\theta_m)$,

$$\theta_m = 2\arcsin\big(-2^m\mu\big), \qquad m = 0, \ldots, n-1, \tag{D.1}$$

for a small real coefficient μ. Because rotations on the same qubit compose by exact angle addition ($R_y(a)R_y(b) = R_y(a+b)$, an $SO(2)$ identity), the net rotation for a column index p with bit set S is $R_y(\Theta(p))$, $\Theta(p) = \sum_{m\in S}\theta_m$, giving the flag qubit's $|0\rangle$-amplitude

$$\mathrm{raw}(p) = -\sin\big(\Theta(p)/2\big) = \sin\Big(\textstyle\sum_{m\in S}\arcsin(2^m\mu)\Big). \tag{D.2}$$

The row-0 correction (entries $\sqrt{2}p$ in place of $2p$) is a second, independently accumulated ladder on the same flag qubit, with per-bit coefficient $\mu(1/\sqrt{2}-1)$, controlled on the same "this is row 0" flag the main construction already computes.

Why this is only approximate. raw(p) in Equation (D.2) would equal the desired linear value μp exactly only if $\sin\big(\sum_m \arcsin(x_m)\big) = \sum_m x_m$, which does not hold in general – sin of a sum is not the sum of sines. The discrepancy is a genuine, quantifiable small-angle error.

Proposition 14 (Entrywise error of the ladder construction)**.** *Let* $J_3(p) := \sum_{m\in S}(2^m)^3$ *for column index* p *with bit set* S*. Then*

$$\big|\mathrm{raw}(p) - \mu p\big| \;\le\; \frac{\mu^3}{6}\,p^3 \;+\; O(\mu^5 N^5). \tag{D.3}$$

Proof. Expand both non-linearities to leading order in the small parameter μ: $\arcsin(y) = y + y^3/6 + O(y^5)$ and $\sin(x) = x - x^3/6 + O(x^5)$. Summing the first expansion over the fired bits and substituting into the second,

$$\sum_{m\in S}\arcsin(2^m\mu) = \mu p + \tfrac{\mu^3}{6}J_3(p) + O(\mu^5 N^5), \tag{D.4}$$

$$\begin{aligned}\sin\Big(\sum_{m\in S}\arcsin(2^m\mu)\Big) &= \sin(\mu p) + \cos(\mu p)\cdot\tfrac{\mu^3}{6}J_3(p) + O(\mu^5N^5)\\ &= \mu p + \tfrac{\mu^3}{6}\big(J_3(p) - p^3\big) + O(\mu^5N^5).\end{aligned} \tag{D.5}$$

Since $p = \sum_{m\in S}2^m$ and every $2^m \le p$, the multinomial expansion of p^3 term-by-term dominates $J_3(p)$, with equality only when S is a single bit (p itself a power of two): $J_3(p) \le p^3$. Taking the worst case $p = N-1$ gives Equation (D.3). □

This is a leading-order bound, valid once the largest single argument, $2^{n-1}\mu$, is small and well inside arcsin's domain – true from $n = 2$ onward at the default $\mu = 2/\|\mathbb{G}\|_S$ (Equation (3.5)), but not at $n = 1$, where $\mu\cdot 2^0 = \mu = 1$ exactly and the bound is not yet meaningful.

Table 14: The published encoding of $\mathbb{G}$ against the paper's claims. Classical exact-mode measurement.

n	N	$\|\mathbb{G}\|_S$	Qubits (claim $2n+3$)	Gates	Bound on $\epsilon_{\mathbb{G}}$	Measured $\epsilon_{\mathbb{G}}$
2	4	6.15×10^1	–	–	–	–
3	8	1.18×10^3	–	–	–	–
4	16	2.04×10^4	–	–	–	–
5	32	3.38×10^5	–	–	–	–
6	64	5.50×10^6	–	–	–	–
7	128	8.88×10^7	–	–	–	–
8	256	1.43×10^9	–	–	–	–

D.3 Measured resources and error

E The Structured Exact Encoding of the Differentiation Matrix

E.1 Proof of Proposition 1, and the two diagonal factors

Proof of Proposition 1. $R_0 = \mathrm{diag}(1/\sqrt{2}, 1, \ldots, 1)$ and $\Lambda = \mathrm{diag}(0, 1, \ldots, N-1)$ are diagonal, so $\big(R_0(2U_{\mathrm{odd}})\Lambda\big)_{kp} = (R_0)_{kk} \cdot 2[p > k,\ p - k \text{ odd}] \cdot p$. This is $2p$ for $k \geq 1$ and $\sqrt{2}\,p$ for $k = 0$ when $p > k$ and $p - k$ is odd, and zero otherwise, which is Equation (4.2). $U_{\mathrm{odd}} = \sum_{d \text{ odd}} S_d$ with $(S_d)_{kp} = [p - k = d]$, the sum running over the $N/2$ odd $d \in [1, N-1]$. □

Lemma 2 (Λ as a sum of Pauli strings)**.**

$$\Lambda = \frac{N-1}{2}\, I - \sum_{j=0}^{n-1} 2^{j-1} Z_j, \tag{E.1}$$

so its LCU has $n+1$ unitary terms, $\lceil \log_2(n+1) \rceil$ selector qubits and $\alpha_\Lambda = N - 1 = \|\Lambda\|_2$.

Proof. On $|p\rangle$ with bits p_j, Z_j has the eigenvalue $z_j = 1 - 2p_j$, so $p = \sum_j 2^j p_j = \sum_j 2^j(1 - z_j)/2 = \frac{N-1}{2} - \sum_j 2^{j-1} z_j$. The weights' magnitudes sum to $\frac{N-1}{2} + \sum_j 2^{j-1} = N - 1$. The negative weights are phases on SELECT's branches, so no value is loaded by rotation. □

Lemma 3 (R_0 as two terms)**.** $R_0 = a\, I + b\,(I - 2\,|0\rangle\langle 0|)$ *with* $a = \frac{1}{2}(1 + 1/\sqrt{2})$ *and* $b = \frac{1}{2}(1 - 1/\sqrt{2})$*, both positive, so* $\alpha_{R_0} = a + b = 1 = \|R_0\|_2$*, with one selector qubit.*

Proof. On $|0\rangle$ the right side is $a - b = 1/\sqrt{2}$; on every other basis state it is $a + b = 1$. □

E.2 The arithmetic SELECT

The ripple-carry adder of Cuccaro et al. [10], on registers a and b of n qubits with a carry-in c and an overflow z, acts as $(c, a, b, z) \mapsto \big(c, a, (a + b + c) \bmod 2^n, z \oplus [a + b + c \geq 2^n]\big)$ on

every input, since its MAJ and UMA identities do not use $c = 0$. Its inverse therefore maps b to $(b - a - c) \bmod 2^n$ and flips z exactly when $b < a + c$: with $c = z = 0$ it is the in-place subtraction $b \leftarrow (b - a) \bmod 2^n$ with the borrow $z = [\, b < a \,]$. The circuit of Figure 1 uses an offset register d of n qubits, the carry and the borrow, $n+2$ ancilla in all: PREPARE applies X to the lowest qubit of d and Hadamards to the others, SELECT is the subtraction with $a = d$ and b the system register, and PREPARE† follows.

Lemma 4 (The odd shifts by subtraction)**.** *The circuit block-encodes U_{odd} exactly at $\alpha = N/2$.*

Proof. PREPARE maps $|0\rangle$ to $\sqrt{2/N} \sum_{d \text{ odd}} |d\rangle$. From $|p\rangle\,|0\rangle$, PREPARE and SELECT give $\sqrt{2/N} \sum_{d \text{ odd}} |(p - d) \bmod N\rangle\,|d\rangle\,|[\, p < d \,]\rangle$. Projecting the borrow on zero keeps the terms with $d \le p$, for which $(p - d) \bmod N = p - d$; projecting the offset register on zero after PREPARE† weights each by $\langle 0|\,\text{PREPARE}^\dagger\,|d\rangle = \sqrt{2/N}$. The block's entry (k, p) is therefore $\frac{2}{N}\,\#\{d \text{ odd} : d = p - k \ge 1\} = \frac{2}{N}[\, p > k,\ p - k \text{ odd} \,]$. □

The terms that wrap around carry a borrow of one and vanish on post-selection, and SELECT is a permutation, so one PREPARE serves all $N/2$ terms. The subtractor costs n MAJ and n UMA gates, each two CNOTs and a Toffoli, and one CNOT: $16n + 1$ CNOTs with a six-CNOT Toffoli, which is the transpiled count at every n computed.

E.3 Proof of Proposition 2

Proof of Proposition 2. Subnormalisation. A product of block encodings is a block encoding of the product at the product of the subnormalisations, so $\alpha_{\mathbb{G}} = \alpha_{R_0}\,\alpha_{2U_{\text{odd}}}\,\alpha_\Lambda = 1 \cdot N \cdot (N - 1)$ by the two lemmas above and Lemma 4 (a block encoding of U_{odd} at $N/2$ is one of $2U_{\text{odd}}$ at N). Its ratio to $\|\mathbb{G}\|_2$ tends to $2j_{-3/4,1}$ by Theorem 2, since $N(N-1)/N^2 \to 1$.

Ancilla. The encodings are composed as $R_0 \cdot \big((2U_{\text{odd}}) \cdot \Lambda\big)$, Λ applied first, each product sharing one ancilla register between its factors and adding one flag that marks the right factor's failure branch. On the right factor's success branch its ancilla are back at zero, the state the left factor needs, and on its failure branch the flag is raised and the branch never reaches the post-selected output, so the product stays exact. The inner product shares $\max\big(n+2, \lceil \log_2(n+1) \rceil\big) = n+2$ qubits and adds a flag; the outer shares $\max(1, n+3)$ with R_0's selector and adds a flag: $n+4$ ancilla in all. Of these, the $n-1$ qubits that carry the uniform superposition are forced by the $N/2$ terms of U_{odd}'s combination, since an LCU of L distinct terms needs $\lceil \log_2 L \rceil$ selector qubits.

Gates. U_{odd}'s PREPARE and PREPARE† are $2n$ one-qubit gates and its subtractor $O(n)$ gates. Λ's PREPARE is a rotation tree on $\lceil \log_2(n+1) \rceil$ qubits, and its SELECT is n gates Z_j and the negative weights' phases, each controlled on the selector. R_0 is $2n$ CNOTs and one n-controlled Z, and the two flags are an X controlled on Λ's selector and one controlled on the inner product's $n+3$ ancilla. An m-controlled gate costs $O(m^2)$ CNOTs in an ancilla-free

synthesis, so the n-controlled gates of R_0 and the flags give $O(n^2)$; with a linear-cost synthesis that borrows an idle qubit [11] they cost $O(n)$, and Λ's SELECT, $O(n \log n)$, dominates. □

Lemma 5 (The norm of the odd shifts)**.** $\|U_{\text{odd}}\|_2 = 1/\big(2 \sin \frac{\pi}{2N+2}\big)$, *so* $\alpha_{U_{\text{odd}}}/\|U_{\text{odd}}\|_2 = N \sin \frac{\pi}{2N+2}$, *which rises to* $\pi/2$.

Proof. Ordering rows and columns by parity, U_{odd} maps odd columns to even rows and even columns to odd rows, and its two blocks are the $M \times M$ upper triangle of ones with its diagonal, $M = N/2$, and the $(M-1) \times (M-1)$ one, padded with zeros. For the $m \times m$ triangle of ones L, $L^{-1} = I - S$ with S the shift above the diagonal, and $(LL^\top)^{-1} = (I - S^\top)(I - S)$ is tridiagonal with $1, 2, \ldots, 2$ on the diagonal and -1 beside it, whose eigenvalues are $4 \sin^2 \frac{(2i-1)\pi}{2(2m+1)}$, $i = 1, \ldots, m$. So $\|L\|_2 = 1/\big(2 \sin \frac{\pi}{4m+2}\big)$, increasing in m, and $m = M$ gives the lemma. □

R_0 and Λ are encoded at their norms, so of the limiting ratio $2j_{-3/4,1}$ the factor $\pi/2$ is U_{odd}'s and the remaining $4j_{-3/4,1}/\pi \approx 1.35$ is the product's: $\|R_0(2U_{\text{odd}})\Lambda\|_2 < \|R_0\|_2 \, \|2U_{\text{odd}}\|_2 \, \|\Lambda\|_2$, because Λ is largest on the last columns while the leading singular vectors of U_{odd} are spread over all of them.

E.4 The norms of the derivative and its powers

Theorem 2 (The norm of $\mathbb{G}$ in the limit)**.** $\|\mathbb{G}\|_2/N^2 \to \|K\| = 1/(2j_{-3/4,1}) \approx 0.4724$ *as* $N \to \infty$, *where* $(Kf)(s) = \int_s^1 t\, f(t)\, dt$ *on* $L^2(0,1)$ *and* $j_{-3/4,1}$ *is the first positive zero of the Bessel function* $J_{-3/4}$.

Proof. Order the rows and columns by parity. $\mathbb{G}$ maps odd columns to even rows and even columns to odd rows, so in that order it is $\left(\begin{smallmatrix} 0 & E \\ O & 0 \end{smallmatrix}\right)$ and $\|\mathbb{G}\|_2 = \max(\|E\|_2, \|O\|_2)$, with, for $M = N/2$ and $0 \le i, j < M$,

$$E_{ij} = \frac{2(2j+1)}{\sqrt{c_{2i}}}\,[\,j \ge i\,], \qquad O_{ij} = 4j\,[\,j > i\,]$$

(row $2i$, column $2j+1$ for E; row $2i+1$, column $2j$ for O; $c_0 = 2$, $c_{k\ge1} = 1$). An $M \times M$ matrix with entries $\frac{1}{M}\kappa(i,j)$ has the operator norm of the integral operator on $L^2(0,1)$ with the step kernel $\kappa(\lfloor Ms \rfloor, \lfloor Mt \rfloor)$, since $J : v \mapsto \sqrt{M} \sum_i v_i \mathbf{1}_{[i/M,(i+1)/M)}$ is an isometry onto the step functions. Here $E/N^2 = \frac{1}{M}\kappa_E$ with $\kappa_E(i,j) = \frac{2j+1}{2M}[\,j \ge i\,]/\sqrt{c_{2i}}$, and $O/N^2 = \frac{1}{M}\kappa_O$ with $\kappa_O(i,j) = \frac{j}{M}[\,j > i\,]$. Both step kernels are bounded and converge to $k(s,t) = t\,[\,t \ge s\,]$ except on a set of measure $O(1/M)$ (the diagonal cells and the first row), and to within $O(1/M)$ elsewhere, so they converge in $L^2([0,1]^2)$. The Hilbert–Schmidt norm bounds the operator norm, so $\|E\|_2/N^2$ and $\|O\|_2/N^2$ both converge to $\|K\|$.

K is Hilbert–Schmidt, so $\|K\|^2$ is the largest eigenvalue λ of K^*K, with $(K^*g)(t) = t \int_0^t g(s)\, ds$. Let $\lambda f = K^*Kf$ and $h(t) = \int_0^t (Kf)(s)\, ds$. Then $\lambda f = t\,h$, $h(0) = 0$, $h'(1) = (Kf)(1) = 0$ and $h'' = (Kf)' = -tf = -t^2 h/\lambda$. With $a = \lambda^{-1/2}$, $h'' + a^2 t^2 h = 0$, whose solutions are

$\sqrt{t}\,J_{\pm 1/4}(at^2/2)$, and $h(0) = 0$ selects $J_{1/4}$. With $z = at^2/2$, the identity $zJ'_\nu(z) = zJ_{\nu-1}(z) - \nu J_\nu(z)$ at $\nu = \frac{1}{4}$ gives

$$h'(t) = \frac{1}{2\sqrt{t}}\big(J_{1/4}(z) + 4zJ'_{1/4}(z)\big) = \frac{2z}{\sqrt{t}}\,J_{-3/4}(z),$$

so $h'(1) = 0$ exactly when $J_{-3/4}(a/2) = 0$. The eigenvalues are $\lambda = 1/a^2$ with $a/2$ the positive zeros of $J_{-3/4}$, the largest at the first, and $\|K\| = 1/(2j_{-3/4,1})$. □

Theorem 3 (The powers of $\mathbb{G}$)**.** *For every $k \geq 1$, $\|\mathbb{G}^k\|_2/N^{2k} \to \|K^k\| > 0$.*

Proof. In the parity ordering $\mathbb{G}^k$ is block diagonal or block anti-diagonal, with blocks that are alternating products of k factors E and O. Under the isometry J of the previous proof, with $J^*J = I$, each block of $\mathbb{G}^k/N^{2k}$ is unitarily equivalent to a product of k operators that converge in norm to K, and so converges to K^k. K is injective, since $\int_s^1 tf(t)\,dt = 0$ for all s forces $f = 0$, so $K^k \neq 0$. □

Corollary 1 (The standard form's norm)**.** *For an equation $\sum_{j\leq k} a_j f^{(j)} = r$ of order k with constant coefficients and $a_k \neq 0$, $\|R_{\mathrm{std}}\|_2 = |a_k|\,\|K^k\|\,N^{2k}(1 + o(1)) = \Theta(N^{2k})$ and $\|H_{\mathrm{std}}\| = \Theta(N^{4k})$.*

Proof. The equation block $\sum_j a_j\mathbb{G}^j$ has norm $|a_k|\,\|K^k\|\,N^{2k}(1 + o(1))$ by Theorem 3, its lower orders being $O(N^{2j})$ with $j < k$. A source block $\mathbb{M}_r\mathbb{D}^{(0)}(x_s)$ is $O(\sqrt{N})$, and a constraint row $\mathbb{B}(x)\mathbb{G}^j$ is rank one with norm $\sqrt{N}\|\tau(x)^\top\mathbb{G}^j\| \leq \sqrt{2N}\,\|\mathbb{G}\|_2^j = O(N^{2j+1/2})$ by Proposition 8, with $j \leq k - 1$, so every constraint row is $O(N^{2k-3/2})$. Since $\|R\|_2^2 = \|A^\dagger A + \sum_i C_i^\dagger C_i\|$ lies between $\|A\|^2$ and $\|A\|^2 + \sum_i \|C_i\|^2$, the rows do not move the leading term. □

The same limits fix the looseness of a power encoded as a product: $\alpha_{\mathbb{G}^k} = \alpha_{\mathbb{G}}^k = (N(N - 1))^k$, so $\alpha_{\mathbb{G}^k}/\|\mathbb{G}^k\|_2 \to 1/\|K^k\|$, a constant for each k, equal to $2j_{-3/4,1}$ at $k = 1$. For R2a, whose equation block is $\mathbb{G}^2$ to leading order, this is the constant towards which the measured $\alpha_R/\|R_{\mathrm{std}}\|_2$ of § 4.6 levels off; the theorem does not by itself bound the whole residual's ratio.

Table 15: **The structured encoding rises to $2j_{-3/4,1} \approx 2.117$ times the norm; the paper's prefactor grows as $j_{-3/4,1}N^3$.** Subnormalisation of the structured encoding (Proposition 2) and the paper's prefactor (Equation (3.5)) of $\mathbb{G}$, relative to $\|\mathbb{G}\|_2$, for $n = 2$ to 10. Exact classical computation.

n	$\|\mathbb{G}\|_2$	Structured	The paper's
2	7.39	1.623	49.9
3	30.1	1.858	468
4	121	1.986	4.05×10^3
5	484	2.051	3.36×10^4
6	1935	2.084	2.73×10^5
7	7739	2.100	2.20×10^6
8	30 957	2.109	1.77×10^7
9	123 827	2.113	1.42×10^8
10	495 308	2.115	1.14×10^9

E.5 Subnormalisation across n

F The Shared Pipeline: Identities and Formulas

F.1 Proof of Proposition 5

F.2 Qubitisation: Chebyshev polynomials of the walk

F.3 The controlled walk

F.4 The imaginary-time filter

$$e^{-\beta x} = I_0(\beta) + 2\sum_{k=1}^{\infty}(-1)^k I_k(\beta)\, T_k(x). \tag{F.1}$$

F.5 GQSP phases

F.6 The readout identity for probabilistic preparation

G The Descriptor Construction in Full

G.1 Block Rows in Full

Table 16 lists every row block of the chain. On one axis the state is $w = (\hat{f}^{(0)}; \ldots; \hat{f}^{(k)}; 0; \ldots; 0)$, one block of N amplitudes per derivative, padded with zero blocks to a power of two so that the block index is a register of n_f qubits, the most significant index of the state. On several axes the field is block 0 and each axis a owns the chain $\partial_a f, \ldots, \partial_a^{k_a} f$ up to the highest order that the equation or a constraint takes along it; a mixed derivative would need a block of its own and is not built. $\tilde{\mathbb{B}}_a$ and $\tilde{\mathbb{D}}_a$ are the banded factors on axis a, tensored with the identity on the other axes.

Table 16: **The chain's row blocks.** Each row block of the descriptor's stacked residual, the blocks of the state it reads and its content. $\mathbb{M}_{a_j}$ is the standard form's multiplication operator (§ 4.2), lifted where the standard form lifts it; the constraint rows are the standard form's (Definition 10) with each power of $\mathbb{G}$ replaced by the block that carries the derivative.

Row block	Reads	Content
The equation	the field and every carried derivative	$\sum_j \mathbb{M}_{a_j} \hat{f}^{(j)} - \mathbb{M}_r \mathbb{D}^{(0)}(x_s) \hat{f}^{(0)}$
Recurrence $m \to m+1$ along axis a	blocks m and $m+1$ of a's chain	$-\tilde{\mathbb{D}}_a \hat{f}^{(m)} + \tilde{\mathbb{B}}_a \hat{f}^{(m+1)}$
Padding block b	block b	$c_{\text{pad}} I$, $c_{\text{pad}} = 1$
A constraint on $\partial_a^m f$	block m of a's chain	a zero row $\mathbb{B}(x^*)$, a collapse row, a datum slice (Equation (4.4)) or a ratio row (§ 6.2)

A Neumann condition reads the derivative block with the zero row $\mathbb{B}(x)$, the same row that a Dirichlet condition applies to the field block, and no $\mathbb{G}$. A source enters the equation row as in the standard form, through the collapse row on the field block. A first-order equation or system is not carried as a chain but in the mass form of Equation (5.7), whose rows are the standard form's equation rows multiplied by $I_C \otimes \tilde{\mathbb{B}}$, with the standard form's constraint rows; its state is the standard form's, one block per component. The system rows of several fields and of a Carleman lift, and their ratio rows, are given in Appendix H.1.

Proposition 15 (The padding decouples)**.** *After ordering the columns,*

$$R_{\text{desc}} = \begin{pmatrix} R_{\text{phys}} & 0 \\ 0 & c_{\text{pad}} I \end{pmatrix},$$

so H_{desc} is block diagonal, every polynomial in H_{desc} maps the physical blocks to themselves, and the padding's singular values all equal c_{pad}.

Proof. No equation, recurrence or constraint row reads a padding block, and each padding row block is $c_{\text{pad}} I$ on its own block. □

The reference, the gap, the filter and the initial state are therefore the physical subspace's, and $c_{\text{pad}} = 1$ adds one to α_R^2 and nothing to the gap the preparation meets. Over the whole spectrum the padding's singular value can lie below the physical subspace's first excited one, but an initial state on the field block never meets it.

G.2 Variable Coefficients and the Padded Block Count

A variable coefficient $a_j(x)$ enters the equation row through the standard form's multiplication operator $\mathbb{M}_{a_j}$, the lift of Paine et al. (Proposition 13), on exactly the axes where the standard form lifts it: an axis with a non-constant coefficient or a source. The descriptor's equation row therefore applies the same operators as the standard form's equation block, which Theorem 1 needs. A multiplication by a polynomial is banded, with bandwidth the polynomial's degree,

so it compiles to the same bands as $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ (Appendix G.6). The square Galerkin form, which truncates the exact product to the first N modes and never lifts, is used only as a control in the gap study of Appendix G.4.

With $K = 1 + \sum_a k_a$ physical blocks, the padded state has

$$\dim w = 2^{\lceil \log_2 K \rceil} \prod_a N_a, \qquad n_f = \lceil \log_2 K \rceil, \tag{G.1}$$

amplitudes, against the standard form's $\prod_a N_a$. On one axis $K = k + 1$:

Order k	Blocks $k+1$	Field qubits n_f	Padded blocks 2^{n_f}	Padding blocks
2	3	2	4	1
3	4	2	4	0
4	5	3	8	3
7	8	3	8	0

A first-order equation takes the mass form and carries no derivative block. The heat equation has $K = 4$, and Laplace's and the wave equation $K = 5$, padded to eight. The padded count equals the unpadded one exactly when K is a power of two and is less than twice it otherwise; the chain's state is at most $2K$ times the standard form's. This is the qubit cost that accompanies the gate counts of § 5.8.

G.3 Kernel Equivalence on Several Axes and Under Rescaling

The proof of Theorem 1 in the body treats one axis without rescaling. Both extensions follow from one identity. Let E map a field's coefficients p to the state whose block of order m along axis a is $\mathbb{G}_a^m p$, with $\mathbb{G}_a$ the differentiation matrix on axis a tensored with the identity on the others, and whose padding blocks are zero. Let S be the diagonal of block weights of § 5.6, equal to $\prod_a \zeta_a^{m_a}$ on a block of order m_a along each axis and to one on the field and padding blocks.

Proposition 16 (Substitution identity)**.** $R_{\mathrm{desc}} S\,(S^{-1} E p) = \big(R_{\mathrm{std}}\, p;\, 0\big)$*, after ordering the rows: the recurrence and padding rows vanish on* Ep*, and every other row block acts on* Ep *as the standard form's row block of the same name acts on* p*.*

Proof. The recurrence of order m along axis a gives $-\tilde{\mathbb{D}}_a \mathbb{G}_a^m p + \tilde{\mathbb{B}}_a \mathbb{G}_a^{m+1} p = (\tilde{\mathbb{B}}_a \mathbb{G}_a - \tilde{\mathbb{D}}_a)\mathbb{G}_a^m p = 0$, and the padding blocks of Ep are zero. The equation row gives $\sum_j \mathbb{M}_{a_j} \mathbb{G}^j p - \mathbb{M}_r \mathbb{D}^{(0)}(x_s) p$ term by term, each term's derivative being read from the block that holds it, which is the standard form's equation block. A constraint on $\partial_a^m f$ applies its row to $\mathbb{G}_a^m p$, which is the standard form's row. The factor S multiplies the columns of each block by its weight and S^{-1} divides the block of Ep by the same weight, so they cancel. □

Theorem 4 (Kernel equivalence, general form)**.** $\ker(R_{\text{desc}}S) = S^{-1}E \ker R_{\text{std}}$*, and* E *is injective, so the two kernels have the same dimension.*

Proof. If $p \in \ker R_{\text{std}}$, Proposition 16 gives $(R_{\text{desc}}S)(S^{-1}Ep) = 0$. Conversely, let $(R_{\text{desc}}S)v = 0$ and $w = Sv$. The padding rows force the padding blocks of w to zero. Along each axis's chain each recurrence gives $w_{(a,m+1)} = \tilde{\mathbb{B}}_a^{-1}\tilde{\mathbb{D}}_a w_{(a,m)} = \mathbb{G}_a w_{(a,m)}$, since $\tilde{\mathbb{B}}_a$ is invertible, so $w = Ew_0$, and Proposition 16 gives $R_{\text{std}}w_0 = 0$. E is injective because its first block is the identity. □

What it does not say. The theorem says that an exact kernel is present in both forms or in neither, with the same dimension; a one-dimensional kernel is one-dimensional in both. It does not say that near-kernels are equally sharp. The ratio of the first excited to the lowest singular value, which the trust threshold of § 7.2 reads, is each form's own and differs between them. For a Carleman lift the kernel is the truncated lift's in both forms, and its uniqueness is the initial-value problem's, not a property of either form (§ 6.5).

G.4 The Gap Mechanism and the Artefacts Ruled Out

Proposition 17 (The quotient on represented states)**.** *On the represented states the Rayleigh quotient of the rescaled descriptor is*

$$\frac{\|R_{\text{desc}}S\,(S^{-1}Ep)\|^2}{\|S^{-1}Ep\|^2} = \frac{\|R_{\text{std}}\,p\|^2}{\sum_{m=0}^{k}\zeta^{-2m}\|\mathbb{G}^m p\|^2}, \tag{G.2}$$

and the j*-th smallest singular value of* $R_{\text{desc}}S$ *is at most the least, over* $(j+1)$*-dimensional subspaces* V *of coefficient vectors, of the largest square root of this quotient on* V*.*

Proof. Proposition 16 gives the numerator, and the denominator is the norm of the represented state, whose block m is $\zeta^{-m}\mathbb{G}^m p$. By the Courant–Fischer theorem the j-th smallest singular value is the least, over $(j+1)$-dimensional subspaces of the state space, of the largest quotient on the subspace; restricting the subspaces to the range of $S^{-1}E$ can only raise it. □

The bound is not tight. States that break the recurrences cost little where $\tilde{\mathbb{B}}$'s least singular value is small, and it falls with N, as about π/N, with its singular vector at the ends of the interval. Measured against the bound at $n = 8$, the descriptor's lowest non-zero singular value is 0.99–1.00 times it on controlled equations with a single zero or none, 0.04 times it on an equation with zeros at both ends, and 0.35–0.72 times it on the two-axis problems at $n = 4$ per axis.

The concentrated states. Let a_k vanish simply at an endpoint. A polynomial of degree N can concentrate within $\mathcal{O}(N^{-2})$ of the endpoint, where the Chebyshev points cluster, and each of its derivatives then grows by a factor $\mathcal{O}(N^2)$. On such a state $\|a_k p^{(k)}\| = \mathcal{O}(N^{-2})\|p^{(k)}\|$ and the lower orders are of the same size, so the square root of the quotient is $\mathcal{O}(N^{-2}\zeta^k)$ against the top block. Unscaled, the lowest non-zero singular value is $\mathcal{O}(N^{-2})$ and the gap $\mathcal{O}(N^{-4})$.

At an interior zero, where the resolution is $\mathcal{O}(N^{-1})$, the gap is $\mathcal{O}(N^{-2})$. At a double zero at an endpoint, $\|a_k p^{(k)}\| = \mathcal{O}(N^{-4})\|p^{(k)}\|$ and the gap is $\mathcal{O}(N^{-8})$. These are heuristics; the rates are measured. Table 17 gives the unscaled gap on controlled equations whose solutions are polynomials where the coefficient vanishes, so that their kernels are exact, built with the square multiplication.

Table 17: **A zero of the leading coefficient closes the unscaled descriptor's gap at the rate the Chebyshev resolution at the zero predicts.** Unscaled gap of the descriptor from $n = 5$ to $n = 9$ on controlled equations, against the standard form's. Planning probes, exact classical computation.

Equation	Leading coefficient	Descriptor, $n = 5 \to 9$	Per qubit	Standard
$f'' + 4f' + 4f$	1	0.458 → 0.458	flat	7.04
$(2+x)f'' + f' - f$	≥ 1	0.434 → 0.432	flat	0.777
$(2+x)f' + f$	≥ 1	1.01 → 1.00	flat	4.90
$(1-x)f'' - xf' + 3f$	zero at $x = 1$	1.5e-5 → 2.2e-10	÷16	2.02
$(1-x)f' + f$	zero at $x = 1$	2.4e-5 → 3.5e-10	÷16	1.6
$(1-x^2)f'' - 2xf' + 20f$	zeros at $x = \pm 1$	9.1e-7 → 8.0e-13	÷33	20.5
$xf'' + (1-x)f' + 3f$	zero at $x = 0$	9.3e-3 → 3.8e-5	÷4	1.85–2.02

The first excited state lies 96–100 % in the top-derivative block, and at $n = 9$ 86–99 % of that block's function lies within 0.02 of the coefficient's zero. Variable coefficients alone do not close the gap; a zero does.

Two axes. The equation constrains a sum of top derivatives. Its lowest modes are equal and opposite in two top blocks (transport, Laplace's equation) or sit in the block the equation weights least (the heat equation's $\partial_x^2 f$ block, weighted 1/25), and only the slice rows penalise them: by $\mathcal{O}(1)$ against $\|f\|$, which is small against the derivative blocks. With R5b's constraints and resolution, the lowest singular value that is not a near-solution falls from 0.29 to 0.033 for transport, from 0.081 to 0.0018 for the heat equation and from 0.10 to 0.0011 for Laplace's equation as n goes from 2 to 4 per axis, while for $f_t + f$, which has one top block, it stays at 0.87–0.86.

The field block's weight. Let P_0 project onto the field block and let the initial state be supported there. Then $\gamma = |\langle\psi_g|\psi_{\text{init}}\rangle| = |\langle P_0\psi_g|\psi_{\text{init}}\rangle| \leq \|P_0\psi_g\| = w_f^{1/2}$ by the Cauchy–Schwarz inequality, with $w_f = \|P_0\psi_g\|^2$ the field block's weight that the build records hold. For an exact kernel, $\psi_g \propto S^{-1}Eb$ with b the solution's coefficients, so $w_f = \|b\|^2 / \sum_m \zeta^{-2m}\|\mathbb{G}^m b\|^2$. Amplitude amplification then costs a factor $1/\gamma \geq w_f^{-1/2}$ (§ 7.4).

What it is not. Each candidate artefact was tested:

Candidate	Test	Outcome
The lift	R3a with the square multiplication	collapses faster: 2.6e-11 at $n = 8$, against 2.7e-8 lifted
The padding	lowest singular value against c_{pad}; the excited state's padding weight	at most 0.74, below $c_{\text{pad}} = 1$, on every padded record; no padding weight
The constraints	controlled equations with none	they collapse too
Precision	lowest singular value against the decomposition's floor $\varepsilon\sigma_{\max}$	six orders of magnitude above it at the smallest
The basis	a descriptor carrying $f^{(j)}$ in the ultraspherical basis $C^{(j)}$	the same rates where a_k vanishes
The reference	dense decomposition against the iterative reference	agree to $1.9\,\varepsilon\sigma_{\max}$

The ultraspherical descriptor keeps a flat gap on regular problems with a norm about four times smaller, and on Legendre's equation gains 160-fold at $n = 9$ over the Chebyshev descriptor, the share of the recurrences, but still collapses.

G.5 Block Rescaling

The rule of § 5.6 returns each axis's weight from the equation's coefficients and the resolution alone:

Problem	ζ
One axis, a_k with no zero on $[-1, 1]$	$2(\bar{a}_j/\bar{a}_k)^{1/(k-j)}$, j the lowest order present, $\bar{a}$ the mean of $\lvert a\rvert$
One axis, a_k with a zero on $[-1, 1]$	$N/2$
Several axes	N_a, times $\sqrt{\max_b \bar{c}_b/\bar{c}_a}$ on an axis of top order two, $\bar{c}_a$ the mean magnitude of its top coefficient

Whether a_k vanishes is decided from its roots, to a tolerance loose enough to find a double root. On R2a the rule gives $\zeta = 2\sqrt{4/1} = 4$, on R2c $\zeta = 2\sqrt{400} = 40$, and on R3a, whose leading coefficient vanishes at both ends, $\zeta = N/2$.

Why each. For a regular equation with constant coefficients, $(\bar{a}_0/\bar{a}_k)^{1/k}$ is the geometric mean of the characteristic roots' magnitudes, the product of the roots being $\pm a_0/a_k$; a solution $e^{\lambda x}$ has $\|f^{(m)}\| \approx |\lambda|^m \|f\|$, so dividing block m by ζ^m balances the blocks at the equation's own scale. The factor two is where the cost's minimum over ζ sits on the regular benchmark problems. Where the leading coefficient vanishes, the concentrated states of Appendix G.4 have quotient $\mathcal{O}(N^{-4}\zeta^{2k})$, so for second-order problems $\zeta \propto N$ makes it of order one; half the resolution is the constant the sweep found. On several axes the resolution is used per axis, as for a singular equation, and an axis of top order two is weighted so that its top-order term $\bar{c}_a\zeta_a^2$ balances the others: the wave equation $f_{tt} = c^2 f_{xx}$ takes (cN, N), where $(N/2, N/2)$ costs 10^4

to 10^6 times more.

The calibration and its check. The cost measured is $\sqrt{\|H\|/\Delta}/\gamma$, the walk's queries at an ideal encoding times amplitude amplification, up to the logarithms both forms share. Sweeping ζ over powers of two on the benchmark problems set the two constants. The same problems then checked the rule, so the check is in-sample. Table 18 gives the window of ζ over which the cost stays within twice its least, and Table 19 the rule's cost against the grid's best.

Table 18: **The block weight need only be right to about a factor of two.** The window of ζ within which the cost $\sqrt{\|H\|/\Delta}/\gamma$ stays within twice its least, at $n = 7$ on one axis and $n = 4$ per axis on two (one weight shown per two-axis problem). Planning probes.

	R2a	R2c	R3a	R4a	R4d	R5a	R5b
Window of ζ	2–16	16–64	32–90	23–90	4–16	8–32	8–16

Table 19: **The a priori rule lands near the best weight on a grid.** The rule's cost as a multiple of the cheapest ζ on a grid of powers of two, with the gap on the physical subspace and the geometric initial state on the field block. A ratio below one is the grid's spacing. Planning probes, in-sample.

Problems	n	Rule's cost / grid's best
R1, R2a–c, R4b–d (regular)	5, 7	0.99–1.17; R2c at $n = 5$: 1.44
R3a, R3b, R4a (a simple zero)	5, 7	1.00–1.04
R3c, R3d (solved for $v = f/\sqrt{1-x^2}$)	5, 7	1.00–1.18 at $n = 5$; 2.07 and 2.22 at $n = 7$
R5a, R5b, R5c	3, 4 per axis	1.00–1.17

A coefficient that dips. $(\delta + x^2)f'' + f = 0$ has no zero but dips to δ. The rule is within 1.3 times the grid's best for $\delta \geq 0.01$, and 2.3–4.7 times at $\delta = 10^{-3}$, where the resolution sees the dip and it behaves like a zero. None of the benchmark problems dips so.

A double zero. The paper's R3c and R3d multiply the associated Legendre operator by $1-x^2$, so their leading coefficient $(1-x^2)^2$ vanishes twice at each end. Unscaled, the descriptor's gap there falls as N^{-8}. Balancing the concentrated states needs $\zeta \propto N^2$, by which $\|H_{\text{desc}}\|$ has grown as $\zeta^4 = N^8$. On a grid of ζ the best cost of the descriptor grows as about $N^{3.7}$–$N^{3.8}$ over $n = 5$ to 8, against the standard form's $N^{2.2}$–$N^{2.3}$, and at $n = 8$ it is 760–850 times the standard form's; no block weight brings the two level. Solved for $v = f/\sqrt{1-x^2}$, whose leading coefficient has simple zeros, the two forms' costs grow alike, as $N^{1.8}$–$N^{1.9}$ at the rule's weight against $N^{2.0}$. The ground state stays accurate in both forms at the double zero, its error set by the solution's branch point; it is the cost that the double zero raises.

What it changes in the circuit. Scaling the columns of block m by ζ^m multiplies the coefficient c_t of every term whose column block has order m by ζ^m (Algorithm 1, line 4). The terms, their selectors and their gates are unchanged, and $\alpha_A = \sum_t |c_t|$ changes with the

coefficients. A general diagonal equilibration, a different weight for each amplitude, would need its own diagonal block encoding, with $\Theta(\dim w)$ gates, and is excluded from every circuit.

G.6 The Term IR and Its Compilation

The system rows of the descriptor (the equation, the recurrences and the padding; the rank-one constraint rows are encoded apart) form one square operator on the block register and the axes' registers. Each row block sits at a block index of its own: a recurrence at the index of its lower block, the equation at the lowest index left, and each padding row at its own. The term IR writes this operator as

$$A_{\text{sys}} = \sum_t c_t \, P_t \otimes \bigotimes_a F_{t,a}, \tag{G.3}$$

with P_t a Pauli word $X^x Z^z$ on the block register (or, for a system's coupling, a matching, § 6.6), and each $F_{t,a}$ a *band*: a diagonal D followed by the truncated shift that takes column q to row $q - \ell$ and discards it where that row does not exist, with D the identity, one Z_j or a reflection Q_m.

Lemma 6 (An affine band with exceptions)**.** *A diagonal whose entries are $v_q = a + bq$ except at a few columns m, where $v_m = a + bm + d_m$, is*

$$\operatorname{diag}(v) = \Big(a + b\,\frac{N-1}{2} + \sum_m \frac{d_m}{2}\Big) I - b \sum_{j=0}^{n-1} 2^{j-1} Z_j - \sum_m \frac{d_m}{2}\, Q_m :$$

$n + 1$ terms and one reflection for each exception.

Proof. $\Lambda = \frac{N-1}{2} I - \sum_j 2^{j-1} Z_j$ (Equation (5.12), and Equation (E.1)), and $|m\rangle\langle m| = (I - Q_m)/2$. □

Lemma 7 (The affine part is tight)**.** *If a and $a + b(N-1)$ have one sign, the combination of Lemma 6 without exceptions has $\alpha = \max(|a|, |a + b(N-1)|) = \|aI + b\Lambda\|_2$.*

Proof. $\alpha = |a + b\frac{N-1}{2}| + |b|\frac{N-1}{2}$, the mean of the end values plus half their range, which is the larger end value in magnitude when the two share a sign. □

Every band of the descriptor has this form. $\tilde{\mathbb{D}}$'s diagonal is 2Λ except at column 1, where the normalisation of T_0 makes it $2\sqrt{2}$; $\tilde{\mathbb{B}}$'s and a constant multiplication's are constant except at their ends; a variable coefficient's multiplication bands and their truncation at the register's edge give the other exceptions. The columns a shift discards are free, and are fitted to keep the combination shortest.

The block register. Terms whose axis factors agree are merged across the row and column blocks they join, so that the block register carries their pattern as one small matrix F, paid for once; the recurrences share $\tilde{\mathbb{D}}$ in this way. F is expanded in Pauli words, $F = \sum_{x,z} f_{xz} X^x Z^z$

with $f_{xz} = 2^{-n_f} \sum_b (-1)^{z \cdot b} F_{b \oplus x, b}$. Where the merged pattern is a single matrix unit on one block, its 2^{n_f} Pauli words are replaced by one term with a flag qubit post-selected to keep that block, at the same α. A large irregular coupling, a Carleman generator above all, is split into matchings instead (§ 6.6).

Proposition 18 (Compilation)**.** *Let $\mathrm{PREPARE}_R$ load $\mathrm{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ and $\mathrm{PREPARE}_L$ load $\sqrt{|c_t|/\alpha_A}$ at each term's joint index of the factored selectors, with $\alpha_A = \sum_t |c_t|$. Then the circuit that applies $\mathrm{PREPARE}_R$, then SELECT, then $\mathrm{PREPARE}_L^\dagger$, post-selected on the selectors, the overflow qubits and the flag, block-encodes A_{sys} at α_A.*

Proof. On the joint index of term t, the factored SELECT applies the tensor product of the operations its registers hold: the term's block-register word and, on each axis, its diagonal and then its shift, which is the band. A joint index that no term uses has zero amplitude in both preparations. The post-selected block is $\sum_t \sqrt{|c_t|/\alpha_A} \cdot \mathrm{sgn}(c_t)\sqrt{|c_t|/\alpha_A}\, U_t = \sum_t c_t U_t/\alpha_A$. With one preparation and its own inverse the signs would square away, and the block would be $\sum_t |c_t| U_t/\alpha_A$. □

The truncated shift. On the output register widened by one overflow qubit, initially zero, add $-\ell$ modulo twice the register's size. A row $q - \ell$ inside $[0, N)$ lands in the lower half, and one outside it lands in the upper half, which the projection of the overflow qubit on $|0\rangle$ discards. The addition is an increment of the qubits from j upwards for each set bit j of $|\ell|$, each a ladder of multi-controlled X gates, and the overflow qubit is the shift's only ancilla.

Rank-one terms. A source $|\hat{r}\rangle \langle \tau(x_s)| / f(x_s)$ is compiled apart and summed with the bands by a further combination: a matrix unit on the block register, the feature map reading $\tau(x_s)$, and a rotation tree loading $\hat{r}$, with two ancilla and no arithmetic. The constraint rows are encoded the same way, each as its own rows, and stacked into the reflection of § 7.3.

The IR check. The terms are built from the problem, the operator library, the residual's layout, its lifts and its block weights, never from the assembled matrices. Summing them back and comparing the sum with the independently assembled system rows is the IR mode of operator verification, possible at any n (Table 8); the compiled circuit is then compared with the terms exactly, or by probe states where it is too large.

G.7 Proof of the Cost Proposition, and the Norms

Ancilla. The registers are those of Appendix G.6: one overflow qubit for each axis that shifts, at most one flag, and the factored selectors. On axis a the diagonal selector indexes the identity, the n_a gates Z_j and the distinct reflections r_a, so it has $\lceil \log_2(1 + n_a + r_a) \rceil$ qubits. The reflections are bounded in number by the problem's bands, and the block-register words and the shifts depend on the equation's structure, not on n, so every other selector has a width fixed by the problem, and the total is $\mathcal{O}(\log n)$. For R2 the reflections are three ($\tilde{\mathbb{B}}$'s diagonal

and second superdiagonal at their first entries and $\tilde{\mathbb{D}}$'s first entry), so the diagonal selector has $\lceil \log_2(n+4) \rceil$ qubits; the shifts are 0, 1 and 2, two qubits; the block-register selector has four; with the overflow qubit and the flag the count is $\lceil \log_2(n+4) \rceil + 8$. It is verified on the compiled circuit of R2's system rows at $n = 2$, 4, 5 and 12.

Gates. SELECT applies each distinct operation once. A Z_j is one gate controlled on $\mathcal{O}(\log n)$ selector qubits; a reflection is one phase controlled on n_a qubits; a shift by a fixed ℓ is $\mathcal{O}(\log |\ell|)$ increments of at most $n_a + 1$ qubits, each $n_a + 1$ multi-controlled X gates of up to n_a controls, $\mathcal{O}(n^2)$ gates with a linear-cost decomposition of each. The block-register operations act on n_f qubits, fixed by the problem, and the preparations are rotation trees over $\mathcal{O}(n)$ joint indices. There are $\mathcal{O}(n)$ distinct operations, the Z_j costing $\mathcal{O}(\log n)$ each and the rest $\mathcal{O}(n^2)$ in all, so the total is $\mathcal{O}(n^2)$.

Exactness. Proposition 18; every term is a unitary, so no part of the encoding is approximate.

The subnormalisation. Take first a second-order chain with polynomial coefficients whose leading coefficient has no zero, at a fixed $\zeta \geq 1$. The recurrences' $\tilde{\mathbb{D}}$ terms share the block-register pattern $\mathrm{diag}(1, \zeta, 0, 0)$, whose Pauli expansion has $\alpha = \frac{1}{2}(|1+\zeta| + |1-\zeta|) = \max(1, \zeta) = \zeta$, its largest entry. With Lemma 7, $\tilde{\mathbb{D}}$'s affine part is encoded at $2(N-1)$, so these terms contribute $2\zeta(N-1)$ to α_A. Every other term is independent of N: $\tilde{\mathbb{D}}$'s exception at column 1, the $\tilde{\mathbb{B}}$ bands weighted by ζ and ζ^2, the coefficients' multiplication bands weighted by at most ζ^2, and the padding. A source's rank-one term has coefficient $\|\hat{r}\| \, \|\tau(x_s)\| / |f(x_s)|$, and $\|\hat{r}\|^2$ grows as N for a fixed smooth source (Equation (4.1)), while $\|\tau(x_s)\|^2 < 2$ (Proposition 8). On the other side, $\zeta\tilde{\mathbb{D}}$ is a block of A_{desc}, so $\|A_{\mathrm{desc}}\|_2 \geq 2\zeta(N-1)$. Hence $\alpha_A / \|A_{\mathrm{desc}}\|_2 \leq 1 + C/N$ without a source and $1 + C'/\sqrt{N}$ with one, with constants that grow with ζ and the coefficients. For the mass form of a scalar equation $f' = af$, $\alpha_A \leq 2(N-1) + \mathcal{O}(1)$ and $\|\tilde{\mathbb{D}} - a\tilde{\mathbb{B}}\|_2 \geq 2(N-1) - |a|(2+\sqrt{2})$, with the same conclusion. For a chain of order three or more the recurrences' pattern need not be encoded at its largest entry, and no limit is claimed; no benchmark problem has one. Where the leading coefficient vanishes, $\zeta = N/2$ makes $\zeta\tilde{\mathbb{D}}$, $\zeta^2\tilde{\mathbb{B}}$ and $\zeta^2 a_2$ all of order N^2 in one row block, and terms of comparable size add in ℓ^1 in α_A, so the ratio tends to a constant, measured near two (§ 5.8).

Optimality. An exact combination whose SELECT indexes L distinct operations needs $\lceil \log_2 L \rceil$ selector qubits [12]. Λ has exactly $n+1$ non-zero Walsh coefficients, so a combination of Pauli-Z strings needs $n+1$ terms for it and $\Omega(\log n)$ selector qubits. The descriptor's count is therefore of the least order among exact combinations that expand their diagonals in Pauli strings, and is not claimed against encodings built otherwise.

The norms of Table 4. Each row block of R_{desc} is a finite sum of blocks weighted by powers of ζ: $\zeta^m\tilde{\mathbb{D}}$, of norm $2\zeta^m(N-1)$; $\zeta^{m+1}\tilde{\mathbb{B}}$, of norm at most $(2+\sqrt{2})\zeta^{m+1}$; $\zeta^j\mathbb{M}_{a_j}$, of norm at most $\sqrt{2}\,\zeta^j \sup|a_j|$ (Proposition 13); the padding, of norm one; and the rank-one constraint

and source rows, of norm $\mathcal{O}(\sqrt{N})$, since $\mathbb{B}(x) = \sqrt{N}\,|0\rangle\langle\tau(x)|$ with $\|\tau(x)\|^2 < 2$. The norm of R_{desc} is at most the sum of these and at least the largest of them. At fixed ζ the largest is $2\zeta^{k-1}(N-1)$, so $\|R_{\text{desc}}\|_2 = \Theta(N)$ and $\|H_{\text{desc}}\| = \Theta(N^2)$. With $\zeta = N/2$ and $k = 2$, $\zeta\tilde{\mathbb{D}}$, $\zeta^2\tilde{\mathbb{B}}$ and $\zeta^2\mathbb{M}_{a_2}$ are each $\Theta(N^2)$, so $\|H_{\text{desc}}\| = \Theta(N^4)$; on two axes with $\zeta_a = N_a$ and $k_a = 2$ the same count gives $\Theta(N^4)$; and with $\zeta \propto N^2$ the block $\zeta^2\tilde{\mathbb{B}}$, of norm at least $2\zeta^2$, gives $\Theta(N^8)$.

G.8 One Query's Success

Proposition 19 (One query on a Haar-random state)**.** *One application of a block encoding of R at α_R, post-selected, succeeds on a Haar-random state of dimension D with mean probability $\|R\|_F^2/(D\alpha_R^2)$.*

Proof. The success probability on ψ is $\|R\psi\|^2/\alpha_R^2$, and $\mathbb{E}\,\psi\psi^\dagger = I/D$ for Haar-random ψ, so its mean is $\operatorname{tr}(R^\dagger R)/(D\alpha_R^2)$. □

In the structured standard form the dense $\mathbb{G}^k$ dominates: for $k = 1$, $\|\mathbb{G}\|_F^2 \sim N^4/2$ (Proposition 9) against $\alpha_{\mathbb{G}}^2 \sim N^4$ on $D = N$ amplitudes, so the mean falls as $1/(2N)$. In the descriptor the dominant $\tilde{\mathbb{D}}$ has $\|\tilde{\mathbb{D}}\|_F^2 = \sum_{q<N}(2q)^2 \sim 4N^3/3$, up to its first entry, against $\alpha^2 \sim 4N^2$, so the mean stays $\Theta(1)$. The pipeline post-selects no single query, its walk being unitary, so this compares the encodings and is not a cost of the preparation.

H Systems and the Nonlinear Route in Full

H.1 Systems in Full

The rows of R7. R7 is $u' = v$, $v' = -\pi^2 u$ on $x \in [-1, 1]$ with $(u, v)(-1) = (0, \pi)$, so $C = 2$, $\mathbb{A} = \left(\begin{smallmatrix} 0 & 1 \\ -\pi^2 & 0 \end{smallmatrix}\right)$ and the interval needs no rescaling ($T/2 = 1$). With the field index the most significant, the equation rows of Equation (6.1) are

$$\begin{pmatrix} \mathbb{G} & -I \\ \pi^2 I & \mathbb{G} \end{pmatrix} \text{ (standard form)}, \qquad \begin{pmatrix} \tilde{\mathbb{D}} & -\tilde{\mathbb{B}} \\ \pi^2\tilde{\mathbb{B}} & \tilde{\mathbb{D}} \end{pmatrix} \text{ (mass form)}, \tag{H.1}$$

the second being the first multiplied on the left by $I_2 \otimes \tilde{\mathbb{B}}$. Two fields fill the one-qubit field register, so there is no padding. In general a padding block $b \geq C$ carries the row $\hat{f}^{(b)} = 0$ of weight one and no other row reads it, so the Hamiltonian is block diagonal between the physical and the padding blocks; the padding's eigenvalue is one, and the gap, the reference and the initial state are those of the physical blocks.

R7's data. The anchor is v, with $y_v = \pi$, and $y_\perp = 0$ because $u(-1) = 0$, so the ratio row divided by y_a is $|u\rangle\langle u| \otimes \mathbb{B}(-1)$: the zero datum holds $u(-1)$ to zero exactly. The regular datum is $v(-1) = \pi$, a collapse row on the v block at $x_s = -1$, which fixes the scale and the readout.

The ratio rows' spectrum. For C components and $\mathbb{F}$ of Equation (6.2):

Proposition 20 (Ratio rows)**.** $\ker \mathbb{F} = \operatorname{span}(y)$*, and the singular values of* $\mathbb{F}$ *are* $\|y\|$ *once,* $|y_a|$ *with multiplicity* $C-2$*, and* 0 *once.*

Proof. If $\mathbb{F}v = 0$ then $v_c = (v_a/y_a)y_c$ for every $c \neq a$, and trivially for $c = a$, so $v \parallel y$; and $\mathbb{F}y = 0$. The $C-1$ non-zero rows, indexed by $c \neq a$, are $[\, y_a I \mid -u \,]$ with $u = (y_c)_{c\neq a}$ (columns $c \neq a$, then a), so $\mathbb{F}\mathbb{F}^\dagger$ restricted to them is $|y_a|^2 I + uu^\dagger$, with eigenvalues $|y_a|^2$ on $u^\perp$ (multiplicity $C-2$) and $|y_a|^2 + \|u\|^2 = \|y\|^2$. □

The rows therefore fix the ratios of the components and leave the scale free, as a zero row does, and their weight against the equation is set by the data: the largest singular value is $\|y\|$ and the smallest non-zero one $|y_a|$, so the anchor with the largest $|y_a|$ conditions them best. For a Carleman lift y is the vector of the data's scaled moments (Appendix H.5).

The ratio rows' circuit. Divided by y_a, the rows are

$$\frac{\mathbb{F}}{y_a} \otimes \mathbb{B}(x^*) = \Big(I - |a\rangle\langle a| - \frac{\|y_\perp\|}{y_a}\, |\hat{y}_\perp\rangle\langle a| \Big) \otimes \mathbb{B}(x^*), \qquad \hat{y}_\perp = y_\perp/\|y_\perp\|, \tag{H.2}$$

three LCU terms, each a field-register operation times the zero row's encoding: the identity; the projector $|a\rangle\langle a|$, a matrix unit placed by X gates around the zero projector; and the rank-one term, the matrix unit $|0\rangle\langle a|$ followed by a rotation tree that prepares $\hat{y}_\perp$ from $|0\rangle$. The subnormalisation is $(2 + \|y_\perp\|/|y_a|)$ times the zero row's, and a datum with $y_\perp = 0$ drops the third term.

H.1.1 The coupling as a sum of matchings

Construction. A matching $\mathcal{M}$ is a set of entries of $\mathbb{A}$ with at most one in each row and column. It acts as a partial permutation of the field register's basis states followed by a diagonal: column c goes to row $\pi(c)$ with weight $v_{\pi(c)}$. The permutation is completed to a permutation of all basis states and built from transpositions, each a cascade of CNOTs around one multi-controlled X, the only gate that takes the selector's control. The weights, divided by the matching's largest magnitude $\alpha_{\mathcal{M}}$, lie in $[-1, 1]$ and are applied as $R_Y(2 \arccos v_r)$ on a flag qubit, whose zero then carries v_r, with weight zero on a row the matching does not reach. The rotations of every matching form one uniformly controlled R_Y on the field register and the matching selector, 2^{n_f+w} CNOTs for all of them, w the selector's width. A matching whose weights share one value is a bare permutation, its sign moved into the coefficient. The coupling is $\sum_{\mathcal{M}} \alpha_{\mathcal{M}} U_{\mathcal{M}}$, block-encoded at $\alpha = \sum_{\mathcal{M}} \alpha_{\mathcal{M}}$.

How many. The matchings are the colour classes of an edge colouring of the bipartite graph whose two sides are $\mathbb{A}$'s rows and columns and whose edges are its non-zero entries. The greedy colouring takes the entries in row order and gives each the least colour that its row and its

column do not yet hold. An entry meets at most $\Delta - 1$ others in its row and $\Delta - 1$ in its column, Δ the largest number of entries in a row or column, so at most $2\Delta - 1$ colours are used. Δ is the least possible, since the entries of one row need distinct colours, and it suffices for a bipartite graph by König's theorem [16]; on every Carleman lift built, the greedy colouring reached Δ. A Pauli-word expansion of the same coupling, $\mathbb{A} = \sum_{x,z} f_{xz} X^x Z^z$, pays for every word with a non-zero coefficient, which for an irregular sparse matrix is most of the 4^{n_f} words.

Table 20 sets the matchings against the tensor layout of Appendix H.4 on the lifts of Burgers' and Navier–Stokes' couplings, each encoded alone.

Table 20: **The matchings are cheaper only on small lifts; the tensor layout's term count does not grow.** The coupling $\mathbb{A}_K$ of the Carleman lift of Burgers (R9c, $a/\nu = 1$) and of the Navier–Stokes vortex pair (R10c, $a/\nu = 10$) on M modes, encoded alone: in the symmetric layout as matchings, and in the tensor layout as $2K - 1$ terms on its physical states. $\alpha/\|\mathbb{A}_K\|$ is the encoding's subnormalisation over the coupling's norm; CNOTs are per query. Every entry is verified by IR to 10^{-10}; "–" marks a symmetric layout not built. Classical construction and verification.

			Symmetric layout (matchings)			Tensor layout		
Problem	M	K	matchings	$\alpha/\|\mathbb{A}_K\|$	CNOTs	terms	$\alpha/\|\mathbb{A}_K\|$	CNOTs
R9c	4	3	unknown	unknown	unknown	unknown	unknown	unknown
R9c	16	3	31	1.34	525 436	5	1.12	17 218
R9c	16	4	unknown	unknown	unknown	unknown	unknown	unknown
R10c	12	2	unknown	unknown	unknown	unknown	unknown	unknown
R10c	20	3	25	3.94	636 214	5	11.4	89 408
R10c	48	2	unknown	unknown	unknown	unknown	unknown	unknown
R10c	80	2	61	7.34	883 336	unknown	unknown	unknown
R10c	316	2	–	–	–	3	23.5	288 120

The matchings' count and cost grow with the entries of $\mathbb{A}_K$, the tensor layout's term count stays at $2K - 1$ and its CNOTs grow slowly with M, so the matchings are cheaper per query only on the smallest lifts. The tensor layout's subnormalisation is close to the norm on Burgers and loose in two dimensions, where the Fourier fold's α is N.

H.1.2 Realification and the real embedding

Realification. Let $\iota : \mathbb{C}^m \to \mathbb{R}^{2m}$, $z \mapsto (\operatorname{Re} z, \operatorname{Im} z)$, a real-linear isomorphism, and let a complex residual be $R = R_r + \mathrm{i}R_i$ with R_r, R_i real. Then $\iota(Rz) = R_\iota\, \iota(z)$ with

$$R_\iota = \begin{pmatrix} R_r & -R_i \\ R_i & R_r \end{pmatrix}, \tag{H.3}$$

so $Rz = 0$ if and only if $R_\iota \iota(z) = 0$, and $\ker R_\iota = \iota(\ker R)$, of real dimension twice the complex one: the solution's real scale and its phase. The data close both. Imposed as real ratio rows on the realified components (Equation (6.2)), they fix the direction of $\iota(z(x^*))$ in $\mathbb{R}^{2m}$, which fixes the phase as well as the ratios, so an initial-value problem has a one-dimensional real

near-kernel, as a real system does. For one mode, $z' = cz$ realifies to Equation (6.3). R7 is $z' = -\mathrm{i}\pi z$ with $z = \pi u + \mathrm{i}v$: the realified state is $(\pi u, v)$, and R7 carries $(u, v) = D^{-1}(\pi u, v)$ with $D = \mathrm{diag}(\pi, 1)$, so its coupling $\left(\begin{smallmatrix} 0 & 1 \\ -\pi^2 & 0 \end{smallmatrix}\right) = D^{-1}\left(\begin{smallmatrix} 0 & \pi \\ -\pi & 0 \end{smallmatrix}\right)D$ is the realification's under a diagonal change of variables, and its kernel the image of the realification's under $D^{-1} \otimes I$.

The real embedding of the Fourier variables. For the fluids, with $\hat{u}_{\pm\mathbf{k}} = (a \mp \mathrm{i}b)/2$ the complex coefficients of $a\cos(\mathbf{k}\cdot\mathbf{x}) + b\sin(\mathbf{k}\cdot\mathbf{x})$, the variables $v = \sqrt{2}\,\hat{u}$ on the wavevectors of both signs are $v = Ew$, with w the real coefficients and E putting $1/\sqrt{2}$ on $\pm\mathbf{k}$ for a cosine and $\mp\mathrm{i}/\sqrt{2}$ for a sine. E is an isometry, $E^\dagger E = I$, and unitary when the real basis holds every kept wavevector, so the complex Fourier–Galerkin operators act on real data through E, and the real system's coupling is $E^\dagger(\cdot)E$. Burgers' sine modes are the case of one axis and sines only.

H.2 The Doubled Space

The doubled residual. On $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$, a term with $k \le 2$ field factors, the i-th carrying the derivative $\mathbb{G}^{j_i}$ and the term a multiplier of degree at most one, becomes $c\,\mathbb{m}_m\big(\mathbb{D}^{(0)}(x_s)^{\otimes(2-k)} \otimes \mathbb{G}^{j_1} \otimes \cdots\big)$, $\mathbb{m}_m$ the fold of its multiplier ($\mathbb{m}_1$, or $\mathbb{m}_x$ for x). A linear term is thereby carried by one copy of the collapse row, which on ψ is the constant $f_q(x_s)/f(x_s)$, and a source by two. An invariant constraint becomes $\mathbb{m}_1\big(\mathbb{D}^{(0)}(x_s) \otimes \mathbb{B}(x^*)\mathbb{G}^j\big)$, and the readout is $\mathbb{m}_1(\mathbb{D}^{(0)} \otimes I)$, with $f_Q = \eta f_q$ ($p = 1$ in $f_Q = \eta^{1/p} f_q$). The equation block maps the N^2 amplitudes of Ψ to the $2N$ coefficients of the folded register on $n + 1$ qubits.

The kernel bound. The equation block has $2N$ rows, so rank at most $2N$, and each constraint block is a fold of a Kronecker product of two rank-one rows, so rank at most one. With c invariant constraints, $\mathrm{rank}\, R \le 2N + c$ and $\dim\ker H_\otimes = \dim\ker R \ge N^2 - 2N - c$, which is at least $N^2/2 = 2^{2n-1}$ once $N \ge 8$ for $c \le 2$. At $n = 2$ the bound is weaker than the paper's count, which the records meet there by measurement. Table 21 gives the measured dimensions; at $n = 2$ to 4 the equation blocks have rank $2N - 3$ (R6a) and $2N - 1$ (R6b), each problem with one constraint.

Table 21: **The doubled space's kernel grows as N^2.** Dimension of the kernel of $H_\otimes$ for the paper's two quadratic equations, R6a ($4f'' + 2(f')^2 + f = 0$) and R6b ($f'' - 2f^2 + x = 0$), against the state's dimension N^2 and the paper's lower bound 2^{2n-1}. Classical computation (SVD of the stacked residual).

n	N^2	2^{2n-1}	R6a	R6b
2	16	8	10	8
3	64	32	unknown	unknown
4	256	128	unknown	unknown
5	1024	512	962	960

Higher degree. § 3.4 states the published construction as quadratic only. A term of degree p in the unknown needs p copies, $|\Psi\rangle = |\psi\rangle^{\otimes p}$, of dimension N^p, its factors folded back into one Chebyshev series by the p-fold product of Proposition 26 (a cubic term by a tripled space, a quartic by a quadrupled one). The register therefore grows with the degree, a limitation also noted in the concurrent work of Paine [7]. The argument of Proposition 4 should carry over, since the corresponding kernel is again a subspace and the p-fold product states are not; this thesis does not use the doubled space beyond $p = 2$. Its alternative for every degree is the Carleman lift of § 6.5.

H.3 Proof of Proposition 4

Proof. (1) As a matrix, $u \otimes u$ is $uu^\top$, of rank one. For independent u and v, $uu^\top + vv^\top$ has rank two, so the sum of two product states is not of the form $w \otimes w$, and the product states are not closed under addition. A kernel is a linear subspace, so a kernel containing $u \otimes u$ and $v \otimes v$ contains their sum, and no kernel is exactly the set of product states.

(2) Write $H_\otimes = \sum_\lambda \lambda \Pi_\lambda$ with $\Pi_0 = \Pi$. Then $P_d(H_\otimes)\psi_{\text{init}} = \Pi\psi_{\text{init}} + \sum_{\lambda>0} P_d(\lambda)\Pi_\lambda\psi_{\text{init}}$, and the second term has norm at most $\max_{\lambda>0} |P_d(\lambda)| \, \|\psi_{\text{init}}\| \to 0$. Since $\Pi\psi_{\text{init}} \neq 0$, normalising gives Equation (6.5). The imaginary-time filter has $P(0) = 1$ and $\max_{\lambda>0} e^{-\beta\lambda} = e^{-\beta\lambda_1} \to 0$; the minimax filter takes the value one at the ground state and at most ε on the excited spectrum, and $\varepsilon \to 0$ as its degree grows at a fixed edge.

(3) For $H_\otimes, P \succeq 0$, $\langle\phi, (H_\otimes + P)\phi\rangle = 0$ if and only if $\langle\phi, H_\otimes\phi\rangle = \langle\phi, P\phi\rangle = 0$, so $\ker(H_\otimes + P) = \ker H_\otimes \cap \ker P$. For this to be $\operatorname{span}\{\psi_* \otimes \psi_*\}$, P must vanish on $\psi_* \otimes \psi_*$ and on no other direction of $\ker H_\otimes$, a condition that refers to ψ_*. □

The scope is the one stated: a penalty built without knowledge of the solution. A penalty that knows the solution isolates it, and is answer-dependent.

H.4 The Carleman Lift

The symmetric layout. A polynomial system $u' = \sum_{p\geq 1} F_p u^{\otimes p}$ on M components, F_p of shape $M \times M^p$, is carried on the monomials $z_\alpha = \prod_i u_i^{\alpha_i}$ with $1 \leq |\alpha| \leq K$, once each: $\sum_{k=1}^{K} \binom{M+k-1}{k}$ components, level by level, the fields first. Differentiating a monomial,

$$z'_\alpha = \sum_i \alpha_i u^{\alpha - e_i} u'_i = \sum_i \alpha_i \sum_p \sum_{j_1,\ldots,j_p} (F_p)_{i,j_1\ldots j_p} \, z_{\alpha - e_i + e_{j_1} + \cdots + e_{j_p}}, \tag{H.4}$$

and $\mathbb{A}_K$ keeps every term whose monomial has degree at most K. A linear system is its own lift, $\mathbb{A}_1 = F_1$. For R8, $y' = -y + y^2$, before the time axis's factor $T/2$,

$$\mathbb{A}_3 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & -3 \end{pmatrix}. \tag{H.5}$$

Proposition 21 (The lift's spectrum)**.** *For a quadratic system with F_1 diagonalisable, $\mathbb{A}_K$ is block upper triangular in its levels, and its eigenvalues are the sums $\lambda_{i_1} + \cdots + \lambda_{i_k}$, $1 \le i_1 \le \cdots \le i_k \le M$, $1 \le k \le K$, of the eigenvalues of F_1.*

Proof. F_1 maps a monomial to monomials of its own degree and F_2 to degree one higher, so level k reads levels k and $k+1$ only. In F_1's eigencoordinates the diagonal block of level k acts on a monomial v^α as $\sum_i \alpha_i \lambda_i v^\alpha$. The change of coordinates is linear and maps the monomials of degree k to polynomials of degree k, so it is a similarity on each level. □

Proposition 22 (The level scale)**.** *Carrying $w = u/s$ multiplies F_p by s^{p-1}, and the lifted generator becomes $D^{-1}\mathbb{A}_K D$, D the diagonal equal to s^k on level k. The truncated solution, the truncation error and ρ are unchanged.*

Proof. $z(u) = Dz(w)$, so $z(w)' = D^{-1}\mathbb{A}_K Dz(w)$, whose entry (α, β) is multiplied by $s^{|\beta|-|\alpha|} = s^{p-1}$ for the terms of F_p. A similarity maps solutions to solutions, and ρ gains $1/s$ from the data and s from F_2. □

With $s = \max(\|u_0\|, 1)$, level k of the data's moments has norm $(\|u_0\|/s)^k \le 1$. Unscaled, data of norm above one grow as $\|u_0\|^k$ with the level, the ground state's weight moves from the fields to the top level, where no initial state independent of the answer reaches it, and the moments inflate the ratio rows' subnormalisation. The readout is scaled back to the physical datum.

The tensor layout. Level k is carried as the whole tensor $u^{\otimes k}$, M^k components in k slots, slot 0 the most significant.

Proposition 23 (The tensor layout's terms)**.** *In the tensor layout the block of $\mathbb{A}_K$ from level $k+p-1$ to level k is*

$$\left(\mathbb{A}_K^{\otimes}\right)_{k,\,k+p-1} = \sum_{j=0}^{k-1} I^{\otimes j} \otimes F_p \otimes I^{\otimes(k-1-j)}, \tag{H.6}$$

a sum of terms "F_p on slots $j, \ldots, j+p-1$", one per degree p and first slot j with $j+p \le K$: $\sum_p (K-p+1)$ terms, $2K-1$ for a quadratic field, whatever M. With V the embedding of the symmetric layout in the tensor layout, $\mathbb{A}_K^{\otimes} V = V \mathbb{A}_K$.

Proof. $\frac{d}{dt} u^{\otimes k} = \sum_j u^{\otimes j} \otimes u' \otimes u^{\otimes(k-1-j)}$ and $u' = \sum_p F_p u^{\otimes p}$, so slot j's factor F_p contracts slots j to $j+p-1$ of level $k+p-1$ into slot j, and one term (p, j) serves every level that occupies its slots. For the embedding, both sides map the moments $z(u)$ to the truncated derivative of $(u, u^{\otimes 2}, \ldots, u^{\otimes K})$; the moments over all u span the monomial space, distinct monomials being linearly independent functions, so the identity holds as matrices. □

The individual terms are not equal on symmetric tensors; only their sum is the symmetric layout's. A native complex preparation in this layout would also need a phase on the collapse row and a complex state loader for the data's moments, neither of which is built.

The tensor layout's circuit. It is built for a Fourier–Galerkin equation in structured form, $u' = L_1 u + \sum \mathrm{out}\big((\mathrm{left}\, u) \cdot (\mathrm{right}\, u)\big)$, every operator a Fourier multiplier, in the variables $v = \sqrt{2}\,\hat{u}$ of § H.1.2. The register is K slots of $d\,m$ qubits, each holding a wavevector's components in two's complement with $2^m \geq 3 \max|k_d| + 1$, so that the product of two kept modes never aliases onto a kept mode, and a level register of $\lceil \log_2(K+1) \rceil$ qubits. A level-k state occupies slots 0 to $k-1$ with kept wavevectors and leaves the others at the zero wavevector, which no truncation keeps: these are the physical states. F_1 on slot j is its multiplier's magnitudes on the kept wavevectors, as one uniformly controlled R_Y on a flag (α the largest magnitude), times the projector onto the levels above j; a symbol $c \prod_d k_d^{a_d} |\mathbf{k}|^{2e}$ has the phase $\mathrm{i}^{\sum_d a_d}$, which joins the term's weight. F_2 on slots j and $j+1$ is the projector onto the levels from $j+2$, the Fourier fold of its input multipliers (Appendix H.6), the output multiplier, which is the Galerkin truncation, and a cyclic shift of the slots above j with the level lowered by one. The terms are combined with weights $T/2$ and, for F_2, $s/\sqrt{2}$. The multipliers vanish off the truncation, so the encoding maps physical states to physical states only, with no leakage. It is verified exactly on Burgers' lift on four modes at $K = 2$, to 1.7×10^{-15} with leakage 1.1×10^{-14}, and by IR on every truncation where IR was run, to 316 modes at $K = 2$ on 100 172 physical states; at $K = 3$ on 148 and 316 modes it is a construction record only.

Where the lift is exact.

Proposition 24 (An exact lift)**.** *If the data lie in a subspace S that F_1 maps into itself and on which $F_2(u \otimes u) = 0$, the truncated lift's first level is the solution exactly, at every K.*

Proof. The solution is $u(t) = e^{F_1 t} u_0 \in S$, with $F_2(u \otimes u) = 0$ throughout. In the tensor layout, which embeds the symmetric one (Proposition 23), the top level evolves by the Kronecker sum of F_1 alone, so from $u_0^{\otimes K}$ it is $u(t)^{\otimes K}$. If level $k+1$ is $u(t)^{\otimes(k+1)}$, the F_2 terms feeding level k apply F_2 to $u \otimes u$ in two adjacent slots, which is zero, so level k evolves by F_1's Kronecker sum and is $u(t)^{\otimes k}$. Down to the first level, $z_1 = u(t)$. □

The Taylor–Green vortex is this case (Appendix H.7).

H.5 The Time Axis, Uniqueness and the Error

Rows and data. A system or a lift on $t \in [0, T]$ is carried on $\xi = 2t/T - 1$, and its equation rows are Equation (6.1) with $\mathbb{A}$, or $\mathbb{A}_K$, holding the factor $T/2$. Its data are the ratio rows $(\mathbb{F}/y_a) \otimes \mathbb{B}(-1)$ at $\xi = -1$ with y the scaled moments $z(u_0/s)$, imposing $z_c(-1) = (y_c/y_a)\, z_a(-1)$. On a padding block $\mathbb{F}/y_a$ is the identity, whose rows meet only that block, so the padding still decouples. The two forms share these rows and differ by the invertible $I \otimes \tilde{\mathbb{B}}_\xi$ on the equation rows, so they share their kernel (Theorem 1).

Proposition 25 (A one-dimensional near-kernel)**.** *On $C^1([-1,1])^C$ the solutions of $z' = \mathbb{A}z$ with $z(-1) \in \mathrm{span}(y)$ are $\{c\, e^{\mathbb{A}(\xi+1)} y : c \in \mathbb{R}\}$, a one-dimensional space.*

Proof. A linear initial-value problem has exactly one solution for each initial value, and the ratio rows leave the initial value one degree of freedom. □

The residual is this operator restricted to polynomials of degree below N_t. Its smallest singular value is the polynomial residual of the true solution, which falls spectrally in N_t for an analytic solution, and the rest of its spectrum approximates the continuum operator's, which has no other kernel. The near-kernel is therefore one-dimensional in both forms and for every $\mathbb{A}$, including a lift whose truncation does not converge: the uniqueness is the initial-value problem's, which the lift carries. Every Carleman build in the records has a one-dimensional kernel; the few not trusted lie at $n_t = 3$, where the time axis does not yet resolve the solution.

The error in two parts. With u the reference, $z^{(K)}$ the exact solution of the truncated lift from the data's moments, and $\hat{u}$ the field decoded from the prepared state, the triangle inequality gives Equation (6.8), both terms on the comparison grid. The lift's term is computed by exponentiating $\mathbb{A}_K$ and does not depend on n_t; the time axis's falls spectrally with n_t. The field's error therefore falls with n_t until the lift's floor and then holds.

H.6 Product Folds

A product of p fields on n qubits has degree at most $p(N-1) < 2^\ell N$, $\ell = \lceil \log_2 p \rceil$, so it is represented exactly on $n+\ell$ qubits, where it is the pointwise product of the fields' values at the nodes. Let $\mathbb{U}$ be the Chebyshev–Gauss transform on $n+\ell$ qubits (coefficients to node values, orthogonal), E the zero-padding from n to $n+\ell$ qubits, and $\Delta_p = \sum_i |i\rangle^{\otimes p}$ the diagonal.

Proposition 26 (The nodal fold is exact and tight)**.** *The fold*

$$\mathbb{m}_1^{(p)} = 2^{p\ell/2}\, \mathbb{U}^\top \Delta_p \left(\mathbb{U}E\right)^{\otimes p} \tag{H.7}$$

satisfies $\langle \tau(x)|_n^{\otimes p} = \langle \tau(x)|_{n+\ell}\ \mathbb{m}_1^{(p)}$ *exactly. For* $p = 2$ *it is block-encoded at* $\alpha = 2$*, and* $\|\mathbb{m}_1^{(2)}\|_2 = 2\sqrt{\lambda_{\max}(Q \circ Q)} \le 2$*, with* $Q = \mathbb{U}E(\mathbb{U}E)^\top$ *the projector onto the padded fields' node values and* $\circ$ *the entrywise product.*

Proof. The normalised bases on n and $n+\ell$ qubits differ by $\varphi_k^{(n)} = 2^{\ell/2}\varphi_k^{(n+\ell)}$, so a field with coefficients b on n qubits has the values $2^{\ell/2}\,\mathbb{U}Eb$ at the $2^\ell N$ nodes. A product of p such fields is represented exactly on $n+\ell$ qubits and its node values are the pointwise product $2^{p\ell/2}\Delta_p \bigotimes_i (\mathbb{U}Eb_i)$; $\mathbb{U}$ is orthogonal, so $\mathbb{U}^\top$ returns the coefficients, which is the first claim. (The same map follows from repeated product-to-sum, $T_jT_k = \frac{1}{2}(T_{j+k} + T_{|j-k|})$.) For $p = 2$, $\ell = 1$, the circuit applies $\mathbb{U}$ to both copies, each padded by one qubit, copies the first into the second with $n+1$ CNOTs, so that the second is zero exactly on the diagonal, post-selects it on zero, and applies $\mathbb{U}^\dagger$ to the first: a block encoding of $\mathbb{m}_1/2$, so $\alpha = 2$. For the norm, $\mathbb{U}^\top$ is orthogonal, so $\|\mathbb{m}_1\| = 2\|Y\|$ with $Y = \Delta(\mathbb{U}E \otimes \mathbb{U}E)$, and $YY^\top = \Delta(Q \otimes Q)\Delta^\top$ has entry (i,j)

equal to Q_{ij}^2. $Q \circ Q$ has non-negative entries and row sums $\sum_j Q_{ij}^2 = Q_{ii} \le 1$, so its largest eigenvalue is at most one. □

The norm rises with n, from 1.58 at $n = 2$ towards about 1.665 (computed to $n = 12$), so $\alpha = 2$ is within 1.21 of it at every n: the encoding is tight to a constant and not optimal. $\mathtt{m}_x$, the fold with the multiplier x, multiplies by the node values $\bar{x}_m = \cos\theta_m$ before $\mathbb{U}^\top$; since θ_m is linear in m, $\mathrm{diag}(\bar{x}) = (D + D^\dagger)/2$ for the diagonal phase $D = \mathrm{diag}(e^{\mathrm{i}\theta_m})$, one phase gate per qubit combined on one ancilla, and α stays 2. The circuit's ancillas are the two padding qubits and $n - 1$ spare qubits that return the register to zero for stacking. $\mathbb{U}$ is the discrete cosine transform, a dense gate counted apart and budgeted as a cited primitive [17]; it is not built gate by gate.

The Fourier fold. The tensor layout's fold is the circular convolution $(c{*}d)_{\mathbf{k}} = \sum_{\mathbf{p}+\mathbf{q}\equiv\mathbf{k}} c_{\mathbf{p}} d_{\mathbf{q}}$, which is $N^{d/2}\,\mathrm{QFT}^\dagger \Delta(\mathrm{QFT} \otimes \mathrm{QFT})$ on d axes: the QFT takes both copies to grid values, a CNOT from each qubit of the first copy to the second leaves the second at zero exactly where the grid points agree, the second copy is post-selected, and the inverse QFT returns the first to coefficients. Its $\alpha = N^{d/2}$ is the convolution's own norm on the register, since nothing restricts its input. The QFT is built.

H.7 Fourier–Galerkin Operators

Burgers. On $x \in [0, 2\pi)$, Equation (6.9) from $u(x, 0) = a \sin x$ keeps an odd solution odd (u_x is even, uu_x and u_{xx} odd), so $u = \sum_k b_k(t) \sin kx$.

Proposition 27 (Burgers' Galerkin system)**.** *Truncated to $k \le M$,*

$$b_k' = -\nu k^2 b_k - \frac{k}{4}\Big[\sum_{j+l=k} b_j b_l - \sum_{|j-l|=k} b_j b_l\Big], \qquad j, l \le M, \tag{H.8}$$

the sums over ordered pairs.

Proof. $u^2 = \frac{1}{2}\sum_{j,l} b_j b_l\big(\cos(j-l)x - \cos(j+l)x\big)$, so $-(u^2/2)_x = \frac{1}{4}\sum_{j,l} b_j b_l\big((j-l)\sin(j-l)x - (j+l)\sin(j+l)x\big)$. Its coefficient of $\sin kx$ takes $-k/4$ from each pair with $j + l = k$ and $+k/4$ from each pair with $j - l = \pm k$, since $(-k)\sin(-kx) = k \sin kx$. With $\nu u_{xx} \mapsto -\nu k^2 b_k$ and the projection onto $k \le M$, this is the system. □

So $F_1 = \mathrm{diag}(-\nu k^2)$ and F_2 is the bracket, each times $T/2$ on the time axis. The data $b(0) = (a, 0, \ldots, 0)$ enter as ratio rows anchored on b_1, which also carries the regular datum. The least damped mode is $k = 1$, so $\rho = a\|F_2\|/\nu$, the factors $T/2$ cancelling, and $\|F_2\|$ grows with M. The reference for the lift and the time axis is the Galerkin system's IVP, integrated with its exact sparse Jacobian, which is stiff at sixteen modes (νk^2 up to 256ν). The truncation's own

error is measured against the Cole–Hopf solution [18], [19]: with $\beta_B = a/(2\nu)$,

$$u(x,t) = \frac{4\nu \sum_{m\geq 1} m\, I_m(\beta_B)\, e^{-\nu m^2 t} \sin mx}{I_0(\beta_B) + 2\sum_{m\geq 1} I_m(\beta_B)\, e^{-\nu m^2 t} \cos mx}, \tag{H.9}$$

from $u = -2\nu\,\partial_x \ln\varphi$, which turns Burgers' equation into the heat equation with $\varphi(x,0) = e^{-\beta_B} e^{\beta_B \cos x}$. The runs, all over $T = 1$, are R9a to R9e at $a/\nu = 0.03$, 0.3, 1, 3 and 10, each on $M = 4$, 8 and 16 modes.

Navier–Stokes. Equation (6.10) is carried on $\cos(\mathbf{k}\cdot\mathbf{x})$ and $\sin(\mathbf{k}\cdot\mathbf{x})$ over the half-plane $k_x > 0$, or $k_x = 0 < k_y$, with $0 < |\mathbf{k}|^2 \leq k_c^2$: $k_c^2 = 4$, 5, 9, 16 and 25 keep $M = 12$, 20, 28, 48 and 80 modes, and 49 and 100, used for the tensor layout's coupling alone, keep 148 and 316. The mean vorticity is invariant and zero, and the stream function is eliminated, $\hat\psi_{\mathbf{k}} = \hat\omega_{\mathbf{k}}/|\mathbf{k}|^2$. With φ_a a basis function of wavevector $\mathbf{p}$, φ_b one of $\mathbf{q}$, and $\mathbf{p}\times\mathbf{q} = p_x q_y - p_y q_x$,

$$J\big(\varphi_a/|\mathbf{p}|^2, \varphi_b\big) = \sigma_a\sigma_b\, \frac{\mathbf{p}\times\mathbf{q}}{|\mathbf{p}|^2}\, h_a(\mathbf{p}\cdot\mathbf{x})\, h_b(\mathbf{q}\cdot\mathbf{x}), \tag{H.10}$$

with $\sigma = -1$, $h = \sin$ for a cosine and $\sigma = 1$, $h = \cos$ for a sine, since $\nabla\varphi_a = \sigma_a \mathbf{p}\, h_a$ and $J(f,g) = \nabla f \times \nabla g$. The product $h_a h_b$ is a sum of the modes $\mathbf{p}\pm\mathbf{q}$, each folded into the half-plane, and the truncation keeps those in the set; every coefficient is rational and computed in exact arithmetic. Mode c's equation is $w_c' = -\nu|\mathbf{k}_c|^2 w_c + \sum_{a\leq b} c_{cab} w_a w_b$, each coefficient times $T/2$, with $\nu = 1$; the data enter as ratio rows anchored on the mode with the largest datum, which carries the regular datum.

Two properties follow. *Modes of equal $|\mathbf{k}|$ do not interact.* In the complex basis the Jacobian's coefficients are $\widehat{J}_{\mathbf{k}} = -\sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{p}\times\mathbf{q})\,\hat\psi_{\mathbf{p}}\hat\omega_{\mathbf{q}}$, and with $\hat\psi_{\mathbf{p}} = \hat\omega_{\mathbf{p}}/|\mathbf{p}|^2$, symmetrised over $\mathbf{p}\leftrightarrow\mathbf{q}$,

$$\widehat{J}_{\mathbf{k}} = -\frac{1}{2}\sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{p}\times\mathbf{q})\big(|\mathbf{p}|^{-2} - |\mathbf{q}|^{-2}\big)\hat\omega_{\mathbf{p}}\hat\omega_{\mathbf{q}}, \tag{H.11}$$

which vanishes for $|\mathbf{p}| = |\mathbf{q}|$; the real basis is a unitary change of the complex one. *Without viscosity the truncation conserves the energy and the enstrophy exactly,*

$$E = \tfrac{1}{4}\sum_m w_m^2/|\mathbf{k}_m|^2, \qquad Z = \tfrac{1}{4}\sum_m w_m^2. \tag{H.12}$$

With P the projection onto the kept modes, $Z' \propto \langle\omega, PJ(\psi,\omega)\rangle = \int J(\psi,\omega^2/2) = 0$ and $E' \propto \langle\psi, PJ(\psi,\omega)\rangle = \int J(\psi^2/2,\omega) = 0$, because $J(f,g) = \partial_x(f g_y) - \partial_y(f g_x)$ is a divergence, whose integral over the periodic square vanishes.

Initial conditions. The Taylor–Green vortex $\omega_0 = 2\sin x \sin y = \cos(x-y) - \cos(x+y)$ occupies two modes with $|\mathbf{k}|^2 = 2$, whose coupling vanishes, so $\omega = \omega_0 e^{-2\nu t}$ exactly in any truncation that holds them, and in the equation itself. Its data lie in a subspace that F_1 preserves and on which F_2 vanishes, so its lift is exact at every order (Proposition 24): R10a

validates the lift, the time axis and the preparation at a known answer. The counter-rotating pair is $\pm e^{-|\mathbf{x}-\mathbf{c}|^2/(2\sigma^2)}$ with $\sigma^2 = \frac{1}{4}$ at $\mathbf{c} = (3\pi/4, \pi)$ and $(5\pi/4, \pi)$, periodised and projected exactly onto the modes: it has no cosine part, and its sine coefficients are proportional to $(-1)^{k_x+k_y+1} \sin(\pi k_x/4)\, e^{-|\mathbf{k}|^2/8}$, so modes with $k_x \equiv 0 \pmod 4$ have zero datum. It spans several $|\mathbf{k}|$, so its triads are active. It is scaled to $\|w_0\| = a$ and run at $a/\nu = 3$, 10 and 30 (R10b to R10d) over $T = 1$, and at $a/\nu = 30$ also over $T = 1/8$. The reference for the lift and the time axis is the Galerkin system's IVP, with its exact sparse Jacobian; the truncation's own error is measured against a pseudo-spectral DNS of Equation (6.10) from the same data, whose slope inside the truncation is the Galerkin system's.

I Verification Protocol

I.1 Modes, sizes and tolerances

I.2 Structured probes

I.3 Per-atom residuals

I.4 The classical reference: eigh against SVD

Table 22: Accuracy of the classical reference, $1 - |\cos|$ against the exact coefficients, for the standard form by $\texttt{eigh}(R_{\text{std}}^\top R_{\text{std}})$ and by SVD of R_{std}, and for the descriptor form.

n	Standard, `eigh`	Standard, SVD	Descriptor
2	1.7×10^{-1}	1.7×10^{-1}	1.6×10^{-2}
3	8.6×10^{-8}	8.6×10^{-8}	5.8×10^{-9}
4	0	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
5	3.7×10^{-15}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
6	1.7×10^{-10}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
7	1.2×10^{-4}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
8	0.73	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$

Table 23: The linear benchmark problems, R1–R5, with the paper's printed invariant constraints.

	Equation	Invariant (printed)	Reference
R1	$f' + 2f = 0$	– ($f(-1) = 1$ regular)	$e^{-2(x+1)}$
R2a	$f'' + 4f' + 4f = 0$	$f(-1) = 0$	$\frac{1}{2}(1+x)e^{-2x}$
R2b	$f'' - 2f' - 3f = 0$	$f(0) = 0$	$\frac{1}{4}\big(e^{3x} - e^{-x}\big)$
R2c	$f'' + 5f' + 400f = 0$	$f'(0) = 0$	$e^{-5x/2}\big(\cos\omega x + \frac{\sqrt{7}}{21}\sin\omega x\big)$, $\omega = \frac{15\sqrt{7}}{2}$
R3a	Legendre, $l = 4$	$f'(0) = 0$	P_4
R3b	Legendre, $l = 5$	$f(0) = 0$	P_5
R3c	associated Legendre, $l = 5$, $m = 1$	$f'(0) = 0$	P_5^1
R3d	associated Legendre, $l = 6$, $m = 1$	$f(0) = 0$	P_6^1
R4a	$(x-1)f'' - xf' + f = (x-1)^2$	$f'(-0.2) = 0$	closed form
R4b	$f'' + 4f' + 4f = e^{-2x}$, $\bar{p} = 3$	$f(-0.73) = 0$	closed form
R4c	$f'' - 5f' + 6f = xe^{2x}$, $\bar{p} = 5$	$f(0.91) = 0$	closed form
R4d	$f'' + 4f' + 20f = xe^{-2x}\sin 4x$, $\bar{p} = 7$	$f'(-0.75) = 0$	closed form
R5a	Laplace, $f_{xx} + f_{yy} = 0$	$f(\pm 1, y) = f(x, -1) = 0$	$\cos\frac{\pi x}{2}\,\sinh\frac{\pi(y+1)}{2}\big/\sinh\pi$
R5b	heat, $f_t = \frac{1}{25}f_{xx}$	$f(t, \pm 1) = 0$, $f_x(t, \pm\frac{3}{4}) = 0$	$e^{-4\pi^2 t/25}\sin 2\pi x$
R5c	wave, $f_{tt} = 4f_{xx}$	$f(\frac{1}{8}, x) = f(\frac{7}{8}, x) = 0$, $f(t, \pm 1) = 0$	$\cos 4\pi t\,\sin 2\pi x$

Table 24: The nonlinear and coupled benchmark problems, R6–R10.

	Equation	Data	Reference
R6a	$4f'' + 2f'^2 + f = 0$ (paper Eq. 29)	$f'(0) = 0$, $x_s = 0$	$1 - x^2/8$
R6b	$f'' - 2f^2 + x = 0$ (paper Eq. 34)	$f(0.02615) = 0$	boundary-value solve to 10^{-12}
R7	$u' = v$, $v' = -\omega^2 u$, $\omega = \pi$	$u(-1) = 0$, $v(-1) = \omega$	$u = \sin\omega(x+1)$
R8	$y' = -y + y^2$, $t \in [0, 3]$	$y(0) = \frac{1}{2}$ ($\rho = 0.5$)	$1/(1+e^t)$
R9	viscous Burgers, periodic, $u_0 = a\sin x$	sine modes $k \le M$, $M = 4, 8, 16$; $a/\nu =$ 0.03–10	Galerkin IVP; Cole–Hopf (truncation); exact lift
R10a	2-D vorticity Navier–Stokes, $\omega_0 = 2\sin x\sin y$	$J \equiv 0$ (Taylor–Green)	exact viscous decay
R10b–d	2-D vorticity Navier–Stokes, counter-rotating vortex pair	$\|w_0\| = a$, $a/\nu = 3, 10, 30$; active triads	Galerkin IVP; DNS (truncation); exact lift

Table 25: State preparation by problem and form. Tier as marked.

	n	Infidelity (std)	Infidelity (desc)	Degree (std)	Degree (desc)	Tier
R1		–	–	–	–	
R2a		–	–	–	–	
R2b		–	–	–	–	
R2c		–	–	–	–	
R3		–	–	–	–	
R4		–	–	–	–	
R5		–	–	–	–	
R7		–	–	–	–	
R8		–	–	–	–	
R9		–	–	–	–	
R10		–	–	–	–	

I.5 The circuit-simulation memory and time model

I.6 Acceptance and the error budget

J Benchmark Catalogue

J.1 Linear problems

J.2 Nonlinear and coupled problems

J.3 Results by problem

K Reproduction Record

K.1 All panels

K.2 The Fig. 5 triage

K.3 Panels not reproduced, and printed errors

K.4 The paper's claims

K.5 The paper's filter and its parameters

The paper realises imaginary time by QSVT on a block encoding of $\tilde{H} = H/\|H\|_F$ [1]. The target $p(x) \approx e^{-tx}$ has indefinite parity, so the paper finds one set of phase angles for its even part and one for its odd part, of maximal degrees d^e and d^o, runs the two opposite-parity QSVT sequences and sums them by an LCU: a mixed-parity circuit. Each sequence uses the unitary dilation

$$U(\tilde{H}) = \begin{pmatrix} \tilde{H} & \sqrt{I - \tilde{H}\tilde{H}^\dagger} \\ \sqrt{I - \tilde{H}^\dagger\tilde{H}} & -\tilde{H}^\dagger \end{pmatrix}, \tag{K.1}$$

Table 26: The paper's solutions at its printed resolution: L^∞ error of the paper-faithful and corrected variants, and the corrected variant's best over n. Classical computation.

Fig.	Problem	Printed n	Faithful	Corrected	Corrected, best over n
3b	R2a	3	1.2×10^{-3}	1.2×10^{-3}	3.2×10^{-13} $(n = 5)$
3c	R2b	3	$1.8 \times 10^{-2\dagger}$	$1.8 \times 10^{-2\dagger}$	2.3×10^{-11} $(n = 5)$
3d	R2c	5	2.2×10^{-3}	2.2×10^{-3}	5.6×10^{-12} $(n = 6)$
4a	R3a	3	1.9×10^{-15}	1.9×10^{-15}	1.9×10^{-15} $(n = 3)$
4b	R3b	3	3.0×10^{-15}	3.0×10^{-15}	2.3×10^{-15} $(n = 4)$
4c	R3c	5	$8.7 \times 10^{-1\dagger}$	4.0×10^{-14}	4.9×10^{-15} $(n = 3)$
4d	R3d	5	$1.2^\dagger$	1.5×10^{-13}	8.0×10^{-15} $(n = 3)$
5a	R4a	3	3.1×10^{-3}	1.8×10^{-6}	3.8×10^{-14} $(n = 4)$
5b	R4b	3	1.9×10^{-2}	7.8×10^{-3}	8.5×10^{-12} $(n = 5)$
5c	R4c	4	7.2×10^{-3}	4.9×10^{-10}	1.6×10^{-13} $(n = 5)$
5d	R4d	4	6.1×10^{-2}	4.9×10^{-7}	5.1×10^{-12} $(n = 5)$
6b	R5a	3	3.9×10^{-3}	1.4×10^{-4}	2.1×10^{-13} $(n = 4)$
6d	R5b	4	1.9×10^{-4}	4.7×10^{-5}	4.6×10^{-12} $(n = 5)$
6f	R5c	5	8.5×10^{-8}	7.3×10^{-8}	6.1×10^{-11} $(n = 6)$
7b	R6a	2	$2.2 \times 10^{-16\dagger}$	$2.2 \times 10^{-16\dagger}$	–
7c	R6b	2	$6.8 \times 10^{-3\dagger}$	$6.8 \times 10^{-3\dagger}$	–

Table 27: Which printed choice causes the Fig. 5 error: the paper-faithful variant over $n = 4, \ldots, 10$, each correction alone, and both together.

Fig.	Problem	Faithful	Exact root	Chebyshev source	Both, best
5a	R4a	3.1×10^{-3}	–	–	3.8×10^{-14} $(n = 4)$
5b	R4b	1.6×10^{-2}	4.0×10^{-2}	3.6×10^{-2}	8.5×10^{-12} $(n = 5)$
5c	R4c	7.2×10^{-3}	4.8×10^{-3}	1.0×10^{-2}	1.6×10^{-13} $(n = 5)$
5d	R4d	6.1×10^{-2}	8.1×10^{-2}	1.1×10^{-1}	5.1×10^{-12} $(n = 5)$

Table 28: The claims of Wu et al. tested in this thesis, and their verdicts.

	Claim	Verdict
A-01	$\mathbb{G}$, $\mathbb{M}$, $\mathbb{n}_1$ and $\mathbb{B}$ as printed at $n = 2$	reproduced
A-02	the printed η_e of Figs 3–5 (11 panels)	within one unit of the last printed digit; 9/11 equal when rounded
A-03	the printed η_e of Fig. 6	not reproduced
A-04	the printed η_e of Fig. 7	not reproduced
A-05	the printed Laplace solution	a typographical error (cosh for cos)
A-06	the $\mathbb{B}$ circuit (Fig. 2): $n + 2$ qubits, prefactor $\sqrt{2^{n+1}}$	confirmed
A-07	the $\mathbb{G}$ circuit (Fig. 9): $2n + 3$ qubits, $O(n^3)$ gates, error decreasing exponentially	–
A-08	the $\mathbb{D}^{(0)}$ circuit (Fig. 8): prefactor $\sqrt{2^{n+1}}/f(x_s)$	confirmed
A-09	the linear H is positive definite with distinct eigenvalues	–
A-10	the doubled-space H has at least 2^{2n-1} zero eigenvalues	–
A-11	the time rule $t \geq \lambda_{\max}/(\lambda_2(n-1))$ suffices	–
A-12	the necessary terms scale as $O(\mathrm{poly}(n))$ and are efficient	true for gates; not for the subnormalisation
A-13	the examples are solved accurately at the printed n	distorted where rounded points or Maclaurin sources enter

built from PREPARE and SELECT subroutines over the unitary decomposition of $\tilde{H}$ with $m = \lceil \log_2 L \rceil$ ancilla for L unitaries. The time satisfies the approximate bound $t \geq \lambda_{\max}/(\lambda_2(n-1))$ (Equation (3.6)). For the paper's two Legendre examples of its supplement, $t = 15$ and $t = 8$ recover the ground state with $(d^e, d^o) = (6, 7)$. Beyond the $O(1/\Delta)\,\mathrm{polylog}(1/\varepsilon)$ preparation time, the paper expects algorithmic thermalisation to reach $O(1/\sqrt{\Delta})$, and notes that GQSP, GQET and GQSVT could improve the circuit.

K.6 The tau-truncation ablation

Table 29: Dropping the top k rows of the equation block (the Chebyshev tau method) gives an exact kernel with accuracy comparable to the paper's least squares.

Problem	n	Least-squares L^∞	σ_0	Tau-truncated L^∞	σ_0
R2a	5	3.2×10^{-13}	3.0×10^{-11}	1.6×10^{-13}	0
R2a	6	2.2×10^{-12}	4.9×10^{-10}	3.8×10^{-12}	0
R2a	7	2.8×10^{-11}	7.8×10^{-9}	1.0×10^{-10}	0
R2b	5	2.3×10^{-11}	5.3×10^{-11}	3.7×10^{-12}	0
R2b	6	1.8×10^{-10}	8.5×10^{-10}	7.4×10^{-11}	0
R2b	7	9.8×10^{-10}	1.4×10^{-8}	8.3×10^{-10}	0
R2c	5	2.2×10^{-3}	1.6×10^{-2}	1.8×10^{-3}	0
R2c	6	5.6×10^{-12}	5.5×10^{-10}	1.8×10^{-11}	0
R2c	7	2.4×10^{-10}	1.1×10^{-8}	1.9×10^{-10}	0

L Software and Reproducibility

L.1 Architecture

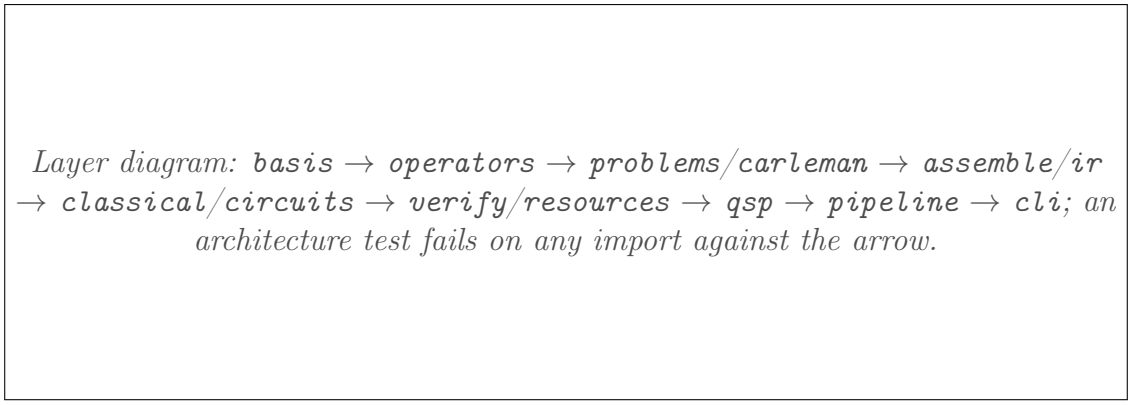


Figure 15: The layers of the `pihm` package and the one direction in which they may import.

L.2 Records and provenance

L.3 Campaigns and computational resources

Table 30: Resource classes for the Setonix campaigns.

Class	Request	Typical runs
S	`work`, 1 CPU, 4 GB, 30 min	construction, IR checks, small exact verification
M	`work`, 16 CPUs, 64 GB, 4 h	exact verification to 20 qubits; mid-size exact emulation
L	`work`, 64–128 CPUs, ∼230 GB, 12–24 h	circuit-simulation ceilings; large emulation; probes to 30 qubits
XL	`highmem`, 1 TB	the largest emulation and probe runs

L.4 Tests

Table 31: The test tiers of the `pihm` package.

Tier	Content	Run
Unit and property	operator identities over random n and x; primitives; arithmetic; IR; filters; phases	every commit
Golden	fixtures from the frozen legacy code and the paper's printed $n = 2$ matrices	every commit
Reproduction	the paper's targets, both variants	every commit
Integration	every problem, both forms, smallest n, all stages	every commit
Documentation	Sphinx build, doctests, spelling	every commit
Claims	one test per claim; writes the claims report	at gates
Slow	larger-n verification; circuit-simulation ceilings	at gates, on Setonix

L.5 Regenerating the results

L.6 Environment

M Research Proposal

Table 32: Deviations from the research proposal of May 2025.

Proposed	Outcome	Reason
Extending QSVT to the generalised construction for complex-valued, non-Hermitian operators		
Nonlinear PDEs: Burgers, nonlinear Schrödinger, complex Ginzburg–Landau		
Data-informed Hamiltonians (not proposed)		

Masters Research Proposal

Dhyan Patel [23340229]
The University of Western Australia

May 25, 2025

Title: Solving Nonlinear PDEs via Quantum Singular Value Transformation

Supervisor: Professor Jingbo Wang

1 Introduction

1.1 Aims

This thesis aims to develop and evaluate quantum algorithms for solving nonlinear partial differential equations (PDEs) using the Quantum Singular Value Transformation (QSVT) framework. Based on recent work that represents differential equations and their boundary conditions as effective Hamiltonians, this work extends the QSVT to the Generalised Quantum Singular Value Transformation (GQSVT). This extension enables the application of this method to complex-valued differential equations which arise naturally in many areas of physics and engineering. Additionally the project will explore how to incorporate data-informed modifications into these Hamiltonians, with the goal of modelling data-rich regimes.

1.2 Significance

Solving complex nonlinear partial differential equations (PDEs) on quantum hardware is a rapidly emerging prospect with broad applications ranging from fluid dynamics and turbulence modelling to quantum field theory. Established algorithms such as the Harrow-Hassidim-Lloyd (HHL) algorithm and variational methods typically rely on discretisation, which inhibits their scalability to stiff, nonlinear, multiscale or dense systems with fine grid sizes. In contrast, the effective Hamiltonian approach uses spectral methods to encode the solution into a Chebyshev polynomial basis, allowing it to surpass performance when fine-grid encoding is required. The quantum singular value transformation (QSVT) provides a flexible and modular approach to quantum linear algebra, allowing operations like matrix inversion under broader conditions than HHL [1]. However, the current method restricts effective Hamiltonians to real values, thereby limiting the class of operators and solutions to purely real outcomes. Moreover, the HHL paper only addresses linear partial differential equations or nonlinear ordinary differential equations

through a process called linearisation [3] and variational algorithms require classical optimisation [7].

This research is significant because it extends the effective Hamiltonian based framework to complex-valued nonlinear PDEs, which are common in frontier areas of various fields. In domains like geophysical inversion, deep turbulence and real-time weather prediction, incorporating data-informed terms into the Hamiltonian provides a promising path towards optimising quantum solutions to problems where complete physical models are unavailable or difficult to create. By enabling hybrid physics and data driven Hamiltonians, this work could expand the applicability of quantum PDE solvers to data-rich, real world problems.

2 Background

2.1 Problem Statement

Many physical systems are described by partial differential equations (PDEs), but solving them remains computationally costly. In the absence of closed-form solutions, many advanced classical algorithms implement numerical methods which are difficult to scale and lose accuracy in systems with a need for more fine grid or time-step discretisations. This is especially true of nonlinear PDEs which are the primary focus of this research.

Quantum algorithms provide a potentially scalable alternative. Quantum Singular Value Transformation (QSVT) allows quantum circuits to encode real-valued polynomials of definite parity. Recent theoretical developments have shown that it is possible to represent differential equations and their boundary conditions as effective Hamiltonians, which are structured matrices that can be processed using QSVT. However, due to QSVT's restriction to real-valued Hermitian Hamiltonians, the range of solvable PDEs is restricted. Systems with damping, gain/loss or complex boundary interactions can lead to complex-valued or non-Hermitian operators.

While novel frameworks such as the Generalised QSVT (GQSVT) have been proposed to handle complex-valued operators and remove parity constraints, they have not been systematically applied to nonlinear PDEs.

The existing effective Hamiltonian approach is largely physics-driven, assuming known governing models. Yet in many high-dimensional, data-rich domains such as geophysical imaging, real-time weather prediction, or turbulence modelling, the governing equations may be incomplete or unavailable. In these settings, data-informed Hamiltonian construction offers a promising path forward towards more flexible and accurate quantum solvers.

This research addresses the gap in current quantum PDE solving frameworks by developing and applying GQSVT based methods for nonlinear, complex-valued PDEs and exploring hybrid physics and data driven constructions for real world data-intensive applications.

2.2 Literature Review

2.2.1 Exploration of Problems

Burgers Equation (Viscous)

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \tag{1}$$

This equation appears in turbulence theory and has a closed form solution in one real spatial dimension with $\nu > 0$. However, complex versions do exist and the equation remains unsolvable analytically outside these circumstances. The nonlinearity term is $u\frac{\partial u}{\partial x}$.

Nonlinear Schrödinger Equation

$$i\frac{\partial \psi}{\partial t} + \nabla^2\psi + |\psi|^2\psi = 0 \tag{2}$$

This equation describes the evolution of complex wave functions in nonlinear media. Its nonlinearity term is $|\psi|^2\psi$.

Complex Ginzburg-Landau Equation

$$\frac{\partial A}{\partial t} = A + (1 + i\alpha)\nabla^2 A - (1 + i\beta)|A|^2 A \tag{3}$$

This equation models phenomena such as superconductivity and fluid convection patterns. The nonlinearity term is $|A|^2A$.

The current physics-informed Hamiltonian method is able to address Burgers equation in its real case, but the Schrödinger and Ginzburg-Landau equations involve are complex-valued fields. This limitation motivates the development of a more robust framework capable of handling a broader class of nonlinear PDEs.

2.2.2 Existing Classical and Quantum Solving Methods

Classical methods for solving PDEs employ mesh based discretisation techniques such as finite difference and finite element methods or spectral methods [11]. These methods approximate the continuous domain onto a discrete grid, converting the PDE into a large system of algebraic equations that can be solved numerically. These methods are well-established and reliable, but for high-resolution or multi-dimensional systems the computational cost increases rapidly. For higher dimensional problems, direct matrix or classical variational methods may be employed [4].

The Harrow-Hassidim-Lloyd (HHL) algorithm was one of the first to demonstrate exponential speedup for solving systems of linear equations, which arise from discretised PDEs [3]. HHL encodes the linear system $A\vec{x} = \vec{b}$ into a quantum state and uses phase estimation to prepare a quantum state proportional to the solution vector $\vec{x}$. However, it is limited to sparse, well-conditioned Hermitian matrices. This makes it impractical for solving PDE problems, particularly nonlinear ones. Importantly, in the 2021 "Grand Unification" paper, it is shown that HHL can be viewed as a special case of QSVT, where

we're looking to find a target polynomial that inverts the singular values of the matrix A.

Variational quantum algorithms use a hybrid quantum-classical loop to optimise a parameterised quantum circuit that approximates the solution to a PDE [7]. This approach also discretises the domains into a grid. While more flexible than HHL, these methods require classical optimisation and suffer from challenges such as barren plateaus and scaling issues for larger systems.

2.2.3 Quantum Singular Value Transformation

The quantum singular value transformation (QSVT) is a powerful framework that generalises quantum linear algebra operations by enabling quantum circuits to apply polynomial transformations to the singular values of block encoded matrices [1]. QSVT enables the application of any parity-constrained polynomial $f(\Sigma)$ to the singular values Σ of a matrix A block encoded within a unitary.

$$U = \begin{bmatrix} A & \cdot \\ \cdot & \cdot \end{bmatrix} \text{ with } \|A\| \leq 1 \tag{4}$$

Assuming we have access to such a unitary U and the ability to apply a sequence of controlled-phase rotations by an angle ϕ, we can construct a unitary $V_{\vec{\phi}}$, dependent on the values of $\vec{\phi}$ we select for our phase rotations, that looks like the following:

$$V_{\vec{\phi}} = \begin{cases} \tilde{\Pi}_{\phi_1} U \left[\prod_{k=1}^{(d-1)/2} \Pi_{\phi_{2k}} U^\dagger \tilde{\Pi}_{\phi_{2k+1}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for odd Poly}^{(\text{SV})}(A) \\ \left[\prod_{k=1}^{d/2} \Pi_{\phi_{2k-1}} U^\dagger \tilde{\Pi}_{\phi_{2k}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for even Poly}^{(\text{SV})}(A) \end{cases} \tag{5}$$

where the sequences of phases $\vec{\phi} = (\phi_0, \phi_1, \ldots, \phi_d)$ is chosen specifically to encode a desired polynomial $f(x)$ of degree d [1]. $\tilde{\Pi}_\phi$ and Π_ϕ represent a controlled phase rotation by ϕ where $\tilde{\Pi}$ and Π are the left and right unitary projectors respectively such that $A := \tilde{\Pi} U \Pi$. By carefully selecting which phase angles to rotate by, we can approximate any real, fixed parity continuous polynomial [6] acting on the singular values of A.

It can be shown that QSVT serves as a unifying framework for many quantum algorithms including quantum search, Hamiltonian simulation, factoring, quantum walks and more [8]. A key limitation of QSVT is that it only supports real-valued polynomials applied to matrices. This restricts its applicability to problems where both the matrix and the function being applied are real-valued [1]. For differential equations, this implies only PDEs with real-valued operators and real-valued solutions can be handled.

For physics-informed partial differential equation (PDE) solvers, QSVT is used in ground-state preparation via Hamiltonian simulation. In this case the target polynomial is $p(x) = e^{-tx}$ [11].

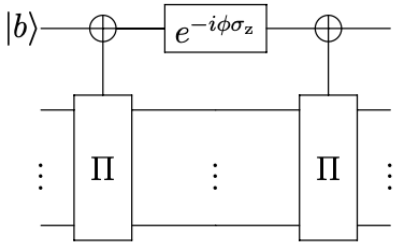


Figure 1: Quantum Circuit Diagram of the implementation of a Projector Controlled Phase Gate, Π_ϕ [2]

2.2.4 Physics-Informed Effective Hamiltonians

This algorithm encodes a differential equation in a latent Chebyshev basis and finds its solution by preparing the ground state of a specifically constructed Hamiltonian [11]. First, consider some differential equation with a differential operator $\mathcal{L}$, solution $u(x)$ and source function $f(x)$:

$$\mathcal{L}[u(x)] = f(x), \qquad x \in [a, b]. \tag{6}$$

We can represent the solution as a state overlap in the Chebyshev basis with a scale factor s, namely

$$u(x) = s\langle T(x)|\psi\rangle \tag{7}$$

Here $|T(x)\rangle$ is known as a "feature map" state, a superposition of computational basis states with Chebyshev coefficients. $|\psi\rangle$ is a parameterised quantum state carrying the unknown amplitudes. Thus, we've effectively represented our solution in a latent Chebyshev basis [11].

To translate differential operators, we use the known differentiation matrices associated with the Chebyshev basis [5]. An example translation is shown below where $\mathbb{G}$ is the generator matrix corresponding to differentiation:

$$\frac{d^2 f(x)}{dx^2} + af(x) = 0 \longrightarrow \mathbb{A} = \mathbb{G}^{\intercal 2} + a\mathbb{I}. \tag{8}$$

Here, $\mathbb{A}$ is the matrix corresponding to the latent space representation of the DE. Crucially, differential operators act simply in the Chebyshev basis, as the following

$$\frac{d}{dx} f(x) = \frac{d}{dx}\langle T(x)|\psi\rangle = \langle T(x)|D^\intercal|\psi\rangle. \tag{9}$$

As a result, all terms in the DE can be rewritten as linear operators acting on $|\psi\rangle$.

To implement constraints, we use $\mathbb{B}(x)$, a data transformation operator dependent on x. For example, for specific constraints such as $f(x_0) = 0$ or $f'(x_1) = 0$, we can use $\mathbb{B}(x_0)$ or $\mathbb{B}(x_1)\mathbb{G}^\intercal$ [11]. From these, we can create our effective Hamiltonian:

$$\mathbb{H} = \mathbb{A}^\intercal \mathbb{A} + \mathbb{B}^\intercal \mathbb{B}. \tag{10}$$

We may add additional boundary constraints as required, which would effectively be more $\mathbb{B}$ matrices. Therefore, by construction, solving the original DE is equivalent to solving the eigen-problem:

$$\mathbb{H}_{\text{eff}}|\psi\rangle = 0, \tag{11}$$

and the desired solution corresponds to the ground state of the effective Hamiltonian. H is a real symmetric positive-definite matrix with distinct eigenvalues and mutually orthonormal eigenvectors [11].

We then block encode the effective Hamiltonian, and use a ground state preparation routine, QSVT, to amplify the amplitude of the ground state. Specifically, we use quantum imaginary time evolution, constructing the polynomial approximation of $e^{-t\mathbb{H}}$. This exponentially suppresses the excited eigen-components, leaving a higher probability of measuring the ground state. Importantly, the algorithm directly prepares the lowest energy solution without classical optimisation [11].

Finally, to measure the quantum state, we use a Chebyshev transformation and Hadamard test to measure the state we have prepared. This controlled overlap test effectively measures the overlap $\langle T(x)|\psi\rangle$. In practice this is done via interferometric measurement [11].

This same framework is extendable to multi-variable and nonlinear DEs. For a function of two variables $f(x, y)$, we use a tensor product Chebyshev basis with two registers, preparing a joint feature state $|T(x)\rangle \otimes |T(y)\rangle$. Take Laplace's equation for example:

$$\frac{d^2 f}{dx^2} + \frac{d^2 f}{dy^2} = 0 \longrightarrow \mathbb{A} = \mathbb{G}_x^{\intercal 2} \otimes \mathbb{G}_y^{\intercal 2}. \tag{12}$$

For nonlinear terms, we add extra registers. For example, a term like $f'(x)^2$ leads to a product of two copies of $|\psi\rangle$ in latent space. We create a new, higher dimensional latent Chebyshev space where $|\Psi\rangle \equiv |\psi\rangle \otimes |\psi\rangle$, so that the nonlinearity term becomes linear in the larger space. This naturally handles all nonlinearity. Inhomogeneous terms are handled via Maclaurin series expansion. For all these cases, we then construct the effective Hamiltonian as usual and solve [11]. This approach allows us to encode the solution over the entire domain, rather than using grid discretisation. Therefore, it is possible to evaluate the solution function at any arbitrary input values. However, as discussed above, QSVT can only handle real values. This restricts the class of differential equations we're able to solve.

Example Solutions:

Below are results from the application of this algorithm on example differential equations [11]. The paper did not attempt to solve nonlinear PDEs, but there are examples of both linear PDEs and nonlinear DEs. $f(x)$ is the known solution and $f_Q^*(x)$ is the quantum solution.

1. Heat Equation

$$\frac{\partial f(t,x)}{\partial t} - k\frac{\partial^2 f(t,x)}{\partial x^2} = 0, k = \frac{1}{25}, \mathcal{BC} = \{f(0,x) = \sin(2\pi x), f(t,\pm 1) = 0\} \tag{13}$$



Figure 2: Solution for the Heat Equation Example. [11]

2. Nonlinear Example 1

$$4\frac{d^2 f(x)}{dx^2} + 2\left(\frac{df(x)}{dx}\right)^2 + f(x) = 0, \mathcal{BC} = \{f(0) = 1, f'(0) = 0\} \tag{14}$$

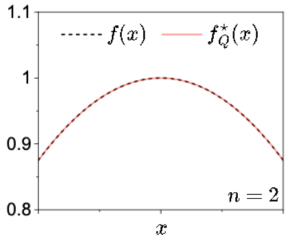


Figure 3: Solution of Nonlinear Example 1 [11]

3. Nonlinear Example 2

$$\frac{d^2 f(x)}{fx^2} - 2f(x)^2 + x = 0, \mathcal{BC} = \{f(-1) = -0.1, f(1) = 0.1\} \tag{15}$$

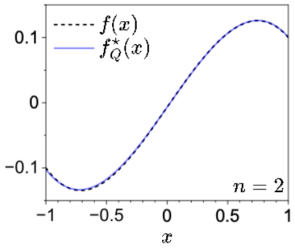


Figure 4: Solution of Nonlinear Example 2 [11]

The model performs well for all examples shown. The errors are almost negligible and decrease as the number of qubits used (shown as n on the figures) increase. The experiments were conducted on classical simulations of quantum hardware and are therefore currently computationally costly, however based on these results it is projected this algorithm will become useful in the early fault-tolerant era [11].

2.2.5 Generalised Quantum Singular Value Transformation

The generalised QSVT (GQSVT) framework extends the standard QSVT algorithm by lifting the restrictions on coefficient phases and polynomial parity [9]. The standard QSVT restricts the target polynomial $p(x)$ to have real coefficients and definite parity. However, the generalised version allows complex polynomials and indefinite parity. These generalisations are inherited from developments in generalised quantum signal processing (GQSP) which is itself an extension of the quantum signal processing framework that QSVT was built on [10]. Eliminating the restrictions on real polynomials and Hamiltonians imposed by QSVT will allow us to solve complex-valued differential equations. GQSVT also promises a simplification in complexity for the crucial and time-consuming step of calculating phase factors. The task was of complexity $\mathcal{O}(d^2)$ for QSVT, GQSVT promises a complexity of $\mathcal{O}(d)$ [9]. This should speed up calculations even for problems which are currently possible via the existing method.

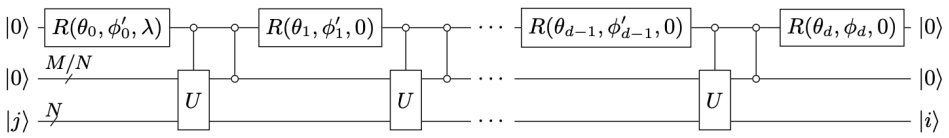


Figure 5: Example circuit for the generalised quantum singular value transformation for a Hermitian matrix A, block encoded as U. R is a rotation matrix analogous to Π for QSVT [9].

3 Research Plan

3.1 Method

This research will be primarily theoretical and computational. The goal is to design, simulate and evaluate novel quantum algorithms for solving nonlinear, complex-valued partial differential equations (PDEs). Development will be conducted using using *Mathematica* for symbolic manipulation and proof-of-concept derivations. Simulations will be created using *Python* based simulators such as *Qiskit* and *Pennylane* and results will be compared to classical solving algorithms. Efforts wil be split into two phases. First, extending the current QSVT algorithm to GQSVT and second, integrating data-informed terms into effective Hamiltonians to enable solving PDEs in data-rich regimes. This research represents a new research direction within QUISA.

Proposed Plan:

1. Reproduce the Current Physics-Informed QSVT Method:
   (a) Implement the standard QSVT framework in *Python*.
   (b) Choose an appropriate problem.
   (c) Implement a physics-informed Hamiltonian PDE solver in *Python*.
   (d) Compare quantum solution against classical benchmarks
2. Extend to GQSVT:
   (a) Create the GQSVT algorithm in *Python*.
   (b) Choose an appropriate problem.
   (c) Implement a new physics-informed Hamiltonian PDE solver in *Python*.
   (d) Compare quantum solution against classical benchmarks
3. Experiment with more data-informed terms in the Hamiltonian

Timeline:

Semester 1 2025: Research Proposal.

Semester 2 2025: Recreation of QSVT Physics-informed method. Begin extension to GQSVT.

Semester 1 2026: Completion of a working GQSVT enhanced algorithm, including simulation and evaluation in comparison to classical algorithms.

Semester 2 2026: Experimentation with data-informed terms.

3.2 Preliminary Work

In semester 1 2025, work consisted of a thorough literature review, achieving an in-depth understanding of QSVT and the physics-informed Hamiltonian approach for PDEs. This was pivotal in developing a thesis that could both build on the newest developments in the field and be achievable within the time-frame of a Master's course. A working *Python* implementation of QSVT has been created and work in underway on implementing the current physics-informed Hamiltonian approach using QSVT.

References

[1] András Gilyén, Yuan Su, Guang Hao Low, and Nathan Wiebe. Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*, pages 193–204, New York, NY, USA, jun 23 2019. ACM. [Online; accessed 2025-05-18].

[2] András Pál Gilyén. *Quantum Singular Value Transformation and Its Algorithmic Applications*. 2019. [Online; accessed 2025-05-25].

[3] Aram W. Harrow, Avinatan Hassidim, and Seth Lloyd. Quantum algorithm for solving linear systems of equations. https://arxiv.org/abs/0811.3171, nov 19 2008.

[4] Joshua Izaac and Jingbo Wang. *Computational Quantum Mechanics*. Springer, feb 15 2019. [Online; accessed 2025-05-25].

[5] Chen Lan, Wei Li, and Huifang Geng. Pseudo-Hermitian Chebyshev differential matrix and non-Hermitian Liouville quantum mechanics. *Journal of Physics A: Mathematical and Theoretical*, 57(46):465302, nov 1 2024. [Online; accessed 2025-05-25].

[6] Guang Hao Low and Isaac L. Chuang. Optimal Hamiltonian Simulation by Quantum Signal Processing. https://arxiv.org/abs/1606.02685, jun 8 2016.

[7] Michael Lubasch, Jaewoo Joo, Pierre Moinier, Martin Kiffner, and Dieter Jaksch. Variational quantum algorithms for nonlinear problems. https://arxiv.org/abs/1907.09032, jul 21 2019.

[8] John M. Martyn, Zane M. Rossi, Andrew K. Tan, and Isaac L. Chuang. Grand unification of quantum algorithms. *PRX Quantum*, 2(4), dec 3 2021. [Online; accessed 2025-05-16].

[9] Danial Motlagh and Nathan Wiebe. Generalized Quantum Signal Processing. *PRX Quantum*, 5(2), jun 28 2024. [Online; accessed 2025-05-25].

[10] Christoph Sünderhauf. Generalized Quantum Singular Value Transformation. https://arxiv.org/abs/2312.00723, dec 1 2023.

[11] Hsin-Yu Wu, Annie E. Paine, Evan Philip, Antonio A. Gentile, and Oleksandr Kyriienko. Quantum algorithm for solving nonlinear differential equations based on physics-informed effective Hamiltonians. https://arxiv.org/abs/2504.13174, apr 17 2025.